

RESEARCH ARCHITECTURE

Layer 1 — Physics:

Baghewala → heavy oil → CSS → temperature → viscosity → mobility → production.

Layer 2 — Artificial Lift:

Viscosity → inflow → SRP → dynamometer → rod floating → failures → pump efficiency.

Layer 3 — Intelligence:

Data → prediction → digital twin → state estimation → optimization → closed-loop recommendations.

Layer 4 — Existing Industry:

What ONGC/SLB/Halliburton/Weatherford/etc. already do → **what is still missing.**

Layer 5 — SIH Strategy:

Novelty → feasibility → demo → measurable KPIs → differentiation → **why judges should select you.**

LAYER 1 – PHYSICS LAYER

Yes. Let's reset and do **only Section 1**, properly, in the documentation format we agreed on.

1. Baghewala Problem Context

1.1 Field and Reservoir

Baghewala Field in Rajasthan produces **heavy crude oil from the Jodhpur Sandstone reservoir**. According to the SIH problem statement, the reservoir presents several conditions that make conventional/primary oil production difficult:

- **High crude-oil viscosity**
- **High asphaltene content**
- **Low reservoir pressure**
- **Low reservoir temperature: 46–48°C**
- Consequently, **poor oil mobility under primary recovery**

These conditions make it difficult for the crude oil to move naturally through the reservoir toward the wellbore.

1.2 What is physically happening?

The fundamental problem can be represented as:

Heavy crude oil



High viscosity



High resistance to flow



Low oil mobility



Poor reservoir inflow



Poor primary production

The **low reservoir pressure** further limits the natural driving force available to move the oil toward the well.

Therefore, simply drilling a well is not sufficient to achieve desirable sustained production.

1.3 Why is temperature important?

For heavy oil, temperature has a strong influence on viscosity.

At the relatively low reservoir temperature reported for Baghewala:

$$T_r = 46 - 48^\circ C$$

the crude remains highly viscous.

The fundamental physical relationship is:

$$T \uparrow \Rightarrow \mu_o \downarrow$$

where:

- T = oil/reservoir temperature
- μ_o = oil viscosity

Therefore, **heating the reservoir can make the oil easier to move.**

This is the fundamental reason thermal recovery is relevant to Baghewala.

1.4 Why is CSS used?

The SIH problem specifically identifies **Cyclic Steam Stimulation (CSS)** as an important thermal recovery method for the field.

The basic physical idea is:

Steam injection



Heat enters reservoir



Oil temperature increases



Oil viscosity decreases



Oil mobility improves



Oil flows more easily toward the well



Production improves

Thus, CSS is fundamentally a **thermal-flow intervention**.

It attempts to modify the reservoir's thermal condition so that the heavy crude becomes more mobile.

1.5 Why does the problem remain difficult?

The improvement produced by heating is not permanent.

During production, the reservoir gradually loses heat:

Production



Reservoir cools



Temperature decreases



Viscosity increases



Mobility decreases



Production declines

Therefore, the well's condition is **dynamic rather than static**.

The operator must repeatedly manage the thermal state of the reservoir through CSS while also maintaining effective artificial lift.

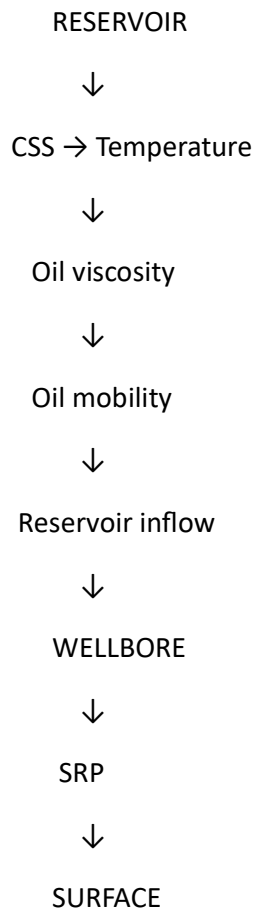
This creates the need for **continuous operational optimization**, rather than a one-time intervention.

1.6 Why artificial lift is also required

Even after thermal stimulation improves oil mobility, the oil still has to be transported from the wellbore to the surface.

The SIH problem therefore identifies **artificial lift**, including **Sucker Rod Pump (SRP)** operation, as another critical component of sustained production.

So the physical system is:



This is the first indication that **CSS and SRP cannot be considered completely independent systems**.

1.7 The fundamental engineering problem

The real challenge at Baghewala can therefore be expressed as:

How can we maintain favourable reservoir thermal conditions and operate the artificial-lift system so that heavy oil is produced efficiently, continuously, and safely?

This is more complicated than simply trying to maximize oil rate.

The operating decisions involve competing objectives:

More steam



Better heating



Potentially better oil mobility



Higher production

BUT

More steam



Higher energy / operating requirement



Higher steam consumption



Higher cost

Similarly:

Higher pump speed



Potentially higher pumping capacity

BUT

If reservoir inflow is insufficient



Poor pump fillage



Unfavourable rod/pump behaviour



Higher mechanical risk

These competing effects are what ultimately create the **optimization problem**.

1.8 FACT vs PHYSICS vs ASSUMPTION

FACT — from the SIH Problem Statement

- Baghewala Field is in Rajasthan.
- Production is from the Jodhpur Sandstone reservoir.
- The crude is heavy oil.
- The reservoir has high crude viscosity.
- High asphaltene content is present.
- Reservoir pressure is low.
- Reservoir temperature is **46–48°C**.
- Oil mobility under primary recovery is poor.
- CSS and artificial lift are important for sustained production.

PHYSICS / ENGINEERING INTERPRETATION

$$T \uparrow \Rightarrow \mu_o \downarrow$$

$$\mu_o \downarrow \Rightarrow \text{mobility} \uparrow$$

$$\text{mobility} \uparrow \Rightarrow \text{reservoir inflow can improve}$$

These relationships explain why thermal stimulation is relevant.

ASSUMPTION

At this stage, we are **not assigning numerical values** to the viscosity-temperature relationship or reservoir-flow response because the SIH problem statement does not provide sufficient data to establish those values.

PROPOSED MODELING VIEW

For our digital twin, we will represent Baghewala as a **dynamic coupled system**:

Thermal state → Viscosity → Mobility → Inflow → Artificial lift → Surface production

1.9 Boundary of Section 1

At this point, we **stop at the physical problem definition**.

We have established:

Baghewala → Heavy oil → High viscosity → Poor mobility → Need for CSS + artificial lift

We have **not yet derived the equations**.

That comes next in:

2. Governing Physics

where we will formally establish the relationships between:

Heat → Temperature → Viscosity → Mobility → Flow

and only then move to the mathematical equations.

Absolutely. I'll keep it as **normal readable text**, not in a code-style block. I'll also avoid the writing-block formatting for these technical sections.

2. Governing Physics

2.1 The Core Physical Chain

The fundamental physics of the Baghewala problem can be represented as:

Steam/Heat Input → Reservoir Temperature ↑ → Oil Viscosity ↓ → Oil Mobility ↑ → Oil Flow ↑

Here:

- **Temperature (T)** represents the thermal state of the reservoir/oil.
- **Oil viscosity (μ_o)** represents the resistance of the crude oil to flow.
- **Oil mobility (λ_o)** represents how easily oil can move through the porous reservoir.
- **Oil flow rate (q_o)** represents the resulting movement of oil toward the well.

The key idea is that **thermal energy changes the properties of the heavy oil, which changes its ability to flow through the reservoir**.

2.2 Heat Transfer → Temperature

During Cyclic Steam Stimulation (CSS), steam is injected into the formation. The steam carries thermal energy into the reservoir.

Therefore:

Steam injection → Heat input → Reservoir temperature increases

The temperature increase is not instantaneous and is not spatially uniform. The region near the wellbore will generally experience stronger thermal influence, while regions farther from the well may remain cooler.

The reservoir's thermal condition is governed by a balance between:

- Heat supplied by injected steam
- Heat conducted into surrounding rock
- Heat carried away by produced fluids
- Heat lost to surrounding formations
- Existing thermal energy stored in the reservoir

For a simplified control-volume representation:

Rate of thermal energy accumulation = Heat input – Heat losses – Heat carried away

The mathematical form of this relationship will be introduced in Section 3, Governing Equations.

2.3 Temperature → Viscosity

This is one of the most important physical relationships in the Baghewala problem.

Heavy-oil viscosity is strongly dependent on temperature.

In general:

Temperature increases → Oil viscosity decreases

Conversely:

Temperature decreases → Oil viscosity increases

Therefore, when CSS heats the reservoir, the crude oil becomes less viscous and can move more easily.

This is particularly important at Baghewala because the SIH problem statement identifies the reservoir temperature as only **46–48°C**, along with high crude-oil viscosity and poor oil mobility.

Important modeling point

We should not assume a numerical viscosity-temperature relationship for Baghewala without supporting data.

The digital twin should ideally calibrate this relationship using:

- Baghewala PVT data
- Laboratory viscosity-temperature measurements
- Field production data
- Validated published field data

If these data are unavailable, an empirical relationship can be used, but it must be explicitly identified as a **modeling assumption**.

2.4 Viscosity → Mobility

Oil mobility describes how easily oil can move through the porous medium.

In general:

Oil mobility is inversely related to oil viscosity.

Therefore:

Viscosity decreases → Mobility increases

and:

Viscosity increases → Mobility decreases

For an oil phase, mobility can be represented as:

$$\lambda_o = (k \times k_{ro}) / \mu_o$$

where:

- λ_o = oil mobility
- k = absolute permeability of the reservoir
- k_{ro} = relative permeability of oil
- μ_o = oil viscosity

This equation shows why reducing viscosity through heating is so important.

If permeability and other conditions remain approximately constant:

$$\mu_o \downarrow \rightarrow \lambda_o \uparrow$$

Thus, heating does not directly “push” the oil toward the well. Instead, it **reduces the resistance to flow**, allowing the existing pressure gradient to move the oil more effectively.

2.5 Mobility → Pressure-Driven Flow

Once oil mobility changes, the ability of the reservoir to transmit oil toward the well also changes.

The basic physical principle is:

Pressure gradient × Oil mobility → Oil flow

Therefore, under comparable pressure conditions:

Mobility increases → Oil inflow can increase

This is the fundamental connection between reservoir heating and production.

CSS therefore does not create additional oil. Instead, it changes the thermal and flow conditions so that the oil already present in the reservoir becomes easier to produce.

2.6 Complete Physical Chain

Combining the relationships:

Steam injection → Reservoir heating → Temperature ↑ → Viscosity ↓ → Mobility ↑ → Reservoir inflow ↑

During the subsequent cooling period, the reverse process occurs:

Heat loss → Temperature ↓ → Viscosity ↑ → Mobility ↓ → Reservoir inflow ↓

This explains why CSS is inherently a **cyclic process**.

The reservoir is heated, production benefits from the improved mobility, and then the reservoir gradually cools and loses some of that benefit.

2.7 Why This Is a Dynamic Problem

The important variables are not independent.

For example:

Temperature at time t → Viscosity at time t → Mobility at time t → Inflow at time t

Furthermore, the current temperature depends on what happened previously.

For example:

- Previous steam injection affects the current reservoir temperature.
- Current temperature affects current viscosity.
- Current viscosity affects current mobility.
- Current mobility affects reservoir inflow.
- Current inflow affects the subsequent production response.

Therefore, Baghewala should be treated as a **dynamic system**, rather than as a collection of independent prediction models.

A simplified representation is:

$$T(t) \rightarrow \mu(t) \rightarrow \lambda(t) \rightarrow q_{in}(t)$$

where each variable changes with time.

2.8 Spatial Nature of the Reservoir

The reservoir is not a single point. Temperature and pressure can vary throughout the formation.

Therefore, more generally:

$$T = T(x, y, z, t)$$

and:

$$P = P(x, y, z, t)$$

Since viscosity depends on temperature:

$$\mu_o = \mu_o(T, x, y, z, t)$$

This means that the region close to the wellbore may be significantly hotter and less viscous than a region farther away.

Proposed simplification for the SIH prototype

Instead of attempting to build a complete three-dimensional reservoir simulator, our first digital-twin implementation can represent the reservoir using a reduced-order thermal state.

For example:

$$T_r(t)$$

or, if sufficient data become available:

[T near-well, T mid-reservoir, T far-reservoir]

This provides a computationally efficient representation while preserving the dominant thermal behaviour required for optimization.

2.9 How the Physics Changes During CSS

A. Injection Phase

Steam is injected into the reservoir.

Steam injection → Thermal energy input → Reservoir temperature increases

As temperature increases:

Temperature ↑ → Viscosity ↓ → Mobility ↑

The objective is to create favourable thermal conditions around the well.

B. Soak Phase

Steam injection is stopped.

The injected heat remains within the formation and redistributes through the reservoir.

The thermal state continues to evolve even though there is no active steam injection.

This phase allows heat to penetrate farther into the formation and interact with the surrounding oil and rock.

C. Production Phase

The well is returned to production.

The heated oil has improved mobility and can move toward the well more easily.

The production response is therefore influenced by:

Reservoir temperature → Oil viscosity → Oil mobility → Reservoir inflow

As production continues, however, heat is gradually lost.

D. Cooling Phase

The reservoir gradually loses thermal energy.

Therefore:

Temperature ↓ → Viscosity ↑ → Mobility ↓ → Inflow ↓

Production consequently tends to decline as the thermal benefit diminishes.

When the production response becomes insufficient, another CSS cycle may be initiated.

2.10 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT — From the SIH Problem Statement

The SIH problem statement identifies:

- Heavy crude oil production at Baghewala
- Jodhpur Sandstone reservoir
- High crude-oil viscosity
- High asphaltene content
- Low reservoir pressure
- Low reservoir temperature of 46–48°C
- Poor oil mobility under primary recovery
- The importance of artificial lift and thermal enhanced oil recovery for sustained production

PHYSICS

The established physical relationships are:

Temperature ↑ → Viscosity ↓

Viscosity ↓ → Mobility ↑

Mobility ↑ + Pressure gradient → Flow can increase

ASSUMPTION

For the initial digital twin, we may represent the reservoir using an effective thermal state rather than a full three-dimensional reservoir model.

This is a **model simplification**, not a statement about the actual physical geometry of Baghewala.

PROPOSED DESIGN

The reservoir component of our digital twin will maintain an estimated chain:

Reservoir thermal state → Oil viscosity → Oil mobility → Reservoir inflow

This estimated reservoir inflow will later become the interface to the wellbore and SRP model.

2.11 Boundary of Section 2

At the end of this section, we have established the physical mechanism:

Heat → Temperature → Viscosity → Mobility → Pressure-driven flow

We have deliberately **not yet introduced the mathematical governing equations**.

The next section is therefore:

3. Governing Equations

There we will mathematically formulate each link in this chain and define **every equation, variable, unit, physical meaning, and modeling assumption**.

3. Governing Equations

This section converts the physical chain established in Section 2 into mathematical relationships:

Heat input → Temperature → Viscosity → Mobility → Pressure-driven oil flow

The equations below form the **physics core of the reservoir component** of our Baghewala digital twin.

3.1 Thermal Energy Balance

The first governing relationship describes how the reservoir temperature changes with time.

For a simplified reservoir control volume:

$$C_{\text{eff}} \frac{dT_r}{dt} = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

Meaning

This equation states:

Rate of thermal energy stored in the reservoir = Heat supplied – Heat lost – Heat carried away by production

Where:

- T_r = effective reservoir temperature, °C or K

- t = time
- C_{eff} = effective thermal capacitance of the reservoir control volume
- $\dot{Q}_{\text{steam}}$ = rate of thermal energy supplied by steam
- $\dot{Q}_{\text{loss}}$ = rate of heat loss to surrounding formations
- $\dot{Q}_{\text{prod}}$ = thermal energy carried away with produced fluids

The thermal capacitance can conceptually be represented as:

$$C_{\text{eff}} = M_{\text{eff}} C_{p,\text{eff}}$$

where:

- M_{eff} = effective mass of rock and fluids being thermally modelled
- $C_{p,\text{eff}}$ = effective specific heat capacity

Physical interpretation

If:

$$\dot{Q}_{\text{steam}} > \dot{Q}_{\text{loss}} + \dot{Q}_{\text{prod}}$$

then:

$$\frac{dT_r}{dt} > 0$$

and the reservoir heats up.

During the cooling period, the opposite can occur:

$$\frac{dT_r}{dt} < 0$$

and reservoir temperature decreases.

3.2 More Detailed Heat-Transfer Representation

The previous equation is a **reduced-order model**.

A more physically detailed reservoir representation would consider temperature variation in space as well as time.

A simplified heat-conduction equation is:

$$\rho_{\text{eff}} C_{p,\text{eff}} \frac{\partial T}{\partial t} = \nabla \cdot (k_T \nabla T) + Q$$

where:

- ρ_{eff} = effective density
- $C_{p,\text{eff}}$ = effective heat capacity
- T = local reservoir temperature
- k_T = effective thermal conductivity
- Q = volumetric heat source
- ∇ = spatial gradient operator

For a full thermal-flow simulator, fluid advection and phase behaviour would also need to be incorporated.

PROPOSED DESIGN

For the SIH prototype, we should **not start with a full 3-D thermal reservoir simulator**.

Instead, we can use the reduced-order energy balance and calibrate its parameters against available field data.

This makes the digital twin computationally fast enough for optimization and what-if simulation.

3.3 Steam Heat Input

The heat supplied by steam can be represented conceptually as:

$$\dot{Q}_{\text{steam}} = \dot{m}_s(h_s - h_{\text{ref}})\eta_{\text{th}}$$

where:

- $\dot{m}_s$ = steam mass injection rate
- h_s = specific enthalpy of injected steam
- h_{ref} = reference enthalpy
- η_{th} = effective thermal-transfer efficiency

The exact formulation depends on whether the model explicitly represents:

- steam quality,
- pressure,
- temperature,
- latent heat,

- sensible heat,
- and heat losses in the injection system.

Important

We should **not assign numerical values to these parameters yet**. They belong in the Baghewala-specific parameter table in Section 5 and should be tied to verified sources.

3.4 Temperature–Viscosity Relationship

The second major governing relationship is the dependence of heavy-oil viscosity on temperature.

A convenient empirical representation can be written as:

$$\ln(\mu_o) = A + \frac{B}{T_K}$$

where:

- μ_o = dynamic oil viscosity
- T_K = absolute temperature in Kelvin
- A, B = empirical calibration parameters

Temperature must be expressed in Kelvin for this particular formulation:

$$T_K = T_{^{\circ}C} + 273.15$$

Therefore:

$$T \uparrow \Rightarrow \mu_o \downarrow$$

provided the fitted relationship has the expected heavy-oil temperature behaviour.

Important modeling restriction

The constants A and B **must not be invented** for Baghewala.

They need to be obtained from:

1. Baghewala PVT/laboratory data, if available;
2. validated field data;
3. or a clearly labelled synthetic calibration for demonstration.

This distinction is critical for our SIH solution because a visually impressive digital twin with arbitrary viscosity parameters would not be scientifically defensible.

3.5 Oil Mobility

Oil mobility is given by:

$$\lambda_o = \frac{k k_{ro}}{\mu_o}$$

where:

- λ_o = oil mobility
- k = absolute permeability
- k_{ro} = oil relative permeability
- μ_o = oil viscosity

Therefore, if permeability and relative permeability remain approximately constant:

$$\mu_o \downarrow \Rightarrow \lambda_o \uparrow$$

This is the mathematical representation of the physical statement:

Heating heavy oil makes it easier to move through the reservoir.

3.6 Darcy's Law

The fundamental pressure-driven flow relationship is Darcy's law.

For one-dimensional radial/linear simplification:

$$q_o = \frac{k k_{ro} A \Delta P}{\mu_o L}$$

where:

- q_o = oil volumetric flow rate
- k = absolute permeability
- k_{ro} = oil relative permeability
- A = effective flow area
- μ_o = oil viscosity
- ΔP = pressure difference driving the flow
- L = characteristic flow distance

Since:

$$\lambda_o = \frac{k k_{ro}}{\mu_o}$$

we can write:

$$q_o = \lambda_o A \frac{\Delta P}{L}$$

This clearly connects the equations:

$$\boxed{T \rightarrow \mu_o(T) \rightarrow \lambda_o \rightarrow q_o}$$

3.7 Reservoir Inflow Relationship

For a reduced-order well model, the reservoir inflow can also be represented using an effective productivity relationship:

$$q_{in} = J_o (P_r - P_{wf})$$

where:

- q_{in} = reservoir inflow rate
- J_o = effective oil productivity index
- P_r = reservoir pressure
- P_{wf} = flowing bottom-hole pressure

The productivity index itself is influenced by reservoir properties and oil viscosity.

Conceptually:

$$J_o \propto \frac{k k_{ro}}{\mu_o}$$

Therefore:

$$T \uparrow \rightarrow \mu_o \downarrow \rightarrow J_o \uparrow \rightarrow q_{in} \uparrow$$

under comparable pressure conditions.

This equation is particularly useful for our digital twin because it provides a direct interface between the **reservoir model** and the later **wellbore/SRP model**.

3.8 Combined Governing System

The most important part of this section is not any individual equation but how they interact.

The complete reduced-order physics chain becomes:

$$C_{\text{eff}} \frac{dT_r}{dt} = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

followed by:

$$\ln(\mu_o) = A + \frac{B}{T_K}$$

then:

$$\lambda_o = \frac{k k_{ro}}{\mu_o}$$

and finally:

$$q_{\text{in}} = J_o (P_r - P_{wf})$$

Therefore:

$$\text{Steam} \rightarrow T_r \rightarrow \mu_o \rightarrow \lambda_o \rightarrow J_o \rightarrow q_{\text{in}}$$

This is the mathematical backbone of Layer 1.

3.9 State-Space Representation for the Digital Twin

For implementation, we can represent the reservoir state as:

$$x_t = [T_r, P_r, \mu_o, q_{\text{in}}]$$

and operational inputs as:

$$u_t = [\dot{m}_s, V_s, t_{\text{soak}}, P_{\text{inj}}]$$

where:

- T_r = reservoir temperature
- P_r = reservoir pressure
- μ_o = oil viscosity
- q_{in} = reservoir inflow
- $\dot{m}_s$ = steam injection rate
- V_s = steam volume
- t_{soak} = soak duration

- P_{inj} = injection pressure

The digital twin can then be represented conceptually as:

$$x_{t+1} = f(x_t, u_t, \theta) + w_t$$

where:

- f = physics model
- θ = model parameters
- w_t = model/process uncertainty

Measurements can be represented as:

$$y_t = h(x_t) + v_t$$

where:

- y_t = measured field data
- h = measurement relationship
- v_t = measurement noise

This structure will later allow the digital twin to combine **physics + real field data + ML**.

3.10 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

The SIH problem identifies the Baghewala reservoir as having low temperature, high viscosity, low pressure and poor oil mobility.

PHYSICS

The governing engineering relationships are:

Thermal energy balance → Reservoir temperature

Temperature → Oil viscosity

Viscosity → Oil mobility

Mobility + pressure gradient → Oil flow

ASSUMPTION

For the initial digital twin:

- The reservoir can be represented by an effective thermal control volume.

- Effective thermal properties can represent the combined rock/fluid system.
- An empirical viscosity-temperature relationship can represent heavy-oil behaviour.
- Darcy-type flow can represent the first-order reservoir inflow response.

These are modeling simplifications and must eventually be calibrated.

PROPOSED DESIGN

The Layer 1 physics engine will use the following computational sequence:

Steam operating conditions

↓

Thermal energy balance

↓

Estimated reservoir temperature

↓

Temperature-viscosity model

↓

Oil mobility

↓

Reservoir inflow/productivity model

↓

Estimated reservoir inflow q_{in}

This q_{in} becomes the input to the next physical layer of the complete well-to-surface model.

3.11 What We Have Established

At this point, we have converted the physical concept into a mathematical chain:

Heat balance

$$C_{\text{eff}} \frac{dT_r}{dt} = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

↓

Viscosity

$$\ln(\mu_o) = A + \frac{B}{T_K}$$

↓

Mobility

$$\lambda_o = \frac{kk_{ro}}{\mu_o}$$

↓

Inflow

$$q_{in} = J_o(P_r - P_{wf})$$

The next section is **4. CSS Physics**, where we will take these equations and explain exactly how they behave across the four CSS stages:

Injection → Soak → Production → Cooling → Next Cycle.

4. CSS Physics

This section explains how the governing physics established in Sections 2 and 3 behaves throughout one complete **Cyclic Steam Stimulation (CSS)** cycle.

The CSS cycle is:

Injection → Soak → Production → Cooling → Next Cycle

The central objective is to temporarily increase the reservoir temperature, reduce heavy-oil viscosity, improve oil mobility, and consequently improve reservoir inflow.

4.1 What is CSS?

Cyclic Steam Stimulation is a thermal recovery method in which the same well is repeatedly used for:

1. **Steam injection**
2. **Soaking**
3. **Oil production**

After production declines and the reservoir cools sufficiently, another steam-injection cycle can be performed.

For Baghewala, CSS is particularly relevant because the SIH problem identifies heavy crude, high viscosity, low reservoir temperature, and poor oil mobility as major production challenges.

The physical purpose of CSS can therefore be summarized as:

Inject heat → reduce viscosity → improve mobility → improve inflow → produce oil → repeat when thermal benefit declines.

4.2 Stage 1 — Steam Injection

During the injection stage, steam is introduced into the reservoir.

The primary effect is the addition of thermal energy:

Steam injection → Thermal energy enters reservoir

The thermal model from Section 3 becomes:

$$C_{\text{eff}} \frac{dT_r}{dt} = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

During active injection:

$$\dot{Q}_{\text{steam}} > 0$$

Therefore, if the heat supplied is greater than the instantaneous heat losses:

$$\frac{dT_r}{dt} > 0$$

and the reservoir temperature increases.

4.2.1 Effect on Oil Viscosity

As the reservoir heats:

$$T_r \uparrow$$

the oil viscosity decreases:

$$\mu_o(T_r) \downarrow$$

Therefore:

$$T_r \uparrow \rightarrow \mu_o \downarrow$$

This is the most important mechanism through which CSS improves heavy-oil mobility.

4.2.2 Effect on Oil Mobility

Recall:

$$\lambda_o = \frac{kk_{ro}}{\mu_o}$$

If viscosity decreases:

$$\mu_o \downarrow \Rightarrow \lambda_o \uparrow$$

Therefore, heating the reservoir increases the ability of oil to move through the porous formation.

4.2.3 Why More Steam Is Not Automatically Better

A critical point for our optimization problem is that:

More steam $\neq$ automatically more profit or maximum oil recovery.

Increasing steam injection may increase thermal input, but it can also increase:

- Steam consumption
- Energy requirement
- Fuel requirement
- Injection cost
- Heat losses
- Water consumption
- Operational constraints

Therefore, the engineering objective is not simply:

Maximize steam injection.

It is closer to:

Find the steam strategy that provides the best production/economic response within operational constraints.

This distinction will later become central to our optimization engine.

4.3 Stage 2 — Soak

After steam injection, the well enters the **soak period**.

During this stage, active steam injection stops.

However, the thermal state of the reservoir continues to evolve.

The injected heat redistributes through:

- Reservoir rock
- Formation fluids
- Near-well region
- Surrounding reservoir volume

Therefore:

$$\dot{Q}_{\text{steam}} \approx 0$$

but:

$$\frac{dT_r}{dt}$$

does not necessarily become zero.

The reservoir is still undergoing heat redistribution and heat loss.

4.3.1 Why Soak Matters

Immediately after injection, the thermal distribution may be highly non-uniform.

The near-well region may have received substantially more heat than the surrounding formation.

The soak period allows the thermal disturbance to evolve before the well is returned to production.

Therefore, soak time becomes an important CSS control variable.

A simplified control vector can be written as:

$$u_{\text{CSS}} = [\dot{m}_s, V_s, t_{\text{soak}}]$$

where:

- $\dot{m}_s$ = steam injection rate
 - V_s = steam volume
 - t_{soak} = soak duration
-

4.4 Stage 3 — Production

After the soak period, the well is opened for production.

The reservoir now has a higher thermal state than before the CSS cycle.

Therefore:

$$T_r^{\text{heated}} > T_r^{\text{initial}}$$

in the effective heated region.

Consequently:

$$\mu_o^{\text{heated}} < \mu_o^{\text{initial}}$$

and:

$$\lambda_o^{\text{heated}} > \lambda_o^{\text{initial}}$$

This creates more favourable conditions for oil to flow toward the well.

4.4.1 Production Response

Using the simplified inflow relationship:

$$q_{\text{in}} = J_o (P_r - P_{wf})$$

and recognizing that:

$$J_o \propto \frac{k k_{ro}}{\mu_o}$$

heating can increase the effective productivity of the well.

Therefore:

Heating → viscosity reduction → higher effective mobility/productivity → improved inflow

The exact production response, however, depends on the actual reservoir and well conditions.

We should therefore avoid assuming that a specific temperature increase will produce a specific oil-rate increase without calibration.

4.5 Production Decline During the Cycle

The thermal benefit does not remain constant.

As production continues, the reservoir loses heat.

Therefore:

$$T_r(t) \downarrow$$

which leads to:

$$\mu_o(t) \uparrow$$

and:

$$\lambda_o(t) \downarrow$$

Consequently:

$$q_{in}(t)$$

can decline.

Conceptually:

Early production

High temperature → Low viscosity → High mobility → Stronger inflow

Later production

Cooling → Higher viscosity → Lower mobility → Reduced inflow

This creates the characteristic decline in production response after thermal stimulation.

4.6 Stage 4 — Cooling

Cooling is not necessarily a separate operational step commanded by the operator.

It is the natural thermal evolution of the reservoir after the heat input has stopped.

The reservoir loses energy through mechanisms such as:

- Conduction into surrounding rock/formations
- Produced fluids carrying thermal energy away
- Other thermal losses represented in the reduced-order model

Therefore:

$$\dot{Q}_{\text{loss}} > 0$$

and the reservoir thermal state gradually decreases.

The physical chain becomes:

$$\begin{array}{c} T_r \downarrow \\ \Downarrow \\ \mu_o \uparrow \\ \Downarrow \\ \lambda_o \downarrow \\ \Downarrow \\ q_{\text{in}} \downarrow \end{array}$$

4.7 Stage 5 — Decision for the Next Cycle

Eventually, the economic and production benefit of the current CSS cycle may become insufficient.

This creates the central decision:

When should the next CSS cycle begin?

A simplistic strategy would be:

Start the next cycle after a fixed number of days.

But this does not account for the actual state of the well.

A better digital-twin approach is to estimate the current state:

$$x_t = [T_r, P_r, \mu_o, q_{\text{in}}]$$

and determine whether another thermal intervention is justified.

For example, the system could evaluate:

- Current estimated reservoir temperature
- Estimated viscosity
- Current production rate
- Production decline rate
- Expected response to additional steam
- Steam requirement
- Expected incremental oil
- Operational constraints
- Economic benefit

This converts CSS scheduling from a **fixed schedule** into a **state-dependent optimization problem**.

4.8 CSS as a Feedback-Control Problem

This is an important concept for our digital twin.

Instead of thinking of CSS as:

Inject → Wait → Produce

we can represent it as a feedback loop:

Measure current well state



Estimate reservoir thermal state



Predict future production



Simulate possible CSS actions



Select the best feasible action



Execute after operator approval



Measure new response



Update digital twin



Optimize again

This is fundamentally different from simply predicting production.

The system becomes a **decision-support and optimization engine**.

4.9 CSS Control Variables

The variables that can potentially be optimized include:

Steam-related controls

- Steam injection rate
- Steam volume
- Injection duration
- Steam pressure
- Steam quality, where reliable measurements are available

Timing-related controls

- Soak duration
- Production start time
- Production duration
- Time between CSS cycles

Reservoir response variables

- Reservoir temperature
- Reservoir pressure
- Oil viscosity
- Oil mobility

- Reservoir inflow
- Expected production rate

Not all of these will necessarily be controllable in the actual field. The final control vector must be restricted to variables that operators can practically change.

4.10 Why CSS Optimization Is Difficult

There are multiple competing effects.

For example:

Higher steam input

- More heat
- Lower viscosity
- Better mobility
- Potentially higher production

But also:

- Higher steam consumption
- Higher energy cost
- Higher operating cost

Similarly, **longer soak time** may allow greater thermal redistribution, but excessive waiting also means lost production time.

Therefore, the optimum is not necessarily:

- Maximum steam
- Maximum temperature
- Maximum soak time
- Maximum production duration

Instead, we need to find a **balanced operating point**.

4.11 CSS State–Control Representation

For the digital twin, we can represent the reservoir state as:

$$x_t = [T_r, P_r, \mu_o, q_{in}]$$

and the CSS control vector as:

$$u_t = [\dot{m}_s, V_s t_{\text{soak}}]$$

The system can then be represented as:

$$x_{t+1} = f(x_t, u_t)$$

where f represents the thermal and flow behaviour of the reservoir.

The digital twin's job is to estimate x_t , simulate different u_t , and determine which feasible control produces the best expected outcome.

4.12 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

The SIH problem identifies CSS as a critical thermal enhanced-oil-recovery mechanism for Baghewala because of the field's heavy crude, high viscosity, low temperature and poor oil mobility.

PHYSICS

The CSS thermal mechanism is:

Steam injection → Heat addition → Temperature increase → Viscosity reduction → Mobility improvement → Better oil flow

During cooling:

Heat loss → Temperature decrease → Viscosity increase → Mobility reduction → Declining inflow

ASSUMPTION

For our reduced-order model:

- The reservoir can initially be represented by an effective thermal state.
- Thermal properties can be represented using calibrated effective parameters.
- The viscosity-temperature relationship can be represented by an empirical model.
- Reservoir inflow can be approximated using the reduced-order productivity relationship.

PROPOSED DESIGN

The digital twin should not use a fixed CSS recipe.

Instead, it should continuously evaluate:

Current state → Candidate CSS strategy → Predicted thermal response → Predicted production → Cost/constraint evaluation → Recommended action

This will later form the basis of the **CSS optimization engine**.

4.13 The Complete CSS Cycle

The complete physical sequence is:

1. Injection

Steam adds thermal energy.



2. Heating

Reservoir temperature increases.



3. Soak

Heat redistributes through the formation.



4. Production

Heated oil has lower viscosity and improved mobility.



5. Cooling

The reservoir progressively loses thermal energy.



6. Decline

Viscosity increases and mobility/inflow can decline.



7. Decision

Determine whether and when another CSS cycle provides sufficient benefit.



8. Next CSS Cycle

Repeat the process with an improved operating strategy.

4.14 Why This Section Matters for Our Winning Solution

The important insight is that **CSS should not be treated as an isolated steam-injection module.**

It creates a changing reservoir state:

CSS → Thermal State → Viscosity → Mobility → Inflow

That changing inflow will subsequently interact with the artificial-lift system.

Therefore, the final well-to-surface model will eventually connect:

CSS thermal state → Reservoir inflow → Wellbore conditions → SRP behaviour → Surface production

That connection is what will allow our solution to move beyond a conventional CSS optimization dashboard toward a true **well-to-surface digital twin.**

Next: 5. Baghewala-Specific Parameters

In the next section, we will create a **Baghewala Parameter Truth Table** containing the actual reported values, their sources, units, and confidence levels. We will explicitly separate **SIH-stated values, OIL-reported values, published values, and assumptions**, especially where publicly available numbers conflict.

5. Baghewala-Specific Parameters

This section is important because **we must not mix Baghewala facts with generic heavy-oil assumptions.**

There are currently **two important sources of field parameters**:

1. **SIH 2026 Problem Statement** — the baseline source for the competition.
2. **Oil India Limited (OIL) public technical information** — useful for additional field parameters and operating data.

Where the values differ, we will **preserve both rather than silently reconcile them.**

5.1 Baghewala Parameter Truth Table

Parameter	Value	Source	Confidence	How we use it
Field	Baghewala Field, Rajasthan	SIH PS	Very High	Fixed field context
Reservoir	Jodhpur Sandstone	SIH PS / OIL	Very High	Reservoir model
Crude type	Heavy crude oil	SIH PS / OIL	Very High	Fluid model
API gravity	17–19° API	SIH PS	Very High	Competition baseline
API gravity	14–17° API	OIL public information	High	Additional field reference
Reservoir temperature	46–48°C	SIH PS	Very High	Primary thermal baseline
BHT	~50°C	OIL	High	Field reference
Reservoir pressure	Low	SIH PS	Very High	Qualitative constraint
Reservoir pressure	~116 ksc	OIL technical presentation	High	Quantitative model reference
Average reservoir depth	—	Not specified in SIH PS	—	Do not assume
Average depth	~1150 m	OIL	High	Wellbore model
Oil viscosity	High	SIH PS	Very High	Core physics
Oil viscosity	~10,000–13,000 cP @ 50°C, shear rate 50 s⁻¹	OIL	High	Viscosity model
Asphaltene	High	SIH PS	Very High	Fluid/operational consideration
CSS	Required/important	SIH PS	Very High	Thermal EOR model
CSS injection period	—	SIH PS does not specify	—	Do not assume

Parameter	Value	Source	Confidence	How we use it
CSS injection period	14–21 days	OIL technical presentation	High	CSS simulation range
Steam injection rate	—	SIH PS does not specify	—	Do not assume
Steam injection rate	~3100 kg/hr	OIL technical presentation	High	CSS simulation range
Steam pressure	—	SIH PS does not specify	—	Do not assume
Steam pressure	~85–97 ksc	OIL technical presentation	High	CSS constraint/reference
Steam temperature	—	SIH PS does not specify	—	Do not assume
Steam temperature	280–305°C	OIL technical presentation	High	Thermal input reference
Steam quality	—	SIH PS does not specify	—	Do not assume
Steam quality	60–70%	OIL technical presentation	High	Thermal model reference
Water consumption	—	SIH PS does not specify	—	Do not assume
Water consumption	~5500 kg/hr	OIL technical presentation	High	Resource/energy model
HSD consumption	—	SIH PS does not specify	—	Do not assume
HSD consumption	~220 kg/hr	OIL technical presentation	High	Energy/cost model
Soak period	—	SIH PS does not specify	—	Do not assume

Parameter	Value	Source	Confidence	How we use it
Soak period	50–60% of injection time	OIL technical presentation	High	CSS simulation range
CSS cycles completed	—	SIH PS does not specify	—	Do not assume
CSS cycles completed	39 as of 22 June 2025	OIL technical presentation	High	Historical calibration/reference
Current production	—	SIH PS does not specify	—	Do not assume
Current production	~655 bbl/day on 10 July 2025	OIL technical presentation	High	Historical benchmark

The SIH problem statement specifically establishes the heavy-oil/Jodhpur Sandstone context, high viscosity, high asphaltene content, low pressure, and **46–48°C** reservoir temperature.

OIL's current Rajasthan-field information reports Jodhpur Sandstone at an average depth of approximately **1150 m**, crude viscosity of **10,000–13,000 cP at 50°C**, and use of SRP for artificial lift. ([Oil India Limited](#))

The more detailed 2025 OIL presentation reports approximately **116 ksc reservoir pressure, 14–17° API, 10,000–13,000 cP at 50°C, ~655 bbl/day production, and 39 completed CSS cycles** for the Jodhpur Sandstone producing zone. (www.slideshare.net)

5.2 The API Gravity Conflict

We have an important discrepancy:

SIH Problem Statement

17–19° API

OIL public technical material

14–17° API

Older OIL documentation also reports approximately **14–18° API**, while another older technical document reports **14–20° API** and viscosity of **5,000–13,000 cP at reservoir conditions**. ([Oil India Limited](#))

What should we use?

For the **SIH presentation**, our primary classification should be:

17–19° API heavy crude — as specified by the SIH problem statement.

For the **digital-twin simulation**, we should not hard-code a single API value.

Instead:

$$API \in [14, 19]$$

can be treated as a **field-reference range**, with the exact value being a calibration parameter when well-specific data are available.

This is much more defensible than pretending the public sources agree perfectly.

5.3 Temperature Conflict

We similarly have:

SIH

$$T_r = 46 - 48^\circ C$$

OIL

$$BHT \approx 50^\circ C$$

These values are actually reasonably close, but they are not necessarily identical measurements.

We should therefore distinguish:

- **Reservoir temperature stated by SIH:** 46–48°C
- **OIL-reported BHT:** approximately 50°C

For the competition model, use **46–48°C as the baseline reservoir condition**, because that is explicitly stated in the problem statement.

The ~50°C OIL value can be used as an external validation/reference range.

5.4 Viscosity — One of Our Most Important Parameters

OIL reports:

$$\mu_o \approx 10,000 - 13,000 \text{ cP @ } 50^\circ C$$

at a reported shear rate of:

$$\dot{\gamma} = 50 \text{ s}^{-1}$$

This is extremely important for our model.

It tells us that Baghewala crude is not simply "slightly viscous."

It is a **very high-viscosity fluid**, which explains why thermal stimulation is so important.

OIL also states that Baghewala's heavy oil has viscosity in the thousands of centipoise and is not regularly producible through conventional methods; SRP combined with thermal EOR is used for production. ([Oil India Limited](#))

5.5 Why the Shear Rate Matters

The reported viscosity is associated with:

$$\dot{\gamma} = 50 \text{ s}^{-1}$$

This matters because heavy crude can exhibit complex rheological behaviour.

Therefore, we should **not treat 10,000–13,000 cP as a universal constant viscosity for the entire well**.

Instead:

$$\mu_o = \mu_o(T, \dot{\gamma}, \text{fluid state})$$

should ideally be used.

For our first prototype, however, we can simplify this to:

$$\mu_o = \mu_o(T)$$

and later add shear dependence if sufficient data are available.

That distinction should be explicitly stated in our methodology.

5.6 Reservoir Depth

OIL reports the Jodhpur Sandstone producing zone at approximately:

$$1150 \text{ m}$$

average depth. ([Oil India Limited](#))

This parameter becomes important later for:

- Hydrostatic pressure
- Wellbore pressure
- Rod-string length

- SRP dynamics
- Temperature distribution
- Fluid lifting requirements
- Pump setting depth

Therefore, depth will become much more important when we move from **Layer 1** → **Layer 2**.

5.7 CSS Operating Parameters

The OIL technical presentation gives the following reference range for a CSS cycle:

Injection duration

14 – 21 *days*

Steam injection rate

~ 3100 *kg/hr*

Steam pressure

~ 85 – 97 *ksc*

Steam temperature

280 – 305°C

Steam quality

60 – 70%

Water consumption

~ 5500 *kg/hr*

HSD consumption

~ 220 *kg/hr*

Soak period

50 – 60% of injection time

These should be treated as **reported operating references**, not universal optimal values.

(www.slideshare.net)

5.8 Why These Parameters Matter to Our Digital Twin

Each parameter affects a different part of our physics model.

Parameter	Twin component affected
Steam rate	Thermal energy input
Steam volume	Total heat input
Steam pressure	Injection constraint / thermal delivery
Steam temperature	Enthalpy/thermal input
Steam quality	Effective thermal energy
Injection duration	Cumulative heating
Soak time	Heat redistribution
Reservoir temperature	Thermal state
Oil viscosity	Fluid-flow resistance
Reservoir pressure	Driving force
Depth	Wellbore/hydrostatic model
Production rate	Model output
Water consumption	Resource cost
HSD consumption	Energy/cost
CSS cycle number	Historical state / cycle-dependent behaviour

This is exactly why a digital twin is useful: these parameters should **not be modelled independently**.

5.9 Historical CSS Evidence

OIL's public material states that CSS has been implemented at Baghewala, including the first pilot at BGW-08 in 2018, and reports multiple CSS cycles across several wells.

(www.slideshare.net)

OIL also describes CSS as a three-stage process:

Injection → Soaking → Production

and its older technical documentation describes injection followed by a shut-in/soak period and then production using SRP. ([Oil India Limited](#))

This supports our Section 4 physical representation.

5.10 Parameter Confidence Classification

For our final documentation, I recommend using **four confidence levels**.

CONFIDENCE A — Direct SIH Problem Statement

Use these as the competition's authoritative baseline:

- Heavy crude
- Jodhpur Sandstone
- 17–19° API
- High viscosity
- High asphaltene
- Low reservoir pressure
- 46–48°C
- Poor oil mobility
- CSS + artificial lift requirement

CONFIDENCE B — OIL Official/Public Technical Information

Examples:

- ~1150 m average depth
- 10,000–13,000 cP @ 50°C
- ~50°C BHT
- ~116 ksc reservoir pressure
- CSS operating ranges
- Historical CSS cycles

CONFIDENCE C — Published/Older Technical Information

Useful for historical context and cross-checking, but should not override the current SIH/OIL values without justification.

CONFIDENCE D — Our Assumptions

Examples:

- Effective thermal capacitance
- Effective heat-loss coefficient
- Viscosity-temperature curve coefficients
- Relative permeability behaviour
- Reservoir thermal radius
- Synthetic sensor values
- Failure thresholds

These must be clearly marked **ASSUMED / SYNTHETIC** in the prototype.

5.11 What We Should NOT Do

We should **not** create a single table saying:

"Baghewala has 15° API, 50°C, 116 ksc, 12,000 cP"

without explaining where each number came from.

That creates a false impression that all measurements belong to the same well, date, and operating condition.

Instead, our model should maintain metadata:

Parameter → Value → Unit → Source → Date → Well/zone → Confidence

For example:

Oil viscosity → 10,000–13,000 cP → @50°C, 50 s⁻¹ → OIL → Jodhpur Sandstone → Confidence B

This is also a strong **data-engineering feature** for our eventual digital twin.

5.12 Final Baghewala Baseline for Our Prototype

For the initial synthetic/physics-calibrated demonstration, I would use:

Reservoir: Jodhpur Sandstone

Temperature baseline: 46–48°C

Oil classification: 17–19° API for SIH baseline

Viscosity reference: 10,000–13,000 cP @50°C, 50 s⁻¹

Depth reference: ~1150 m

Reservoir pressure reference: ~116 ksc, explicitly tagged as OIL-reported rather than SIH baseline

Steam rate reference: ~3100 kg/hr

Steam temperature: 280–305°C

Steam pressure: ~85–97 ksc

Steam quality: 60–70%

Injection duration: 14–21 days

Soak: 50–60% of injection time

These values should be treated as **calibration/reference ranges**, not claimed optimal settings. (www.slideshare.net)

5.13 The Most Important Takeaway

We now have a defensible parameter hierarchy:

SIH Problem Statement

→ defines the **competition problem**

OIL technical information

→ provides **real field operating ranges**

Physics equations

→ explain how these parameters interact

Our assumptions

→ fill only the gaps necessary to build the prototype

Digital Twin

→ continuously estimates the actual state instead of assuming every parameter is fixed.

That gives us a much stronger foundation than simply feeding the numbers into an ML model.

Next: 6. Assumptions & Simplifications — where we will explicitly decide what our Baghewala digital twin will model, what it will simplify, what it will ignore, and why those simplifications are scientifically defensible.

6. Assumptions & Simplifications

This section defines **what our Baghewala digital twin will assume, what it will simplify, and what it will deliberately not claim to model.**

This is important because a winning solution should not pretend to be a complete reservoir simulator. It should demonstrate that we understand the real physics and have made **controlled engineering simplifications** to achieve a fast, usable digital twin.

6.1 Why Simplification Is Necessary

A complete physical model of Baghewala would potentially need to represent:

- 3-D reservoir geometry
- Rock properties
- Multiphase flow
- Steam condensation
- Heat conduction
- Heat convection
- Pressure distribution
- Temperature distribution
- Heavy-oil rheology
- Asphaltene behaviour
- Wellbore hydraulics
- SRP dynamics
- Sucker-rod elasticity
- Pump behaviour
- Surface equipment
- Operational constraints

Building all of this from first principles would require extensive field data and significant computational resources.

For an SIH prototype, the better approach is:

Retain the dominant physics, simplify secondary phenomena, and use ML/data-driven models where first-principles modelling becomes impractical.

6.2 Assumption 1 — Reduced-Order Reservoir Representation

ASSUMPTION

Instead of representing the entire reservoir in a high-resolution 3-D grid, we initially represent the heated reservoir using an **effective thermal control volume**.

The primary thermal state is:

$$T_r(t)$$

where T_r is the effective reservoir temperature.

Why?

The main Layer 1 objective is to capture:

$$\text{Steam} \rightarrow T_r \rightarrow \mu_o \rightarrow \lambda_o \rightarrow q_{\text{in}}$$

rather than reproduce every spatial detail of the reservoir.

Limitation

A single temperature cannot represent the true spatial thermal distribution.

Therefore, this model is a **reduced-order approximation**, not a replacement for a full reservoir simulator.

6.3 Assumption 2 — Effective Thermal Properties

We use effective properties such as:

- Effective thermal capacitance
- Effective thermal conductivity
- Effective heat-loss coefficient

instead of explicitly modelling every individual rock and fluid component.

For example:

$$C_{\text{eff}} = M_{\text{eff}} C_{p,\text{eff}}$$

where the effective parameters represent the combined thermal behaviour of the modelled reservoir volume.

Purpose

This reduces the number of unknown parameters that must be calibrated.

6.4 Assumption 3 — Temperature Is the Dominant Driver of Viscosity

The first version of the model assumes:

$$\mu_o = \mu_o(T)$$

That means temperature is treated as the primary variable controlling oil viscosity.

This is a deliberate simplification.

In reality, heavy-oil rheology may also depend on:

- Shear rate
- Pressure
- Composition
- Asphaltene behaviour
- Fluid history
- Temperature history

The OIL-reported viscosity itself is associated with a particular temperature and shear-rate condition, so we should not treat the reported viscosity as a universal constant.

PROPOSED EVOLUTION

Version 1:

$$\mu_o = f(T)$$

Later version:

$$\mu_o = f(T, P, \dot{\gamma}, \text{fluid state})$$

The second version can be introduced if sufficient data become available.

6.5 Assumption 4 — Darcy-Type Flow

For the first-order reservoir inflow model, we assume that Darcy-type flow is an adequate approximation:

$$q_o = \frac{kk_{ro}A}{\mu_o} \frac{\Delta P}{L}$$

This captures the dominant relationship:

Higher mobility + pressure gradient → higher flow potential.

Why this is reasonable

We are not trying to replace a commercial reservoir simulator.

We need a computationally lightweight model capable of answering:

“If the thermal state changes, what happens to expected reservoir inflow?”

That is sufficient for the first version of the digital twin.

6.6 Assumption 5 — Effective Relative Permeability

The first version does not attempt to build a detailed temperature-dependent multiphase relative-permeability model.

Instead, k_{ro} is treated as an effective parameter or calibrated function.

Therefore:

$$\lambda_o = \frac{k k_{ro}}{\mu_o}$$

is primarily driven by changes in μ_o .

Limitation

Actual thermal EOR behaviour can involve:

- Water saturation changes
- Steam condensation
- Relative-permeability changes
- Multiphase flow
- Changing fluid composition

These effects should be added if the available data justify the increased model complexity.

6.7 Assumption 6 — Effective Reservoir Pressure

The initial model uses an effective reservoir pressure:

$$P_r(t)$$

rather than resolving a complete pressure field throughout the reservoir.

The pressure-driven inflow is then represented by:

$$q_{\text{in}} = J_o(P_r - P_{wf})$$

where P_{wf} is flowing bottom-hole pressure.

Why?

The digital twin eventually needs to connect reservoir inflow to the wellbore and SRP system.

A reduced-order pressure representation provides that interface without requiring a full pressure-grid simulation.

6.8 Assumption 7 — Steam Input Is Represented Through Thermal Energy

Rather than modelling every detail of steam behaviour in the first prototype, steam injection is converted into an effective thermal-energy input:

$$\dot{Q}_{\text{steam}} = f(\dot{m}_s, h_s, \eta_{\text{th}})$$

where:

- $\dot{m}_s$ = steam mass rate
- h_s = steam enthalpy
- η_{th} = effective thermal-transfer efficiency

This allows the digital twin to ask:

“How much thermal energy is entering the reservoir?”

without initially requiring a full steam-condensation simulator.

6.9 Assumption 8 — Heat Loss Is Represented by an Effective Term

Instead of explicitly modelling every surrounding formation layer, the initial model uses:

$$\dot{Q}_{\text{loss}}$$

as an effective heat-loss term.

A simple implementation could be:

$$\dot{Q}_{\text{loss}} = U_{\text{eff}}(T_r - T_{\text{sur}})$$

where:

- U_{eff} = effective heat-loss coefficient

- T_r = reservoir temperature
- T_{sur} = surrounding formation temperature

This parameter can later be calibrated using the observed cooling behaviour after CSS.

6.10 Assumption 9 — CSS Stages Are Explicit States

The digital twin will explicitly distinguish:

1. Injection
2. Soak
3. Production
4. Cooling

This is preferable to treating CSS as one continuous process.

For example:

Injection

$$\dot{Q}_{\text{steam}} > 0$$

Soak

$$\dot{Q}_{\text{steam}} \approx 0$$

while thermal redistribution continues.

Production

$$q_o > 0$$

with heat being removed through produced fluids.

Cooling

$$\dot{Q}_{\text{loss}} > 0$$

with declining thermal state.

This state-based structure will later make optimization much easier.

6.11 Assumption 10 — Field Measurements Are Imperfect

Real field measurements should not be treated as perfect ground truth.

Measurements may contain:

- Sensor noise
- Missing values
- Communication gaps
- Calibration errors
- Time synchronization errors
- Outliers

Therefore, the digital twin should distinguish between:

Physical state

and

Measured observation

Mathematically:

$$y_t = h(x_t) + v_t$$

where:

- x_t = actual/estimated physical state
- y_t = measured data
- v_t = measurement noise

This becomes particularly important when the model is updated continuously from field data.

6.12 Assumption 11 — Synthetic Data Will Be Clearly Labelled

A major issue for the SIH prototype is likely to be limited access to real-time Baghewala SCADA and sensor data.

Therefore, if synthetic data are required, they will be generated from:

- Physics equations
- Publicly reported Baghewala ranges
- Reasonable noise models
- Controlled operational scenarios

But we must **never present synthetic data as actual OIL field data**.

For example:

Correct:

“Physics-calibrated synthetic Baghewala scenario.”

Incorrect:

“Real-time Baghewala well data.”

This distinction is essential for technical credibility.

6.13 Assumption 12 — Model Parameters Are Calibrated, Not Arbitrary

Parameters such as:

- A, B in the viscosity-temperature model
- C_{eff}
- U_{eff}
- J_o
- k_{ro}

should not simply be chosen because they make the graph look realistic.

They should be estimated using:

1. Field data, where available.
2. Laboratory/PVT data.
3. Published Baghewala information.
4. Engineering correlations.
5. Synthetic calibration only where necessary.

Every parameter in the final model should therefore carry metadata:

Parameter → Value → Unit → Source → Calibration method → Confidence

6.14 Assumption 13 — No Fully Autonomous Control

For the SIH prototype, the digital twin should **recommend** operating changes rather than directly control the field.

The architecture should be:

Sensors/SCADA



State estimation



Digital twin



Prediction



Optimization



Recommended action



Operator approval



Field operation

This is safer and more realistic than claiming fully autonomous control of steam injection or artificial lift.

6.15 What We Deliberately Do NOT Model in Layer 1

The following are outside the current reservoir-physics layer:

- Detailed sucker-rod dynamics
- Pump fillage
- Rod floating
- Dynamometer-card classification
- Motor torque
- Surface motor behaviour
- Pump mechanical failure
- Wellhead equipment
- Surface flowline behaviour

These belong to the **wellbore/artificial-lift layers** of the complete well-to-surface digital twin.

This separation is important because Layer 1 should remain focused on:

Reservoir thermal and flow behaviour.

6.16 Assumption Hierarchy

Our modeling hierarchy should therefore be:

Level 1 — High confidence

Use established physics:

Energy balance

Temperature-viscosity relationship

Mobility

Darcy-type flow

Level 2 — Calibrated simplifications

Use effective parameters:

Effective thermal capacity

Effective heat loss

Effective productivity

Level 3 — Data-driven corrections

Use ML where physics becomes insufficient:

Production correction

Viscosity correction

Thermal response correction

Level 4 — Future advanced model

If sufficient data become available:

3-D thermal reservoir model / PINN / surrogate reservoir simulator

This gives us a clear upgrade path without overengineering the first prototype.

6.17 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

The SIH problem establishes that Baghewala has heavy crude, high viscosity, high asphaltene content, low reservoir pressure, low reservoir temperature and poor oil mobility.

PHYSICS

The model retains the dominant physical chain:

Heat → Temperature → Viscosity → Mobility → Flow

ASSUMPTION

We simplify the reservoir using:

- Effective thermal state
- Effective pressure
- Effective heat-loss parameters
- Effective relative permeability
- Temperature-dominated viscosity
- Darcy-type inflow

PROPOSED DESIGN

The digital twin will be **hybrid rather than purely physics-based**:

Physics model

-

Field data

-

ML correction

=

Hybrid Baghewala Digital Twin

This is preferable to either:

Pure ML

or

Full high-fidelity reservoir simulation

for the SIH prototype.

6.18 Final Simplified Layer-1 Model

The complete Layer-1 model is therefore:

CSS operating inputs

↓

Steam thermal-energy model

↓

Effective reservoir temperature T_r

↓

Viscosity model $\mu_o(T)$

↓

Mobility model λ_o

↓

Pressure/inflow model

↓

Reservoir inflow q_{in}

↓

Interface to Layer 2

The most important point is that **every simplification is explicit**.

We are not claiming:

“This is the exact physical reservoir.”

We are claiming:

“This is a computationally efficient, physics-consistent reduced-order representation of the dominant thermal-flow behaviour, designed to be calibrated and corrected using field data.”

That distinction will make our technical justification much stronger during the SIH evaluation.

Next: 7. What Physics Can Predict

There we will identify exactly what this Layer-1 physics model can reliably predict — **temperature, viscosity, mobility, inflow and production response** — and distinguish those predictions from quantities that require the later SRP/wellbore layer.

7. What Physics Can Predict

The purpose of Layer 1 is not to predict every detail of the well. Its purpose is to predict the **reservoir-side response to CSS operations** and provide a physically meaningful input to Layer 2.

The central prediction chain is:

CSS operating conditions → Reservoir thermal state → Oil temperature → Oil viscosity → Oil mobility → Reservoir inflow → Production potential

The SIH problem statement establishes that Baghewala has **high crude viscosity, low reservoir pressure, low reservoir temperature of about 46–48°C, and poor oil mobility under primary recovery.**

7.1 Reservoir Temperature / Thermal State

FACT

The Baghewala reservoir is reported to have a relatively low temperature of approximately **46–48°C**. This is important because the heavy crude has very high viscosity at reservoir conditions, limiting its mobility.

CSS introduces thermal energy into the reservoir through steam injection.

PHYSICS

The first quantity our Layer 1 model should predict is therefore:

How hot is the effective reservoir region at any point in the CSS cycle?

During injection:

Steam → thermal energy enters reservoir → reservoir temperature increases

During soak:

Steam injection stops → heat redistributes within the reservoir

During production/cooling:

Heat is removed/lost → reservoir temperature decreases

Therefore, temperature is a **dynamic state variable**, not a fixed parameter.

EQUATION

Our reduced-order thermal balance is:

$$C_{\text{eff}} dT_r/dt = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

where:

- T_r = effective reservoir temperature
- C_{eff} = effective thermal storage capacity of the heated reservoir region
- $\dot{Q}_{\text{steam}}$ = thermal energy supplied by steam
- $\dot{Q}_{\text{loss}}$ = heat lost to surrounding formation/overburden/underburden
- $\dot{Q}_{\text{prod}}$ = thermal energy carried away with produced fluids

Steam energy can be represented as:

$$\dot{Q}_{\text{steam}} = \dot{m}_s (h_s - h_{\text{ref}}) \eta_{\text{th}}$$

where:

- $\dot{m}_s$ = steam mass injection rate
- h_s = steam enthalpy
- h_{ref} = reference enthalpy
- η_{th} = effective thermal efficiency

ASSUMPTION

Instead of modelling the complete 3-D reservoir temperature field, V1 uses an **effective heated reservoir volume** represented by one or several thermal states.

This makes the model computationally fast enough for optimization and real-time decision support.

PROPOSED DESIGN

The digital twin should continuously estimate:

$$T_r(t)$$

and compare the predicted thermal state against whatever temperature-related field measurements are available.

The model should therefore be capable of answering:

“Given the steam injected and the current CSS stage, what is the estimated thermal state of the reservoir?”

7.2 Oil Viscosity

FACT

High crude viscosity is one of the central reasons for poor oil mobility at Baghewala. The SIH problem statement specifically identifies high viscosity as a major challenge.

The important physical effect of CSS is therefore not simply “heating the reservoir.”

It is:

Heating → viscosity reduction → improved mobility

PHYSICS

For heavy oil, viscosity is strongly temperature dependent.

As temperature increases:

$$T \uparrow \rightarrow \mu_o \downarrow$$

As the reservoir cools:

$$T \downarrow \rightarrow \mu_o \uparrow$$

Therefore, once the thermal state has been predicted, the model can estimate the corresponding oil viscosity.

EQUATION

A temperature-viscosity relationship can be represented using a calibrated correlation such as:

$$\ln(\mu_o) = A + B/T_K$$

where:

- μ_o = oil viscosity
- A, B = empirical calibration parameters
- T_K = absolute temperature in Kelvin

with:

$$T_K = T_C + 273.15$$

The important point is that **A and B must be calibrated using actual fluid-property data.**

They should not simply be invented for the prototype.

ASSUMPTION

For the first version:

$$\mu_o = \mu_o(T)$$

That means temperature is treated as the dominant variable controlling viscosity.

A higher-fidelity version could eventually use:

$$\mu_o = f(T, P, \text{shear rate, fluid composition})$$

This could become particularly important if the heavy crude exhibits significant non-Newtonian or shear-dependent behaviour.

PROPOSED DESIGN

The digital twin should have a calibrated **temperature-to-viscosity module**.

For every predicted reservoir temperature:

$$T_r(t) \rightarrow \mu_o(t)$$

This becomes one of the most important state transformations in the entire Layer 1 model.

7.3 Oil Mobility

FACT

The SIH problem explicitly identifies poor oil mobility as a major production limitation at Baghewala.

Mobility is therefore the bridge between **fluid properties** and **reservoir flow**.

PHYSICS

Oil mobility increases when oil viscosity decreases.

Therefore:

Temperature $\uparrow \rightarrow$ viscosity $\downarrow \rightarrow$ mobility $\uparrow$

and:

Temperature $\downarrow \rightarrow$ viscosity $\uparrow \rightarrow$ mobility $\downarrow$

EQUATION

Oil mobility can be represented as:

$$\lambda_o = (k k_{ro}) / \mu_o$$

where:

- λ_o = oil mobility
- k = absolute reservoir permeability

- k_{ro} = relative permeability to oil
- μ_o = oil viscosity

This equation is extremely important for our digital twin because it directly connects the thermal subsystem to the flow subsystem.

For a fixed reservoir permeability:

$$\mu_o \downarrow \rightarrow \lambda_o \uparrow$$

ASSUMPTION

In the first model, the dominant changing quantity is assumed to be **oil viscosity**.

Therefore:

k and $k_{ro} \rightarrow$ treated as effective/calibrated parameters

while:

$\mu_o \rightarrow$ dynamic variable

This is a simplification.

In reality, relative permeability can change with saturation and multiphase conditions.

PROPOSED DESIGN

The digital twin should calculate mobility after every thermal-state update:

$$T_r \rightarrow \mu_o \rightarrow \lambda_o$$

This gives us a physically interpretable metric for answering:

“Has the reservoir actually become easier for oil to flow through after steaming?”

This is more meaningful than looking only at steam volume.

7.4 Reservoir Inflow

FACT

Baghewala has low reservoir pressure and poor oil mobility under primary recovery, making natural reservoir inflow insufficient for sustained production.

PHYSICS

Reservoir inflow depends primarily on two things relevant to our Layer 1 model:

1. **Pressure driving force**
2. **Oil mobility**

Therefore:

Pressure difference $\uparrow \rightarrow$ inflow $\uparrow$

and:

Mobility $\uparrow \rightarrow$ inflow $\uparrow$

CSS mainly improves the second component by reducing viscosity.

EQUATION

A first-order inflow relationship can be written as:

$$q_{in} = J_o(P_r - P_{wf})$$

where:

- q_{in} = reservoir inflow rate
- J_o = effective oil productivity/inflow index
- P_r = reservoir pressure
- P_{wf} = flowing bottomhole pressure

The productivity index is related to mobility:

$$J_o \propto (k k_{ro}) / \mu_o$$

Therefore:

$$CSS \rightarrow T \uparrow \rightarrow \mu_o \downarrow \rightarrow \lambda_o \uparrow \rightarrow J_o \uparrow \rightarrow q_{in} \uparrow$$

This is the mathematical representation of the main physical benefit of CSS.

ASSUMPTION

For V1:

- reservoir pressure is represented by an effective state $P_r(t)$
- bottomhole pressure is represented by $P_{wf}(t)$
- productivity is represented through an effective inflow relationship

We are not yet solving the complete reservoir pressure field.

PROPOSED DESIGN

The digital twin should calculate:

$$q_{in}(t)$$

for every simulated CSS scenario.

This gives the optimization engine a physically meaningful quantity to compare between different steam schedules.

7.5 First-Order Production Potential

This is where we need to be very careful.

FACT

The physical problem at Baghewala involves both **thermal EOR and artificial lift**. The SIH statement explicitly identifies artificial lift and thermal EOR as critical for sustained production.

Therefore, reservoir inflow is **not identical to surface production**.

PHYSICS

Layer 1 can estimate:

How much oil the reservoir is capable of supplying to the wellbore under the current thermal and pressure state.

That is the **reservoir production potential**.

For example:

CSS cycle A

T_r = lower
 μ_o = higher
 λ_o = lower
 q_{in} = lower

versus:

CSS cycle B

T_r = higher
 μ_o = lower
 λ_o = higher
 q_{in} = higher

Thus the model can compare the reservoir-side effectiveness of different CSS strategies.

EQUATION

At the simplest level:

$$q_{\text{potential}} \approx q_{\text{in}}$$

But this is only a **reservoir-side approximation**.

Actual surface production should eventually be represented as:

$q_{\text{surface}} = f(q_{\text{in}}, \text{wellbore hydraulics, artificial lift, pump performance, operating constraints})$

That second part belongs to **Layer 2**.

ASSUMPTION

For Layer 1:

q_{in} is treated as the reservoir-side production potential.

We deliberately do **not** claim that it is the actual measured surface production rate.

PROPOSED DESIGN

The digital twin should output two conceptually different quantities:

Reservoir-side prediction:

$q_{\text{in}} \rightarrow$ “available inflow potential”

and later, after Layer 2:

Surface-side prediction:

$q_{\text{surface}} \rightarrow$ “expected produced rate”

This distinction will be extremely important in the final SIH architecture.

7.6 What Can Be Predicted Directly vs Indirectly?

Quantity	Prediction from Layer 1	Method	Confidence
Reservoir temperature T_r	Direct	Thermal balance	High, if thermal parameters are calibrated
Oil viscosity μ_o	Derived	$\mu(T)$ correlation	Medium–High
Oil mobility λ_o	Derived	Mobility equation	Medium
Reservoir inflow q_{in}	Derived	Inflow/productivity model	Medium
Surface oil rate	Not directly	Requires Layer 2	Low from Layer 1 alone

Quantity	Prediction from Layer 1	Method	Confidence
Pump performance	Not captured	Requires SRP model	None in Layer 1
Rod floating	Not captured	Requires rod/pump dynamics	None in Layer 1
Mechanical failures	Not captured	Requires equipment/failure model	None in Layer 1

This table defines the **boundary of Layer 1**.

7.7 Prediction Chain of Layer 1

The complete Layer 1 prediction pathway is:

CSS operating conditions

↓

Steam thermal energy

↓

Reservoir thermal state $T_r(t)$

↓

Oil viscosity $\mu_o(t)$

↓

Oil mobility $\lambda_o(t)$

↓

Reservoir productivity $J_o(t)$

↓

Reservoir inflow $q_{in}(t)$

↓

Production potential

This is the first physics backbone of our digital twin.

7.8 What Does This Allow the Digital Twin to Answer?

With Layer 1 alone, the system can answer questions such as:

Question 1

“If we inject more steam, what happens to reservoir temperature?”

→ Thermal model.

Question 2

“If reservoir temperature reaches a certain level, what happens to oil viscosity?”

→ Temperature-viscosity model.

Question 3

“Does the reduction in viscosity actually improve oil mobility?”

→ Mobility model.

Question 4

“Does increased mobility translate into higher reservoir inflow?”

→ Inflow model.

Question 5

“Which CSS schedule should produce a better reservoir response?”

→ Simulate multiple candidate schedules and compare predicted states.

This is where Layer 1 starts becoming useful for **optimization**, rather than merely being a physics explanation.

7.9 Uncertainty in These Predictions

The predictions will never be exact because several quantities are uncertain.

Parameter uncertainty

Examples:

- Effective reservoir thermal capacity
- Heat-loss coefficient
- Permeability
- Relative permeability

- Viscosity-temperature relationship
- Effective reservoir pressure

Measurement uncertainty

Real field sensors may contain:

- noise
- missing values
- drift
- calibration errors
- delayed measurements

This can be represented conceptually as:

$$\mathbf{y}_t = \mathbf{h}(\mathbf{x}_t) + \mathbf{v}_t$$

where $\mathbf{v}_t$ represents measurement noise.

Model-form uncertainty

Our reduced-order model is deliberately simpler than the real reservoir.

Therefore:

Prediction = Physics model + calibrated parameters + measurement correction + uncertainty

This is why the final digital twin should report not only a prediction but also a **confidence/uncertainty range**.

7.10 Final Output of Layer 1

At the end of Layer 1, the digital twin should maintain a reservoir-side state such as:

$$\mathbf{x}_t = [T_r, P_r, \mu_o, \lambda_o, q_{in}]$$

and receive CSS controls such as:

$$\mathbf{u}_t = [\text{steam rate}, \text{steam volume}, \text{soak time}, \text{cycle timing}]$$

The model then predicts:

$$\mathbf{x}_{(t+1)} = \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}) + \mathbf{w}_t$$

where:

- $\mathbf{x}_t$ = current estimated reservoir state

- u_t = CSS operating controls
- θ = calibrated physical parameters
- w_t = process/model uncertainty

The most important output is:

“Given the current reservoir state and a proposed CSS action, what reservoir response should we expect?”

That output becomes the **input to Layer 2**.

Layer 1 Boundary

Layer 1 CAN predict:

Steam → Temperature → Viscosity → Mobility → Reservoir Inflow → Production Potential

Layer 1 CANNOT predict alone:

Pump fillage → Rod loads → Rod floating → SRP dynamics → Surface production → Mechanical failures

Those require the next physical layer.

7 → 8 transition

So now we have established what the physics model **can predict**.

The next section in our exact sequence is:

8. What Physics Cannot Capture Alone

This is important because it will define **why the SIH solution needs Layer 2, data engineering, ML, and eventually the full well-to-surface digital twin**.

8. What Physics Cannot Capture Alone

This section is critical because it defines the **boundary of the Layer 1 reservoir-physics model**.

Layer 1 can explain:

Steam → Heat → Temperature → Viscosity → Mobility → Reservoir Inflow

But the SIH problem is explicitly **well-to-surface optimization**, not only reservoir optimization. The actual production system contains the wellbore, sucker rod pump, rods, motor, surface equipment, operating constraints, and failures.

Therefore, a reservoir-only physics model cannot explain the complete observed production behaviour.

8.1 Reservoir Inflow $\neq$ Actual Surface Production

FACT

The Baghewala problem involves heavy oil production using **thermal stimulation and artificial lift**. The SIH problem statement identifies artificial lift and thermal EOR as critical because of the heavy oil's poor mobility and low reservoir pressure.

PHYSICS

Layer 1 predicts approximately:

Reservoir $\rightarrow$ Wellbore inflow

But the actual production path is:

Reservoir $\rightarrow$ Perforations $\rightarrow$ Wellbore $\rightarrow$ Pump $\rightarrow$ Tubing $\rightarrow$ Surface facilities $\rightarrow$ Measured production

Therefore:

$q_{in} \neq q_{surface}$

in general.

Even if the reservoir is capable of supplying a large quantity of oil, the pump may not be able to lift it efficiently.

EQUATION

Conceptually:

$q_{surface} = f(q_{in}, P_{wf}, \text{pump performance, wellbore hydraulics, surface constraints})$

Layer 1 only provides:

q_{in}

The remaining transformation belongs to the next layer.

ASSUMPTION

We deliberately do not force Layer 1 to predict surface production directly.

PROPOSED DESIGN

Use:

Layer 1 output → Layer 2 input

where:

$q_{in}(t) \rightarrow \text{wellbore/SRP model} \rightarrow q_{surface}(t)$

8.2 Pump Fillage Cannot Be Determined from Reservoir Physics Alone

FACT

Baghewala uses sucker rod pumping for heavy-oil production according to OIL's public technical information.

The reservoir model does not contain the mechanical information necessary to determine how completely the pump fills during each stroke.

PHYSICS

Pump fillage depends on factors such as:

- Available reservoir inflow
- Pump displacement
- Pump speed
- Fluid compressibility
- Gas interference
- Fluid viscosity
- Pressure conditions
- Valve behaviour
- Rod/tubing dynamics

Therefore, even if Layer 1 predicts increased inflow:

$q_{in} \uparrow$

it does not automatically mean:

pump fillage $\uparrow$ proportionally

PROPOSED DESIGN

Layer 2 should contain a pump-performance model that converts reservoir inflow into expected pump filling and displacement.

This is one of the first major interfaces between **reservoir physics and artificial-lift physics**.

8.3 Rod Floating Cannot Be Captured by Reservoir Physics Alone

This is one of the most important boundaries for our SIH solution.

FACT

The SIH problem specifically identifies **rod floating detection** as part of the challenge.

This means the solution cannot stop at reservoir temperature and oil viscosity.

PHYSICS

Rod behaviour is a mechanical/dynamic problem.

The rod string experiences:

- gravitational forces
- buoyancy
- viscous/frictional forces
- acceleration
- tension/compression
- pump loads
- tubing interaction
- dynamic effects

The reservoir thermal model does not contain these mechanical states.

EQUATION

A simplified rod dynamic system could eventually be represented conceptually as:

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{C} \dot{\mathbf{x}} + \mathbf{K} \mathbf{x} = \mathbf{F}_{\text{fluid}} + \mathbf{F}_{\text{pump}} + \mathbf{F}_{\text{gravity}} + \mathbf{F}_{\text{other}}$$

where:

- **M** = effective rod-string mass
- **C** = damping
- **K** = effective stiffness
- **x** = rod displacement
- **F_{fluid}** = fluid-related force
- **F_{pump}** = pump load

- **F_gravity** = gravitational contribution

This is **not part of Layer 1**.

It belongs to the mechanical/wellbore model.

PROPOSED DESIGN

Layer 2 should eventually estimate rod-string behaviour from available measurements such as:

Surface dynamometer data + pump operating data + production data + pressure/temperature data

and detect abnormal states such as:

normal pumping → abnormal loading → possible rod floating

8.4 Surface Dynamometer Behaviour

FACT

Reservoir physics does not directly provide a surface dynamometer card.

A dynamometer card represents the relationship between:

Rod position ↔ Rod load

over a pumping cycle.

PHYSICS

The shape of a dynamometer card depends on:

- pump loading
- rod-string mechanics
- fluid loading
- tubing effects
- friction
- pump condition
- operating speed

Therefore, two wells with similar reservoir inflow can potentially exhibit very different dynamometer behaviour.

PROPOSED DESIGN

This creates an important opportunity for our team's:

AI/ML + data engineering

skills.

A future Layer 2 model can learn:

Dynamometer signal → operational state → anomaly/failure probability

while the physics model provides the underlying reservoir state.

8.5 Mechanical Failures

FACT

A reservoir thermal model cannot directly determine whether a mechanical component is about to fail.

For example, it cannot independently identify:

- rod buckling
- rod wear
- pump valve failure
- tubing problems
- abnormal friction
- mechanical sticking
- surface equipment problems

PHYSICS

These phenomena involve mechanical dynamics and equipment-specific behaviour.

They depend on variables that Layer 1 does not contain.

Therefore:

Reservoir temperature ≠ equipment health

and:

Reservoir inflow ≠ pump health

PROPOSED DESIGN

Equipment-health prediction should therefore be treated as a separate subsystem using:

Physics + sensor data + ML anomaly detection

rather than trying to force the reservoir model to explain mechanical failures.

8.6 Wellbore Hydraulics

PHYSICS

Between reservoir and surface, the produced fluid travels through the wellbore.

During this process, the system experiences:

- pressure losses
- temperature changes
- fluid density changes
- viscosity changes
- multiphase effects
- tubing restrictions
- pump pressure changes

Therefore:

Reservoir inflow → wellbore flow → pump → surface

is itself a dynamic physical system.

EQUATION

A simplified pressure balance can conceptually be written as:

$$P_{\text{surface}} = P_{\text{wf}} - \Delta P_{\text{friction}} - \Delta P_{\text{hydrostatic}} - \Delta P_{\text{other}}$$

where:

- **P_{wf}** = flowing bottomhole pressure
- **$\Delta P_{\text{friction}}$** = pressure loss due to friction
- **$\Delta P_{\text{hydrostatic}}$** = hydrostatic pressure contribution
- **ΔP_{other}** = other pressure losses

The actual model can become much more sophisticated depending on available data.

PROPOSED DESIGN

Layer 2 should eventually contain a **wellbore/artificial-lift model** that converts:

reservoir inflow → pump intake conditions → surface production

8.7 Surface Constraints

This is another major reason why optimization cannot be done using reservoir physics alone.

PHYSICS

A theoretically optimal CSS operation may not be operationally feasible.

Examples include:

- maximum allowable steam pressure
- boiler capacity
- water availability
- fuel consumption
- pump operating limits
- surface handling capacity
- pressure constraints
- safety limits

Therefore:

Best reservoir response ≠ best field operation

PROPOSED DESIGN

The optimization engine should eventually solve:

maximize production + maximize recovery + minimize energy/water/cost + maintain safety

subject to:

operational constraints

This turns our system from a simple simulator into a **constrained decision-support system**.

8.8 Data-Driven Phenomena

Some phenomena may be too complicated or poorly characterized to model accurately from first principles within an SIH prototype.

Examples:

- abnormal pump behaviour
- subtle rod-floating signatures
- equipment degradation
- unknown operational interactions
- measurement anomalies
- production-rate deviations

PHYSICS

Physics tells us **what should happen**.

Data tells us **what actually happened**.

The difference is valuable.

For example:

Physics prediction: 100 units/day

Observed production: 72 units/day

The discrepancy could indicate:

- pump inefficiency
- wellbore restriction
- mechanical problem
- inaccurate model parameters
- sensor error
- unmodelled reservoir behaviour

PROPOSED DESIGN

This creates the hybrid architecture:

Physics model + Data-driven correction

rather than:

Physics OR AI

This is one of the strongest technical directions for our SIH solution.

8.9 Why Pure Physics Is Not Enough

The limitation can be summarized as:

Phenomenon	Layer 1 Physics	Requires Additional Model
Reservoir temperature	✓	—
Oil viscosity	✓	Calibration
Oil mobility	✓	Calibration
Reservoir inflow	✓	Pressure/inflow model
Pump fillage	✗	SRP/pump model
Rod floating	✗	Rod dynamics + ML
Dynamometer card	✗	Well/pump mechanics
Pump failure	✗	Equipment model + ML
Wellbore pressure losses	Limited	Wellbore hydraulics
Surface production	✗	Integrated well-to-surface model
Operational constraints	✗	Optimization layer
Anomaly detection	✗	ML/data engineering
Operator decision	✗	Decision-support layer

This table is important because it defines exactly **why our architecture needs multiple layers**.

8.10 The Critical Insight for SIH

The mistake would be to build:

“A model that predicts how much steam to inject.”

That is too narrow.

The stronger interpretation of the problem is:

“A coupled digital twin that understands how thermal stimulation changes reservoir inflow, how that inflow interacts with artificial lift, how the mechanical system responds, and what operating action should be taken next.”

That is the difference between a **CSS simulator** and a **well-to-surface digital twin**.

8.11 Physics–Data Boundary

The architecture should therefore be divided conceptually into:

Physics explains the system

Steam → Temperature → Viscosity → Mobility → Inflow

Data describes the real system

Sensors → SCADA → Production → Pump/Dynamometer → Historical operations

ML learns what physics misses

Observed behaviour – Physics prediction → Data-driven correction/anomaly detection

Optimization decides what to do

Current state + constraints + candidate actions → Best operating recommendation

This gives us the eventual architecture:

Physics + Data + ML + Optimization

rather than simply “AI for oil production.”

8.12 Layer 1 Final Boundary

At the end of Layer 1, we should have a clean interface:

INPUT

CSS operating conditions:

Steam rate + Steam volume + Soak time + Cycle timing

↓

LAYER 1 — RESERVOIR THERMAL-FLOW MODEL

↓

OUTPUT

Reservoir temperature

↓

Oil viscosity

↓

Oil mobility



Reservoir inflow potential



LAYER 2 INPUT

Wellbore + SRP + Rod-string + Pump behaviour

This boundary is extremely important.

We are **not abandoning physics** when we move to Layer 2.

We are extending the physics model from:

Reservoir scale

to:

Wellbore + artificial-lift scale.

8.13 Why This Matters for Our Winning Solution

Our team has AI/ML, deep learning, computer vision, data engineering, and AR/VR capabilities.

Layer 1 gives the solution its **engineering credibility**.

But Layer 1 alone would not exploit the full strength of the team.

The next layers allow us to bring in:

Reservoir physics

→ **Wellbore/SRP physics**

→ **Sensor/data engineering**

→ **ML anomaly detection**

→ **Optimization**

→ **Digital twin**

→ **AR/VR visualization**

That is the route from a basic academic model to a **competition-grade industrial decision-support platform**.

8 → 9 Transition

We have now established:

What Layer 1 can predict
and
where Layer 1 stops being sufficient.

The final part of Layer 1 is therefore:

9. Interface to Layer 2 — Reservoir Inflow Becomes the Input to SRP/Wellbore Physics

Here we will define **exactly what information Layer 1 must pass to Layer 2**, what Layer 2 must return, and how the two layers become a closed-loop well-to-surface digital twin.

9. Interface to Layer 2 — Reservoir Inflow Becomes the Input to SRP/Wellbore Physics

This is the **final section of Layer 1**.

The purpose of this interface is to make sure that the reservoir model does not operate independently from the artificial-lift model.

The key idea is:

Layer 1 tells us how much fluid the reservoir wants to deliver. Layer 2 determines how effectively the well and SRP system can lift that fluid to the surface.

So the overall system becomes:

Reservoir → Wellbore → SRP → Surface

9.1 Why an Interface Is Required

FACT

The SIH problem is not only about reservoir heating. It specifically concerns the **well-to-surface optimization of CSS and SRP operations** in heavy-oil wells.

The Baghewala problem combines:

- heavy crude
- high viscosity
- low reservoir pressure
- thermal stimulation
- artificial lift
- operational optimization.

Therefore, the reservoir and artificial-lift models cannot be treated as independent modules.

PHYSICS

Layer 1 produces:

Reservoir inflow

Layer 2 receives that inflow and determines:

Pump intake → pump displacement → wellbore flow → surface production

So:

Layer 1 = supply side

Layer 2 = lifting side

9.2 The Fundamental Interface Variable

The most important quantity passed from Layer 1 to Layer 2 is:

$q_{in}(t)$ — reservoir inflow rate

It represents the rate at which the reservoir is capable of supplying fluid to the wellbore under the current thermal and pressure conditions.

Layer 1 calculates:

$$q_{in} = J_o(P_r - P_{wf})$$

where:

- q_{in} = reservoir inflow
- J_o = effective productivity index
- P_r = reservoir pressure
- P_{wf} = flowing bottomhole pressure

This becomes the principal input to Layer 2.

9.3 But q_{in} Alone Is Not Enough

A strong digital twin should not pass only one number.

Layer 1 should provide a **reservoir state package**.

For example:

Layer 1 Output	Meaning	Used by Layer 2
----------------	---------	-----------------

T_r	Reservoir temperature	Fluid properties
-------	-----------------------	------------------

Layer 1 Output Meaning		Used by Layer 2
μ_o	Oil viscosity	Wellbore/pump hydraulics
λ_o	Oil mobility	Inflow interpretation
P_r	Reservoir pressure	Pressure boundary condition
q_{in}	Reservoir inflow	Pump/wellbore model
dT_r/dt	Heating/cooling trend	Future fluid-property prediction
$d\mu_o/dt$	Viscosity trend	Dynamic hydraulic prediction
Confidence	Prediction reliability	Decision weighting

This is much more powerful than simply sending:

“Production = X.”

The digital twin should know **why** production is expected to change.

9.4 Layer 1 → Layer 2 Data Contract

We can define a formal interface.

Layer 1 output

Reservoir State Vector

$x_{res} = [T_r, P_r, \mu_o, \lambda_o, q_{in}]$

along with:

$u_{CSS} = [\text{steam rate, steam volume, soak time, cycle stage}]$

and uncertainty:

σ_{res}

This becomes the input to Layer 2.

9.5 What Layer 2 Does With This Information

Layer 2 receives:

T_r

μ_o

P_r

q_{in}

and combines them with artificial-lift operating conditions such as:

- pump speed
- stroke length
- pump geometry
- rod-string characteristics
- tubing conditions
- pump intake pressure
- surface pressure
- motor/load information
- available sensor measurements

The resulting chain becomes:

Reservoir inflow

↓

Wellbore fluid conditions

↓

Pump intake

↓

SRP displacement

↓

Rod/pump dynamics

↓

Surface production

9.6 The Important Feedback Loop

The interface should be **bidirectional**.

It should not be:

Layer 1 → Layer 2

and stop there.

Instead:

Layer 1 → Layer 2 → observations → Layer 1 correction

This creates the digital-twin feedback loop.

Forward direction

CSS changes reservoir conditions:

Steam → T_r → μ_o → q_{in}

Then:

q_{in} → SRP → $q_{surface}$

Reverse direction

Actual production and pump measurements tell us whether our reservoir prediction was correct.

For example:

Predicted q_{in} = 100

but observed system behaviour indicates only:

70 units/day

The discrepancy becomes information for model updating.

9.7 Physics–Data Feedback

This can be expressed as:

Predicted state

→ compare with

Measured state

→ calculate

Prediction error

→ update model

For example:

$$e_q = q_{\text{measured}} - q_{\text{predicted}}$$

Similarly:

$$e_T = T_{\text{measured}} - T_{\text{predicted}}$$

These errors can be used for:

- parameter calibration
- anomaly detection
- model correction
- confidence estimation

9.8 Why This Is a Digital Twin Rather Than a Simulator

A simulator normally works like:

Input → Simulation → Output

For example:

Steam rate → predicted production

A digital twin should work like:

Real system → data → state estimation → simulation → prediction → comparison → model update → recommendation

Therefore our architecture should be:

Physical Well



Sensors / SCADA



Data Engineering



State Estimation



Digital Twin



Prediction



Optimization



Recommended Action



Physical Well

That closed loop is much closer to the actual meaning of a digital twin.

9.9 Layer 1 and Layer 2 Have Different Responsibilities

This separation should remain very clear.

Layer 1 — Reservoir Thermal-Flow Physics

Responsible for:

- CSS thermal response
- reservoir temperature
- oil viscosity
- mobility
- reservoir pressure
- inflow potential

Layer 2 — Wellbore/SRP Physics

Responsible for:

- pump behaviour
- pump displacement
- pump fillage
- rod-string dynamics
- rod loads
- wellbore pressure losses
- surface production

- artificial-lift operating response

Layer 3 — Data/ML

Responsible for:

- anomaly detection
- sensor validation
- rod-floating classification
- production forecasting
- model correction
- failure prediction

Layer 4 — Optimization

Responsible for:

- CSS schedule
- steam rate
- soak duration
- SRP settings
- production target
- energy/water/fuel constraints

This modular structure prevents us from building one enormous, uninterpretable AI model.

9.10 The Most Important Coupling Variable: Bottomhole Pressure

There is another important connection between the layers.

Layer 1 uses:

$$q_{in} = J_o(P_r - P_{wf})$$

Therefore **P_{wf}** affects reservoir inflow.

But P_{wf} is also influenced by the artificial-lift system.

That means:

SRP operation → P_{wf} → reservoir inflow

while:

reservoir inflow → pump loading → SRP behaviour

This creates a **two-way coupling**.

So the actual architecture is not simply:

Reservoir → Pump

It is:

Reservoir ⇌ Wellbore/SRP

This is a major conceptual point for our final digital twin.

9.11 Coupled Mathematical Representation

We can represent the interaction as:

Reservoir model

$$q_{in} = f_{res}(T_r, \mu_o, P_r, P_{wf})$$

Well/SRP model

$$q_{surface} = f_{SRP}(q_{in}, P_{wf}, N, S, \text{rod/pump parameters})$$

where:

- **N** = pumping speed
- **S** = stroke length

The coupling condition is:

P_{wf} and q_{in} must be mutually consistent.

Therefore, the two models must eventually be solved together.

9.12 Example of the Coupling

Suppose CSS increases reservoir temperature.

Then:

T_r ↑

↓

μ_o ↓

↓

mobility ↑

↓

q_{in} ↑

Now the pump receives more available fluid.

But suppose the SRP is already operating near its effective capacity.

Then:

q_{in} ↑

does not necessarily produce the same proportional increase in:

q_{surface}

Instead, the system may experience:

- increased pump loading
- incomplete/altered pump filling
- changed rod loads
- changed operating conditions

Therefore the optimal decision might not be:

“Inject more steam.”

It could instead be:

“The reservoir has sufficient mobility; adjust artificial-lift operation to utilize the available inflow.”

That is precisely why the problem is **well-to-surface optimization**.

9.13 The Interface Should Carry Uncertainty

Another important design decision:

Layer 1 should not send:

q_{in} = 100

as if that value were perfectly known.

Instead:

q_{in} = 100 ± uncertainty

For example, conceptually:

q_{in} = 100 with confidence = 82%

Layer 2 can then account for this uncertainty when making decisions.

This becomes especially useful when:

- sensor data are missing
- parameters are poorly calibrated
- reservoir conditions change
- the model encounters a previously unseen operating regime

9.14 What Happens When Layer 1 and Layer 2 Disagree?

This is actually useful information.

Suppose:

Layer 1 predicts high inflow

but:

Layer 2 predicts poor surface production

Then the system should investigate the reason.

Possible explanations include:

1. Pump limitation
2. Wellbore pressure loss
3. Mechanical issue
4. Rod/pump abnormality
5. Incorrect reservoir parameter
6. Sensor problem
7. Unmodelled multiphase behaviour

Therefore, disagreement between layers becomes an **anomaly signal**.

This is a very strong opportunity for our ML subsystem.

9.15 Final Layer 1 → Layer 2 Interface

The final interface can therefore be defined as:

Layer 1 provides

Reservoir state

- T_r
- P_r
- μ_o
- λ_o
- q_{in}
- state trends
- uncertainty

Layer 2 provides

Artificial-lift state

- pump operating state
- pump fillage
- rod loads
- pump displacement
- P_{wf}
- surface production
- mechanical-health indicators

The coupling is

$P_{wf} \leftrightarrow q_{in}$

and:

$q_{in} \leftrightarrow$ SRP behaviour

9.16 Complete Layer 1 Architecture

We can now close Layer 1:

CSS Controls

↓

Steam Thermal Input



Reservoir Thermal Model



T_r



Viscosity Model



μ_o



Mobility Model



λ_o



Inflow Model



q_{in}



INTERFACE

Reservoir State → Layer 2

9.17 The Key Takeaway of Layer 1

Layer 1 establishes the **physical cause-and-effect chain**:

CSS does not directly create production. CSS changes the thermal state of the reservoir, which changes oil viscosity and mobility, which changes reservoir inflow.

The artificial-lift system then determines how effectively that available inflow can be converted into surface production.

Therefore:

CSS optimization cannot be separated from SRP optimization.

And that is the fundamental reason our final system must be a **coupled well-to-surface digital twin** rather than a standalone CSS model.

Layer 1 is now complete

We have established:

- 1. Baghewala Problem Context**
- 2. Governing Physics**
- 3. Governing Equations**
- 4. CSS Physics**
- 5. Baghewala-Specific Parameters**
- 6. Assumptions & Simplifications**
- 7. What Physics Can Predict**
- 8. What Physics Cannot Capture Alone**
- 9. Interface to Layer 2**

Absolutely. We'll start **Layer 2, Section 1** and keep the same disciplined structure we used for Layer 1.

LAYER 2 — WELLBORE + SUCKER ROD PUMP (SRP) PHYSICS

2.1 SRP System Architecture

2.1.1 What is the SRP system?

FACT

Sucker Rod Pumping (SRP) is an artificial-lift method in which a surface pumping unit reciprocates a rod string connected to a downhole pump.

For Baghewala, this is particularly important because the field produces **heavy, high-viscosity crude under relatively low reservoir pressure**, making natural flow insufficient for sustained production. The SIH problem specifically identifies artificial lift as a critical component of the production challenge.

The basic purpose of SRP is therefore:

Convert mechanical energy at the surface into reciprocating motion of a downhole pump so that reservoir fluid can be lifted to the surface.

2.1.2 The Physical Flow Path

For our digital twin, we should visualize the well as a chain of interconnected subsystems:

Reservoir



Perforations / Formation-Well Interface



Wellbore



Downhole Pump



Rod String



Tubing



Wellhead



Surface Flowline



Production Facility

The reservoir supplies the fluid, while the SRP system provides the lifting mechanism.

This immediately gives us an important distinction:

Layer 1

“How much fluid can enter the well?”

Layer 2

“How effectively can the SRP system lift that fluid?”

2.1.3 Main Components of an SRP System

For the digital twin, we should explicitly model the major physical components.

Component	Physical role	Digital-twin role
Reservoir	Supplies formation fluid	Layer 1 boundary
Perforations	Fluid enters wellbore	Inflow interface
Pump intake	Receives reservoir fluid	Boundary for pump model
Downhole pump	Converts reciprocating motion into fluid displacement	Pump model
Rod string	Transmits surface motion to pump	Rod dynamics
Tubing	Provides flow path to surface	Wellbore hydraulics
Pumping unit	Generates reciprocating motion	Surface operating input
Motor	Supplies mechanical energy	Energy/load model
Gearbox	Converts motor speed/torque	Mechanical transmission
Surface sensors	Measure operating behaviour	Data/ML layer

Component	Physical role	Digital-twin role
Wellhead	Surface pressure/flow boundary	Surface constraint

This component-level decomposition is important because **each component produces different kinds of data and failure signatures.**

2.1.4 Surface Pumping Unit

At the surface, the prime mover drives the pumping unit.

The simplified energy pathway is:

Motor

↓

Gearbox

↓

Crank

↓

Walking beam

↓

Horsehead

↓

Polished rod

↓

Rod string

↓

Downhole pump

The pumping unit converts **rotational motion** into **reciprocating vertical motion**.

Therefore:

Rotational motor motion → reciprocating rod motion

This is the fundamental mechanical principle of an SRP.

2.1.5 The Polished Rod

The polished rod is the surface member directly connected to the rod string.

Its important functions are:

- transmit reciprocating motion
- transmit mechanical load
- provide the surface measurement point
- interact with stuffing-box/sealing equipment

For our digital twin, the polished rod is particularly valuable because **surface measurements can provide information about downhole behaviour.**

This is why dynamometer-based monitoring becomes important later.

2.1.6 Rod String

The rod string connects the surface pumping unit to the downhole pump.

Its basic function is:

Transmit displacement and force between the surface pumping unit and the downhole pump.

But the rod string is not a perfectly rigid connection.

It has:

- mass
- elasticity
- damping
- inertia
- friction
- fluid interaction

Therefore, surface movement and downhole pump movement are **not necessarily identical at every instant.**

This becomes extremely important when we study **rod floating** later.

2.1.7 Downhole Pump

The downhole pump is where the mechanical motion of the rod string is converted into fluid displacement.

A conventional rod pump contains two principal valve functions:

Traveling valve

Moves with the plunger.

Standing valve

Remains associated with the pump barrel/intake and controls fluid entry.

The valves allow the pump to perform a repeated suction-and-discharge cycle.

Conceptually:

Pump stroke → valve sequence → fluid intake → fluid displacement → upward movement through tubing

We will derive the complete pumping cycle in **Section 2.2**.

2.1.8 Tubing

The tubing provides the primary flow path for produced fluid from the downhole pump toward the surface.

The fluid must travel through the tubing against:

- hydrostatic pressure
- frictional pressure losses
- surface pressure
- changing fluid properties

This becomes particularly important for Baghewala because heavy oil can have very high viscosity.

Therefore:

Higher viscosity → higher hydraulic resistance → greater lifting difficulty

This creates a direct connection between Layer 1 and Layer 2.

Layer 1 predicts:

μ_o ↓ after heating

which can subsequently influence:

wellbore hydraulic resistance ↓

and potentially improve lifting conditions.

2.1.9 Pumping Cycle

The SRP does not produce fluid continuously in the same way as a centrifugal pump.

It operates through repeated reciprocating cycles.

A simplified cycle is:

Upstroke

Rod string moves upward.

↓

Plunger moves upward.

↓

Pressure conditions change inside pump.

↓

Traveling and standing valves change state.

↓

Fluid is displaced into the tubing.

Downstroke

Rod string moves downward.

↓

Plunger moves downward.

↓

Valve states reverse.

↓

Pump chamber is prepared for the next intake/displacement sequence.

The exact valve sequence and pressure conditions will be developed in **Section 2.2**.

2.1.10 The Most Important Concept: Pump Displacement

An SRP has a theoretical displacement determined by its geometry and operating speed.

A simplified theoretical displacement relationship is:

$$Q_{\text{theoretical}} = A_p \times S \times N$$

with appropriate unit conversion depending on the chosen units.

Where:

- $Q_{\text{theoretical}}$ = theoretical pump displacement rate
- A_p = effective plunger area
- S = stroke length
- N = strokes per unit time

However:

Theoretical displacement is not the same as actual production.

Why?

Because the pump may not fill completely.

This introduces:

Pump fillage

The fraction of the theoretical pump chamber volume that is actually filled with fluid.

Therefore:

$$Q_{\text{actual}} \approx Q_{\text{theoretical}} \times \eta_{\text{fill}}$$

where:

- η_{fill} = effective pump fillage/displacement efficiency

This will become one of the most important variables in Layer 2.

2.1.11 Why Pump Fillage Matters for Baghewala

This is where the heavy-oil problem becomes more interesting.

Layer 1 may predict:

$$CSS \rightarrow T \uparrow \rightarrow \mu_o \downarrow \rightarrow q_{\text{in}} \uparrow$$

But the SRP does not automatically convert all of that inflow into surface production.

The actual system could behave as:

q_{in} ↑

↓

Pump filling changes

↓

Pump load changes

↓

Rod dynamics change

↓

q_{surface} changes

Therefore:

Reservoir improvement and pump performance must be evaluated together.

This is precisely why the final system needs a coupled digital twin.

2.1.12 SRP Operating Variables

For our digital twin, the key controllable SRP variables should eventually include:

Stroke length — S

Distance travelled by the polished rod/pump during a stroke.

Pumping speed — N

Number of pumping cycles per unit time.

Pump/plunger geometry

Determines theoretical displacement.

Surface operating conditions

Such as motor/load and surface pressure.

These variables determine how aggressively the artificial-lift system attempts to produce fluid.

2.1.13 The Core SRP Relationship

At a simplified level:

SRP operating parameters



Theoretical displacement



Pump filling



Actual displacement



Wellbore lifting



Surface production

Mathematically:

$$Q_{\text{surface}} \approx A_p S N \eta_{\text{fill}} \eta_{\text{system}}$$

where:

- A_p = effective pump/plunger area
- S = stroke length
- N = pumping speed
- η_{fill} = pump fillage efficiency
- η_{system} = combined effective lifting/system efficiency

This is a **reduced-order engineering representation**, not yet the final SRP model.

2.1.14 Coupling With Layer 1

Now we connect both layers.

Layer 1 produces:

T_r



μ_o



λ_o



q_{in}

Layer 2 receives:

$q_{in} + \mu_o + P_r + P_{wf}$

and determines:

Pump behaviour + Rod behaviour + Wellbore flow



$q_{surface}$

So the complete chain becomes:

CSS



Reservoir heating



Temperature



Viscosity



Mobility



Reservoir inflow



SRP



Pump fillage



Rod/pump dynamics



Wellbore lifting



Surface production

This is the backbone of our eventual well-to-surface digital twin.

2.1.15 Why SRP Cannot Be Optimized Independently

Suppose the SRP is operated at a very high speed.

Theoretical displacement increases:

$$N \uparrow \rightarrow Q_{\text{theoretical}} \uparrow$$

But the reservoir may not be capable of supplying that amount of fluid:

$$Q_{\text{theoretical}} > q_{\text{in}}$$

Then the pump may experience inadequate filling.

Therefore:

Higher pumping speed does NOT necessarily mean higher production.

Conversely, if CSS increases reservoir inflow:

$$q_{\text{in}} \uparrow$$

but SRP speed remains unnecessarily low:

$$Q_{\text{theoretical}} < q_{\text{in}}$$

then some reservoir production potential may remain unused.

This establishes a fundamental optimization problem:

Match artificial-lift capacity to reservoir inflow capacity.

2.1.16 The SRP Operating Point

We can therefore think of the well as having an operating point where:

Reservoir supply $\approx$ Pump demand

Conceptually:

Reservoir capacity

$$q_{\text{in}}$$

versus

Pumping capacity

q_{pump}

The useful operating region occurs when the two are appropriately matched.

If:

$$q_{\text{pump}} \gg q_{\text{in}}$$

→ potential underfilling / inefficient operation.

If:

$$q_{\text{pump}} \ll q_{\text{in}}$$

→ artificial lift may not exploit available inflow.

Therefore, our optimization engine should eventually search for:

SRP settings that best match the current reservoir state.

2.1.17 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

- Baghewala is a heavy-oil field where artificial lift is an important part of production.
- The SIH problem explicitly connects the field challenge with artificial lift and thermal EOR.

PHYSICS

- SRP converts surface rotational motion into reciprocating rod motion.
- Rod motion drives the downhole pump.
- Pump valves regulate intake and displacement.
- Pump geometry and operating speed determine theoretical displacement.
- Actual production depends on pump filling and hydraulic/system limitations.
- Reservoir inflow and pump capacity must be dynamically matched.

EQUATION

First-order pump displacement:

$$Q_{\text{theoretical}} = A_p S N$$

Effective production:

$$Q_{\text{actual}} \approx A_p S N \eta_{\text{fill}} \eta_{\text{system}}$$

These equations are deliberately simplified at this stage.

ASSUMPTION

For V1:

- SRP can be represented using a reduced-order pump-displacement model.
- Pump fillage is represented by an effective efficiency term.
- Wellbore hydraulics are initially represented using effective pressure-loss relationships.
- Detailed rod-string dynamics will be introduced separately rather than hidden inside one empirical equation.

PROPOSED DESIGN

Build Layer 2 as **three coupled submodels**:

2A. Wellbore hydraulics

Reservoir inflow → pump intake → tubing/surface flow.

2B. SRP/pump model

Stroke + speed + pump geometry → displacement + fillage.

2C. Rod mechanics

Surface motion/load → rod-string response → downhole pump behaviour.

Then combine them into:

Reservoir ⇌ Wellbore ⇌ SRP ⇌ Surface

2.1.18 What Section 2.1 Gives Us

We now have the architectural skeleton of Layer 2:

Surface Motor

↓

Pumping Unit

↓

Polished Rod

↓

Rod String

↓

Downhole Pump

↓

Pump Intake

↓

Reservoir

with produced fluid travelling:

Reservoir → Pump → Tubing → Wellhead → Surface

And the critical coupling is:

Layer 1: q_{in}

↓

Layer 2: Pump/Wellbore response

↓

$q_{surface}$

The Next Section

Now that the SRP architecture is established, the next section should be:

2.2 SRP Pumping Cycle Physics

There we will go stroke-by-stroke through:

Upstroke → pressure changes → standing/traveling valve behaviour → fluid intake/displacement → downstroke → valve reversal → next cycle

and then connect the cycle to:

pump fillage → production → rod loads → dynamometer card.

2.2 SRP Pumping Cycle Physics

Now that we have established the SRP architecture, we need to understand **what physically happens during one complete pumping cycle**.

This section is important because later we will use the pumping cycle to understand:

- pump fillage
- production rate
- rod loads
- dynamometer cards
- rod floating
- pump inefficiency
- abnormal operating conditions

The basic cycle is:

Upstroke → Fluid displacement → Downstroke → Pump refill → Next cycle

2.2.1 The Downhole Rod Pump

FACT

An SRP system uses a reciprocating rod string to operate a downhole pump. The pump periodically admits formation fluid and then displaces it upward into the tubing.

The SIH problem is concerned with SRP operation in heavy-oil wells at Baghewala, where artificial lift is required because reservoir conditions make natural production difficult.

PHYSICS

A simplified rod-pump system contains:

- **Plunger**
- **Pump barrel**
- **Standing valve**
- **Traveling valve**
- **Rod string**
- **Tubing**

The plunger is connected to the rod string.

Therefore:

Surface pumping unit → Rod string → Plunger

The movement of the plunger creates pressure changes that control fluid movement through the valves.

2.2.2 Two Fundamental Pumping Motions

The SRP operates through two fundamental motions:

Upstroke

The plunger moves upward.

Downstroke

The plunger moves downward.

These two motions together form **one complete pumping cycle**.

If the pump operates at:

N strokes/minute

then approximately:

N complete cycles occur every minute.

2.2.3 Upstroke — Beginning of the Cycle

During the upstroke, the rod string pulls the plunger upward.

Conceptually:

Rod moves ↑

↓

Plunger moves ↑

↓

Pressure inside the pump changes

↓

Valve states change

↓

Fluid is displaced into the tubing

At the same time, the pressure reduction below the plunger can allow formation fluid to enter the pump chamber through the appropriate valve pathway.

The exact pressure differences depend on the instantaneous pump position, fluid properties and operating conditions.

2.2.4 Standing and Traveling Valves

The two valve functions are central to understanding the pumping cycle.

Standing valve

The standing valve controls fluid entry from the wellbore/reservoir side into the pump chamber.

Traveling valve

The traveling valve moves with the plunger and controls fluid movement between the pump chamber and tubing.

The valves are designed to permit fluid movement predominantly in the desired direction.

Conceptually:

Reservoir → Pump chamber → Tubing → Surface

rather than allowing fluid to simply move back and forth.

2.2.5 Upstroke Valve Behaviour

During the upward movement of the plunger, the pressure relationship across the valves changes.

The simplified sequence is:

Plunger moves upward

↓

Pressure below plunger decreases

↓

Formation fluid enters pump chamber

At the same time, fluid already above the plunger can be displaced upward into the tubing.

Thus the pump performs two simultaneous functions from the overall system perspective:

1. **Accept new fluid into the pump chamber**
2. **Lift previously captured fluid upward**

This is the fundamental displacement mechanism of the rod pump.

2.2.6 Downstroke

During the downstroke:

Rod moves ↓

↓

Plunger moves ↓

↓

Pressure inside pump chamber changes

↓

Valve configuration changes

↓

Pump chamber is prepared for the next pumping cycle

The exact valve opening/closing depends on the pressure differential across each valve.

This is why valve behaviour should ultimately be modelled using **pressure conditions**, rather than simply assuming that a valve is always open at a particular geometric position.

2.2.7 Pressure Difference Controls Valve State

The important physical principle is:

A valve opens or closes according to the pressure difference across it and its mechanical characteristics.

For a simplified valve:

If:

$$P_{\text{upstream}} > P_{\text{downstream}} + \Delta P_{\text{open}}$$

the valve can open.

If the pressure relationship reverses sufficiently:

$$P_{\text{downstream}} > P_{\text{upstream}}$$

the valve tends to close.

Here:

ΔP_{open}

represents the effective pressure differential required for opening.

In a real pump, valve behaviour is more complicated because of:

- valve mass
- fluid inertia
- mechanical geometry
- wear
- leakage
- viscosity
- pressure transients

These details can be added later.

2.2.8 Why Pressure Matters So Much in Heavy Oil

This is especially important for Baghewala.

Layer 1 gives us:

CSS $\rightarrow$ $T_r \uparrow$

$\downarrow$

$\mu_o \downarrow$

$\downarrow$

Mobility $\uparrow$

$\downarrow$

$q_{\text{in}} \uparrow$

But Layer 2 sees the consequences through:

Fluid viscosity + pressure differences + pump operation

Lower viscosity generally makes fluid movement easier.

Therefore, CSS does not merely improve reservoir inflow.

It can also change the **hydraulic environment in which the pump operates**.

This is one of the important couplings between the two layers.

2.2.9 Pump Chamber Filling

During a pumping cycle, the pump chamber must receive sufficient fluid.

Let:

V_{available}

be the theoretical volume available for fluid during the pumping cycle.

Let:

V_{filled}

be the actual volume occupied by produced fluid.

Then pump fillage can be represented as:

$$F = V_{\text{filled}} / V_{\text{available}}$$

where:

- **F** = pump fillage fraction
- **V_{filled}** = actual fluid volume entering the pump
- **V_{available}** = theoretical available pump volume

Therefore:

$$0 \leq F \leq 1$$

in the idealized representation.

A value close to:

$$F = 1$$

means the pump is nearly completely filled.

A lower value indicates incomplete filling.

2.2.10 Theoretical Pump Displacement

For a simplified single-cylinder representation:

$$V_{\text{stroke}} = A_p S$$

where:

- **V_stroke** = theoretical displaced volume per stroke
- **A_p** = effective plunger area
- **S** = stroke length

If the pump performs **N strokes per unit time**:

$$Q_{\text{theoretical}} = A_p S N$$

with appropriate unit conversion.

This is the maximum geometric displacement under ideal conditions.

2.2.11 Actual Pumped Volume

The actual displacement is lower than or equal to the ideal geometric displacement because of incomplete filling and other losses.

A first-order representation is:

$$Q_{\text{actual}} = A_p S N F \eta$$

where:

- **F** = pump fillage fraction
- **η** = effective pump/system efficiency

This equation is useful for our digital twin because it separates:

Geometry and operating control

A_p, S, N

from

Physical performance

F, η

This distinction becomes extremely important for optimization.

2.2.12 What Determines Pump Fillage?

Pump fillage is not simply a pump setting.

It depends on the interaction between the reservoir and pump.

Important factors include:

1. Reservoir inflow

$q_{in} \uparrow \rightarrow$ potentially greater pump supply

2. Pump speed

$N \uparrow \rightarrow$ less time available for filling each cycle

3. Pump displacement

A larger displacement demands more fluid per cycle.

4. Fluid viscosity

Higher viscosity can make fluid movement more difficult.

5. Gas interference

Free gas can reduce effective liquid filling.

6. Pressure conditions

The pressure differential determines how easily fluid enters the pump.

7. Leakage

Valve or plunger leakage reduces effective displacement.

Therefore:

$$F = f(q_{in}, N, S, \mu_o, P, \text{gas, pump condition, geometry, ...})$$

This is not yet our final equation; it is the physical dependency structure.

2.2.13 The Critical Reservoir–Pump Mismatch

This gives us an important operating scenario.

Suppose:

Pump demand > reservoir supply

Then the pump attempts to displace more fluid than the reservoir can supply.

Potential result:

Incomplete pump filling

↓

Lower actual displacement



Lower production efficiency

This is why simply increasing SRP speed is not necessarily beneficial.

2.2.14 The Opposite Situation

Suppose:

Reservoir supply > effective pump capacity

Then the artificial-lift system may become the limiting component.

In simplified form:

$$q_{in} > q_{pump_capacity}$$

The reservoir can provide more fluid than the pump system can efficiently handle.

Therefore, increasing CSS may produce diminishing surface-production gains unless SRP operation is also optimized.

This is a central idea for our final optimization engine.

2.2.15 One Complete Cycle as a State Machine

For the digital twin, we can represent the pumping cycle as discrete states:

State 1 — Upstroke

Rod position ↑

State 2 — Valve transition

Pressure differential changes.

State 3 — Fluid displacement/intake

Fluid is transferred through the appropriate valve path.

State 4 — Top of stroke

Rod velocity approaches zero and direction reverses.

State 5 — Downstroke

Rod position ↓

State 6 — Valve transition

Pressure conditions reverse.

State 7 — Pump refill/preparation

Pump chamber prepares for the next cycle.

State 8 — Bottom of stroke

Direction reverses again.



Next cycle

This state-machine representation will later be useful for simulation.

2.2.16 Surface Motion vs Downhole Motion

An important engineering complication is that:

Surface rod motion $\neq$ perfectly identical downhole plunger motion

because the rod string is elastic.

The rod string can:

- stretch
- compress
- vibrate
- experience dynamic loading

Therefore:

Surface displacement $\rightarrow$ rod-string dynamics $\rightarrow$ downhole displacement

This is why a surface sensor can contain information about downhole behaviour, but cannot simply be treated as a direct measurement of the pump.

This becomes central to **dynamometer-card analysis** later.

2.2.17 Mechanical Energy During the Cycle

The surface pumping unit supplies mechanical energy.

Conceptually:

Motor power

↓

Gearbox

↓

Crank/beam mechanism

↓

Rod-string motion

↓

Pump work

↓

Fluid lifting

The useful hydraulic power is approximately related to:

$$P_{\text{hydraulic}} \approx \Delta P \times Q$$

where:

- **$P_{\text{hydraulic}}$** = hydraulic power
- **ΔP** = pressure increase provided to lift the fluid
- **Q** = production rate

The actual motor energy requirement will be higher because of mechanical and hydraulic inefficiencies.

This becomes important later when our optimizer considers:

Production vs energy consumption

rather than maximizing production alone.

2.2.18 Connection to CSS

Now we can connect the pumping cycle back to Layer 1.

Before CSS

Reservoir:

T low $\rightarrow$ μ high $\rightarrow$ q_{in} low

Pump:

fluid supply limited → potential underfill

After effective CSS

Reservoir:

$T \uparrow \rightarrow \mu \downarrow \rightarrow q_{in} \uparrow$

Pump:

greater fluid availability → potentially improved fillage

But this improvement is **not guaranteed**.

If SRP settings, pressure conditions or mechanical problems are limiting production, the increase in reservoir inflow may not translate proportionally to surface production.

Therefore:

CSS changes the boundary conditions of the SRP system.

This is the fundamental coupling between Layers 1 and 2.

2.2.19 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

- Baghewala is a heavy-oil field where artificial lift is required as part of the production system.
- SRP uses reciprocating motion to operate a downhole pump.

PHYSICS

- SRP operates through repeated upstroke/downstroke motion.
- Pressure differentials control valve behaviour.
- Pump geometry determines theoretical displacement.
- Actual displacement depends on pump filling and system efficiency.
- Rod-string elasticity causes surface and downhole motion to differ.
- Reservoir inflow and pump capacity must remain dynamically compatible.

EQUATION

Theoretical displacement:

$$Q_{\text{theoretical}} = A_p S N$$

Pump fillage:

$$F = V_{\text{filled}} / V_{\text{available}}$$

First-order actual production:

$$Q_{\text{actual}} = A_p S N F \eta$$

These are reduced-order equations and will require refinement/calibration.

ASSUMPTION

For V1:

- Valve behaviour can initially be represented using pressure-differential logic.
- Pump fillage can be represented using an effective fillage state.
- Detailed valve dynamics are not initially required.
- Detailed rod elasticity will be introduced in the rod-dynamics section.
- Gas effects can initially be represented as a correction rather than a complete multiphase pump model.

PROPOSED DESIGN

Represent each pumping cycle using:

Position → Velocity → Pressure → Valve state → Pump fillage → Displacement → Load

This creates a physically interpretable SRP simulation rather than a black-box production predictor.

2.2.20 What We Have Established

The fundamental SRP causal chain is now:

Surface rotation



Reciprocating rod motion



Plunger motion



Pressure differential



Valve state

↓

Pump chamber filling

↓

Fluid displacement

↓

Wellbore lifting

↓

Surface production

And the cross-layer coupling is:

CSS

↓

Reservoir temperature

↓

Oil viscosity

↓

Reservoir inflow

↓

Pump supply

↓

Pump fillage

↓

Surface production

This is the physical foundation for the next major problem.

2.2.21 Why This Section Matters for Our SIH Solution

The most important insight is:

Production is not determined only by how much the pump can theoretically displace. It is determined by how much fluid is actually available, how completely the pump fills, how

efficiently the rod/pump system operates, and how effectively the fluid can be lifted to surface.

That gives us the next modelling target:

2.3 Pump Fillage

Here we will go deeper into:

What exactly is pump fillage?

- **Why incomplete fillage occurs**
- **How CSS affects fillage**
- **How SRP speed affects fillage**
- **How viscosity affects fillage**
- **How pump fillage can be estimated from measurements**
- **How fillage affects production**
- **How ML can estimate fillage when direct measurement is unavailable**
- **How fillage becomes an optimization variable for our digital twin.**

2.3 Pump Fillage

Pump fillage is one of the **most important variables in the entire SRP layer** because it connects three things that we need to optimize together:

Reservoir inflow → Pump capacity → Actual production

A pump may have sufficient mechanical displacement capacity, but if the pump chamber does not fill adequately, that capacity cannot be converted into production.

2.3.1 What Is Pump Fillage?

FACT

In an SRP system, the downhole pump operates cyclically. The amount of fluid actually entering the pump chamber determines how much of the pump's theoretical displacement becomes useful liquid production.

PHYSICS

Define pump fillage as:

$$F = V_{\text{filled}} / V_{\text{available}}$$

where:

- **F** = pump fillage fraction
- **V_{filled}** = actual volume of liquid entering the pump chamber
- **V_{available}** = theoretical pump-chamber volume available for filling

Therefore:

$$0 \leq F \leq 1$$

in our simplified model.

Interpretation:

- **F ≈ 1** → pump is substantially filled
- **F < 1** → incomplete filling
- **F → 0** → severe underfilling

The important point is:

Pump fillage is a measure of how effectively reservoir fluid supply matches the pump's displacement demand.

2.3.2 Why Fillage Matters

The theoretical pump displacement from Section 2.2 was:

$$Q_{\text{theoretical}} = A_p S N$$

But actual liquid production is closer to:

$$Q_{\text{actual}} \approx A_p S N F \eta$$

where:

- **A_p** = plunger area
- **S** = stroke length
- **N** = pumping speed
- **F** = pump fillage
- **η** = other effective system efficiency

This immediately gives us a powerful result.

Suppose the pump geometry and operating speed remain unchanged:

$A_p, S, N = \text{constant}$

Then:

$F \uparrow \rightarrow Q_{\text{actual}} \uparrow$

and:

$F \downarrow \rightarrow Q_{\text{actual}} \downarrow$

Therefore, **pump fillage is a direct production-efficiency variable.**

2.3.3 What Causes Incomplete Pump Fillage?

Pump fillage is determined by the interaction between the reservoir, fluid, pump and operating conditions.

The major causes we should consider are:

1. Insufficient reservoir inflow

If:

$q_{\text{in}} < \text{pump demand}$

the pump cannot obtain enough fluid during the available filling period.

2. Excessive pumping speed

If pumping speed increases:

$N \uparrow$

the number of cycles per unit time increases.

The available filling time per cycle decreases:

$t_{\text{cycle}} = 1/N$

Therefore, excessively aggressive pumping can cause incomplete filling.

3. High oil viscosity

This is particularly relevant to Baghewala.

The SIH problem identifies high crude viscosity as a major production challenge.

Higher viscosity means greater resistance to fluid movement.

Conceptually:

$\mu_o \uparrow \rightarrow \text{fluid mobility} \downarrow \rightarrow \text{inflow/filling response} \downarrow$

CSS attempts to reverse this:

$T \uparrow \rightarrow \mu_o \downarrow \rightarrow \text{mobility} \uparrow$

which can improve fluid availability to the pump.

4. Low reservoir pressure

The Baghewala problem also identifies low reservoir pressure.

Lower pressure differential can reduce the ability of the reservoir to deliver fluid into the well/pump.

Therefore:

$\Delta P \downarrow \rightarrow q_{in} \downarrow \rightarrow \text{potential fillage} \downarrow$

5. Gas interference

If gas enters the pump chamber, the effective liquid volume can be reduced.

The pump may appear to have a chamber filled with material, but not all of that volume corresponds to producible liquid.

For our V1 model, this can initially be represented through a correction factor rather than a full multiphase pump model.

6. Valve leakage or malfunction

If valves do not seal correctly:

effective displacement $\downarrow$

Even if the pump chamber appears to cycle normally.

7. Plunger leakage

Clearance between plunger and barrel can allow fluid to bypass the plunger.

This reduces effective pumping efficiency.

8. Wellbore pressure conditions

The pressure at the pump intake and pressure in the tubing influence the valve pressure differential and therefore fluid movement.

2.3.4 The Reservoir–Pump Matching Problem

This is one of the most important ideas for our digital twin.

The reservoir has a supply capability:

q_{in}

The pump has a displacement demand:

$q_{theoretical}$

The pump performs efficiently only when these are appropriately matched.

Conceptually:

Case A — Pump demand too high

$q_{theoretical} \gg q_{in}$

Potential consequence:

Incomplete filling

↓

F ↓

↓

Actual production lower than theoretical

Case B — Pump and reservoir well matched

$q_{theoretical} \approx \text{available inflow}$

Potential result:

F high

↓

Efficient utilization of pump capacity

Case C — Pump capacity too low

$$q_{\text{theoretical}} \ll q_{\text{in}}$$

The reservoir may be capable of supplying more fluid than the artificial-lift system is effectively extracting.

Potential result:

Reservoir production potential underutilized

2.3.5 Pump Fillage and CSS

This is where Layer 1 and Layer 2 become tightly coupled.

Before effective thermal stimulation:

T_r low

↓

μ_o high

↓

mobility low

↓

q_{in} low

↓

pump supply limited

↓

F potentially low

After successful CSS:

T_r ↑

↓

μ_o ↓

↓

mobility ↑

↓

$q_{in} \uparrow$

↓

greater fluid availability

↓

F may increase

But notice the word:

may

CSS does not guarantee higher fillage.

If the limiting factor is instead:

- pump condition
- valve leakage
- excessive speed
- wellbore restriction
- rod/pump problem

then increasing reservoir inflow may not solve the production problem.

This is exactly why the final system must diagnose **where the bottleneck is**.

2.3.6 A More Useful Fillage Model

For the prototype, we should not initially attempt a highly detailed multiphase pump simulation.

Instead, we can define:

$F = f(q_{in}, N, \mu_o, P_{intake}, P_{tubing}, pump_condition, gas_fraction)$

where:

- **q_{in}** = reservoir inflow
- **N** = pumping speed
- **μ_o** = oil viscosity
- **P_{intake}** = pump intake pressure
- **P_{tubing}** = tubing pressure

- **pump_condition** = pump health/state
- **gas_fraction** = effective gas interference

This is the **dependency model**.

The exact function can then be calibrated using field data.

2.3.7 First V1 Mathematical Formulation

For a competition prototype, we can use a reduced-order relationship such as:

$$F = \text{clip}[f(q_{\text{in}} / q_{\text{theoretical}}, \mu_o, \Delta P, \text{pump_state}), 0, 1]$$

where:

$$q_{\text{in}} / q_{\text{theoretical}}$$

represents the basic supply-to-demand ratio.

The function $f(\dots)$ can initially be:

- physics-derived
- empirically calibrated
- or hybrid physics + ML

The clip operation simply ensures that predicted fillage remains physically meaningful between 0 and 1.

We should **not claim this as an established Baghewala equation**. It is a proposed modelling architecture.

2.3.8 Supply-to-Demand Ratio

A particularly useful diagnostic variable is:

$$R_{SD} = q_{\text{in}} / q_{\text{theoretical}}$$

where:

- **R_{SD}** = reservoir-supply to pump-demand ratio.

Interpretation:

$$R_{SD} \gg 1$$

Reservoir potentially supplies more than pump displacement.

Potentially:

Pump-limited

$$R_{SD} \approx 1$$

Supply and pump demand are reasonably matched.

Potentially:

Efficient operating region

$$R_{SD} \ll 1$$

Pump is attempting to displace more fluid than the reservoir can supply.

Potentially:

Underfilled / reservoir-limited

This ratio could become a **real-time operating indicator** in our dashboard.

2.3.9 Why Pump Speed Optimization Matters

Consider:

$$Q_{\text{theoretical}} = A_p S N$$

If:

$$N \uparrow$$

then:

$$Q_{\text{theoretical}} \uparrow$$

But reservoir inflow does not necessarily increase at the same rate.

Therefore:

$$N \uparrow$$

can produce:

$$R_{SD} \downarrow$$

and potentially:

$$F \downarrow$$

So there may be an optimal pumping speed:

$$N^*$$

at which production is maximized without causing severe underfill or excessive mechanical loading.

This is an optimization problem, not simply a “run the pump faster” problem.

2.3.10 Why Stroke Length Also Matters

Similarly:

$$Q_{\text{theoretical}} = A_p S N$$

Increasing stroke length:

$S \uparrow$

increases theoretical displacement.

But again:

q_{in}

may not increase proportionally.

Therefore:

$S \uparrow \rightarrow \text{pump demand} \uparrow$

could result in:

$F \downarrow$

if the reservoir cannot supply enough fluid.

So our optimizer should consider **stroke length and pumping speed jointly**, rather than optimizing either variable independently.

2.3.11 Pump Fillage as a Hidden State

A very important design decision:

We may not always have a direct measurement of F .

Therefore, pump fillage can be treated as a **latent/hidden state**.

We observe:

- surface load
- polished-rod position
- motor current

- production rate
- pressure
- pumping speed
- stroke length
- temperature

and infer:

F

This is an ideal problem for our team's AI/ML capability.

2.3.12 Physics + ML Fillage Estimation

Instead of asking ML to directly predict production, we can formulate a more physically meaningful task:

Inputs

q_{in} estimate

μ_o

P_{wf}

pumping speed

stroke length

surface load

position

pressure

temperature

ML output

Estimated pump fillage F

Then:

$$Q_{\text{actual}} = A_p S N F \eta$$

This has a major advantage.

The ML model is not simply saying:

“Production will be 650 bbl/day.”

It is saying:

“Given the current reservoir and SRP state, estimated pump fillage is 78%, which explains the expected production.”

That is much more interpretable.

2.3.13 Fillage Estimation From Dynamometer Data

Later, when we introduce the dynamometer card, we can use its shape as an important source of information.

Conceptually:

Surface load + position

↓

Dynamometer card

↓

Card-shape features

↓

Pump operating state

↓

Estimated fillage

This can potentially be implemented using:

- feature-based ML
- time-series models
- CNNs on normalized dynamometer-card images
- physics-informed neural networks
- hybrid classifiers/regressors

For SIH, the best approach would likely be a **hybrid model**, rather than a pure deep-learning model.

2.3.14 Fillage and Rod Floating

This section also establishes the foundation for a later major problem.

If the pump experiences poor filling and unusual fluid-loading conditions, the rod/pump dynamics can change.

However:

Low fillage should not automatically be labelled rod floating.

These are different physical phenomena.

A good system must distinguish:

Low fillage

from:

Rod floating

from:

Valve/pump malfunction

from:

Sensor anomaly

That distinction will become important in the rod-floating section.

2.3.15 Fillage and Energy Efficiency

Pump fillage also matters for energy efficiency.

Suppose:

N ↑

and therefore motor power consumption increases.

But:

F ↓

because the reservoir cannot supply the pump adequately.

Then the system may consume more energy without obtaining proportional additional oil.

Therefore:

Production optimization ≠ energy optimization

The optimizer should eventually consider:

Production + Energy + Equipment stress

simultaneously.

2.3.16 Fillage as an Optimization Objective

Instead of simply maximizing:

q_{surface}

we can formulate a more intelligent objective:

maximize production

while maintaining:

$F \geq F_{\min}$

and respecting:

- motor limits
- rod-load limits
- pump limits
- pressure limits
- CSS constraints
- energy constraints

This prevents the optimizer from recommending an operating point that produces high theoretical displacement but poor actual pump performance.

2.3.17 Example of the Digital Twin Decision

Imagine the twin estimates:

Reservoir inflow = 80 units/day

and the current SRP configuration has:

Theoretical displacement = 120 units/day

Then:

$R_{SD} = 80/120 = 0.67$

This indicates that the pump is demanding substantially more fluid than the reservoir is currently supplying.

The system might recommend:

Reduce pumping speed

rather than:

Increase pumping speed

Now suppose CSS is applied and the model estimates:

Reservoir inflow = 110 units/day

with the same pump:

$R_{SD} = 110/120 = 0.92$

The system could now predict better filling.

This demonstrates why the **CSS and SRP optimization problems cannot be separated**.

2.3.18 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

- Baghewala has high crude viscosity and low reservoir pressure, both of which contribute to poor oil mobility and difficult production.
- SRP is part of the artificial-lift system used for heavy-oil production.

PHYSICS

- Pump fillage represents how much of the theoretical pump volume is actually occupied by producible fluid.
- Pump displacement depends on pump geometry, stroke and pumping speed.
- Fillage depends on the balance between reservoir supply and pump demand.
- Higher viscosity and insufficient pressure support can reduce fluid delivery to the pump.
- Increasing pumping speed does not necessarily increase actual production.

EQUATION

$$F = V_{\text{filled}} / V_{\text{available}}$$

$$Q_{\text{theoretical}} = A_p S N$$

$$Q_{\text{actual}} \approx A_p S N F \eta$$

$$R_{SD} = q_{\text{in}} / q_{\text{theoretical}}$$

ASSUMPTION

For V1:

- Fillage can be represented as an effective state between 0 and 1.
- The supply-to-demand ratio is a useful first-order predictor.
- Gas interference, leakage and other secondary effects can initially be represented through correction terms.
- Detailed multiphase pump simulation is deferred.

PROPOSED DESIGN

Create a **Pump Fillage Estimator** combining:

Physics-based supply/demand ratio

-

Measured SRP signals

-

ML correction

to estimate:

$F(t)$

and its uncertainty.

2.3.19 What Pump Fillage Gives Our Digital Twin

After this section, Layer 2 gains an important intermediate state:

Reservoir inflow

↓

Pump demand

↓

Pump fillage

↓

Actual displacement

↓

Surface production

This is much more informative than directly predicting production.

It also gives us a measurable optimization target:

Operate the SRP so that the pump efficiently utilizes the fluid available from the reservoir without creating unnecessary mechanical or energy costs.

2.3.20 Layer 2 State Expansion

After Section 2.3, our coupled state can conceptually become:

$x = [T_r, P_r, \mu_o, q_{in}, F, q_{surface}]$

with controls:

$u = [\text{steam rate, steam volume, soak time, stroke length, pumping speed}]$

The next important question is:

How does the rod string mechanically respond to the pumping cycle and pump loading?

That leads directly to:

2.4 Rod-String Mechanics

There we will build the physics of:

**surface motion → rod elasticity → tension/compression → inertia → damping →
downhole plunger motion → rod loads**

and this will set up the later sections on:

dynamometer cards → rod floating → mechanical anomaly detection.

2.4 Rod-String Mechanics

Now we move from **pump fluid behaviour** to the **mechanical behaviour of the sucker-rod string**.

This section is critical because the rod string is the physical connection between:

Surface pumping unit ↔ Downhole pump

and it is also the foundation for understanding:

- rod loads
- surface dynamometer cards
- rod compression/tension

- rod deformation
- rod floating
- mechanical anomalies
- pump loading

The central idea is:

The surface pumping unit does not directly control the downhole plunger. It controls the top of an elastic, massive mechanical system.

2.4.1 What Is the Rod String?

FACT

In an SRP system, the sucker-rod string connects the surface polished rod to the downhole pump.

Its purpose is to transmit:

1. **displacement**
2. **force**
3. **mechanical energy**

between the surface pumping unit and the downhole pump.

PHYSICS

The rod string is not a perfectly rigid body.

It has:

- mass
- stiffness
- elasticity
- damping
- inertia
- friction
- buoyancy effects

Therefore, when the polished rod moves:

Surface motion → rod deformation → downhole response

rather than:

Surface motion = instantaneous downhole motion

This distinction is extremely important.

2.4.2 Why Rod Mechanics Matter in Heavy Oil

Baghewala produces heavy crude with high viscosity and low reservoir pressure.

That affects the rod system indirectly.

The complete chain becomes:

CSS

↓

Temperature ↑

↓

Viscosity ↓

↓

Reservoir inflow ↑

↓

Pump loading changes

↓

Rod loading changes

Therefore, the thermal operation can eventually influence the mechanical behaviour of the rod string.

This is one of the important **cross-layer couplings** in the digital twin.

2.4.3 Forces Acting on the Rod String

The rod string experiences several forces.

1. Rod weight

The rods have gravitational weight.

$F_{\text{weight}} = m_r g$

where:

- m_r = rod-string mass
- g = gravitational acceleration

This is a major component of the static rod load.

2. Buoyancy

The rods are immersed in fluid.

The surrounding fluid creates a buoyancy effect that reduces their effective weight.

A simplified effective-weight representation is:

$$W_{\text{eff}} = W_{\text{air}} - F_b$$

where:

- W_{eff} = effective rod weight in fluid
- W_{air} = rod weight in air
- F_b = buoyancy force

The exact buoyancy calculation depends on rod geometry and fluid density.

3. Fluid load

The pump and fluid impose additional forces on the rod string.

These depend on:

- fluid density
 - pressure
 - viscosity
 - pump conditions
 - acceleration
-

4. Inertial force

The rod string accelerates and decelerates during every stroke.

Therefore:

$$F_{\text{inertia}} = m_r a$$

where:

- m_r = effective rod mass
- a = acceleration

At high pumping speeds, inertial effects become increasingly important.

5. Elastic force

The rod string stretches and compresses.

The corresponding restoring force can be represented by:

$$F_{\text{elastic}} = k_r \Delta x$$

where:

- k_r = effective rod-string stiffness
 - Δx = rod deformation
-

6. Damping/friction

Energy is dissipated through:

- fluid interaction
- mechanical friction
- rod/tubing interaction
- material damping

A simplified damping term is:

$$F_{\text{damping}} = c_r v$$

where:

- c_r = effective damping coefficient
 - v = rod velocity
-

2.4.4 First-Order Rod Dynamic Equation

Combining the major mechanical effects gives a simplified mass-spring-damper representation:

$$\mathbf{M}_r \ddot{\mathbf{x}} + \mathbf{C}_r \dot{\mathbf{x}} + \mathbf{K}_r \mathbf{x} = \mathbf{F}_{\text{ext}}(\mathbf{t})$$

where:

- $\mathbf{M}_r$ = effective rod-string mass
- $\mathbf{C}_r$ = effective damping
- $\mathbf{K}_r$ = effective stiffness
- $\mathbf{x}$ = rod displacement/deformation state
- $\mathbf{F}_{\text{ext}}(\mathbf{t})$ = external force/load acting on the system

This is a **reduced-order model**.

The real sucker-rod string is distributed along the well depth, so a high-fidelity model would treat the rod as a continuous/discretized elastic system.

For SIH V1, the reduced-order model is more practical.

2.4.5 Rod Stretch

One of the simplest mechanical effects is elastic stretch.

For a simplified uniform rod:

$$\Delta L = FL/(EA)$$

where:

- ΔL = change in rod length
- F = axial force
- L = rod length
- E = Young's modulus
- A = rod cross-sectional area

This equation shows:

Higher load → greater deformation

and:

Longer rod → greater deformation

while:

Larger area or higher stiffness → lower deformation

In a real rod string, the cross-sectional area can vary between rod sections, so the full model should account for multiple rod segments.

2.4.6 Why Rod Stretch Matters

Suppose the surface polished rod moves upward by:

$\Delta x_{\text{surface}}$

The downhole plunger may not move by exactly the same amount at the same instant.

Part of the surface displacement can temporarily be stored as elastic deformation.

Therefore:

$x_{\text{downhole}} \neq x_{\text{surface}}$

at every instant.

This means that directly using surface position as pump position can introduce error.

2.4.7 Dynamic Rod Behaviour

The rod string is repeatedly accelerated and decelerated.

Therefore its behaviour is dynamic rather than purely static.

The simplified acceleration term is:

$F_{\text{inertia}} = M_r \ddot{x}$

As pumping speed increases:

$N \uparrow$

↓

acceleration/deceleration $\uparrow$

↓

inertial loading can increase

Therefore, high pumping speed can create mechanical consequences even if reservoir inflow is adequate.

This creates another optimization tradeoff:

Higher speed may increase theoretical production but can also increase dynamic rod loading.

2.4.8 Surface Load

The surface equipment experiences a load transmitted through the rod string.

Conceptually:

$$F_{\text{surface}} = F_{\text{static}} + F_{\text{fluid}} + F_{\text{inertia}} + F_{\text{friction}} + F_{\text{dynamic}}$$

The exact decomposition depends on the chosen model.

The surface load is extremely valuable because it provides indirect information about downhole conditions.

This is the foundation of the **dynamometer card**.

2.4.9 Position–Load Relationship

A conventional surface dynamometer card plots:

Rod load vs polished-rod position

rather than simply plotting load against time.

Conceptually:

$x(t)$

and

$F(t)$

are measured.

Then:

$$F = f(x)$$

over one complete pumping cycle.

The resulting closed-loop curve contains information about:

- pump loading
- rod behaviour
- fluid loading
- pump fillage

- abnormal operation

We will study this formally in **Section 2.5**.

2.4.10 Rod String as a Mechanical Filter

An important conceptual insight for our digital twin is that the rod string acts somewhat like a **mechanical transmission/filter**.

The surface observes:

Surface motion + surface load

But the actual downhole system experiences:

Downhole pump motion + downhole load

The rod string transforms one into the other through:

- elasticity
- inertia
- damping
- friction

Therefore:

Surface signal → rod dynamics → downhole state

This means we can potentially infer downhole behaviour without placing a sensor directly at every downhole location.

That is highly valuable in an industrial monitoring system.

2.4.11 Distributed Rod String

A real rod string may consist of multiple sections with different dimensions and mechanical properties.

Instead of one lumped rod, we can represent:

Rod 1 → Rod 2 → Rod 3 → ... → Rod n

Each segment has:

- mass
- length

- cross-sectional area
- stiffness
- damping

For a more detailed model, each segment can be represented by a mechanical state.

Conceptually:

$$\mathbf{M}_i \ddot{\mathbf{x}}_i + \mathbf{C}_i \dot{\mathbf{x}}_i + \mathbf{K}_i (\mathbf{x}_i - \mathbf{x}_{i-1}) = \mathbf{F}_i$$

This allows the model to capture wave propagation and dynamic deformation more realistically.

However, this is more computationally expensive.

2.4.12 V1 vs High-Fidelity Model

For the SIH prototype, we should use a hierarchy.

V1 — Reduced-order model

Treat the rod string as an effective:

mass + stiffness + damping system

Advantages:

- fast
- easy to calibrate
- suitable for optimization
- suitable for real-time digital twin

V2 — Segmented rod model

Divide the rod string into multiple sections.

Advantages:

- better representation of distributed dynamics
- better load prediction

V3 — High-fidelity rod-wave model

Model distributed wave propagation and detailed dynamic behaviour.

Advantages:

- higher physical fidelity

Disadvantage:

- more data
- more computation
- more difficult calibration

For SIH, **V1 + data-driven correction** is likely the better initial strategy.

2.4.13 Rod Mechanics and Pump Fillage

The rod string and pump fillage are coupled.

Suppose pump fillage decreases:

F ↓

↓

Fluid loading on pump changes

↓

Rod load pattern changes

↓

Surface dynamometer signature changes

Therefore:

Pump fillage → rod loads → surface signal

Conversely:

Surface load/position → inferred rod/pump behaviour → estimated fillage

This creates an opportunity for state estimation.

2.4.14 Rod Mechanics and Rod Floating

This is the connection to one of our most important future sections.

Rod floating is not simply:

“Rod moves strangely.”

It is a mechanical state associated with the interaction of:

- rod weight
- buoyancy
- fluid forces
- pump forces
- acceleration
- rod elasticity
- operating conditions

Therefore, to detect rod floating reliably, we need to understand the underlying rod mechanics first.

That is why rod floating should **not** be treated as a generic black-box classification problem.

2.4.15 A Simplified Rod-Load Model

For the first digital twin version, we can represent the surface rod load as:

$$\mathbf{F_rod(t)} = \mathbf{F_static} + \mathbf{F_fluid(t)} + \mathbf{F_inertia(t)} + \mathbf{F_elastic(t)} + \mathbf{F_friction(t)}$$

where:

- **F_static** = effective weight/buoyancy contribution
- **F_fluid** = fluid/pump loading
- **F_inertia** = acceleration-dependent force
- **F_elastic** = rod deformation contribution
- **F_friction** = friction/damping contribution

This is not intended to be a final exact field equation.

It is our **model decomposition**.

2.4.16 Why Temperature and Viscosity Appear Again

Layer 1 may seem separate from rod mechanics, but it isn't.

Recall:

$$T_r \uparrow \rightarrow \mu_o \downarrow$$

Now consider the rod system.

Changing viscosity can affect:

- fluid movement
- pressure losses
- pump loading
- damping/frictional effects
- fluid forces

Therefore:

CSS → temperature → viscosity → hydraulic conditions → pump loading → rod loading

This is a powerful example of **cross-domain coupling**.

It is exactly the kind of coupling that a well-to-surface digital twin should capture.

2.4.17 Mechanical Constraints

The SRP optimizer cannot simply maximize speed or stroke.

It must respect mechanical constraints.

Examples:

$$\mathbf{F_rod} \leq \mathbf{F_max}$$

$$\mathbf{N} \leq \mathbf{N_max}$$

$$\mathbf{S} \leq \mathbf{S_max}$$

$$\mathbf{Torque} \leq \mathbf{Torque_max}$$

and potentially:

$$\mathbf{rod\ stress} \leq \mathbf{allowable\ stress}$$

These constraints prevent the optimizer from recommending an operationally unsafe state.

The exact limits should come from actual equipment specifications rather than arbitrary values.

2.4.18 Energy Connection

The mechanical power required by the pumping system is related to force and velocity:

$$\mathbf{P_mech} = \mathbf{F\ v}$$

where:

- **P_mech** = instantaneous mechanical power
- **F** = mechanical load
- **v** = rod velocity

Average power depends on the entire pumping cycle.

This gives another important optimization dimension:

Production ↑

but if:

Energy consumption ↑ disproportionately

the operation may not be economically optimal.

Therefore, eventually we should optimize:

production + energy efficiency + equipment health

rather than production alone.

2.4.19 FACT vs PHYSICS vs ASSUMPTION vs PROPOSED DESIGN

FACT

- SRP uses a sucker-rod string to transmit reciprocating motion to the downhole pump.
- Baghewala's heavy-oil and low-pressure conditions make artificial lift an important part of production.

PHYSICS

- Rods have mass, stiffness and elasticity.
- Rods experience tension/compression, inertia, buoyancy, fluid loading and damping.
- Surface movement and downhole pump movement are not perfectly identical.
- Rod loads vary throughout every pumping cycle.
- Rod mechanics influence the surface dynamometer signature.

EQUATION

Simplified rod dynamics:

$$\mathbf{M}_r \ddot{\mathbf{x}} + \mathbf{C}_r \dot{\mathbf{x}} + \mathbf{K}_r \mathbf{x} = \mathbf{F}_{\text{ext}}(\mathbf{t})$$

Simplified elastic deformation:

$$\Delta L = FL/(EA)$$

Inertial force:

$$F_{\text{inertia}} = M_r \ddot{x}$$

Mechanical power:

$$P_{\text{mech}} = Fv$$

ASSUMPTION

For V1:

- The rod string can initially be represented by an equivalent mass-spring-damper system.
- Effective parameters can be calibrated from available operating data.
- Detailed distributed wave propagation is deferred.
- Friction and damping can initially be represented through effective coefficients.
- Actual equipment-specific mechanical limits must be obtained from field/equipment data.

PROPOSED DESIGN

Build a **Rod Dynamics Module** that takes:

Surface motion + operating speed + pump loading + fluid properties

and estimates:

downhole plunger motion + rod load + mechanical state

The module should provide both:

physics prediction

and

features for ML anomaly detection.

2.4.20 Layer 2 State After Rod Mechanics

Our state vector is becoming richer:

$$x = [T_r, P_r, \mu_o, q_{in}, F, x_{rod}, v_{rod}, F_{rod}, q_{surface}]$$

where:

- T_r = reservoir temperature
- P_r = reservoir pressure
- μ_o = oil viscosity
- q_{in} = reservoir inflow
- F = pump fillage
- x_{rod} = rod displacement state
- v_{rod} = rod velocity
- F_{rod} = rod load
- $q_{surface}$ = surface production

This is beginning to look like a genuine **stateful digital twin**, rather than a collection of independent formulas.

2.4.21 The Key Insight

The most important takeaway from rod mechanics is:

The rod string is not merely a connector between the surface unit and the pump. It is a dynamic mechanical system whose response contains information about what is happening downhole.

That means the same signal used to operate the SRP can also become a **diagnostic signal**.

This sets up our next section perfectly.

2.5 Dynamometer Card Physics

In the next section we will build:

What is a dynamometer card?

- How it is generated from load and position
- What each region of the card physically represents
- How pump fillage changes the card
- How rod loads appear
- How abnormal cards differ from normal cards
- How the card can be converted into ML features

→ **Why the dynamometer card is one of the most valuable data sources for detecting rod floating and pump problems.**

2.5 Dynamometer Card Physics

2.5.1 What is a Dynamometer Card?

FACT

A **dynamometer card** is a plot of **polished-rod load versus polished-rod position** during one complete pumping cycle of an SRP well.

It is one of the most valuable surface measurements for understanding what is happening downhole because the downhole pump itself is generally not directly observable during operation.

For our digital twin, the dynamometer card becomes an important bridge:

Surface measurements → Rod-string mechanics → Downhole pump condition → Pump fillage / abnormality

PHYSICS

During one pumping cycle:

- The polished rod moves upward.
- The rod string experiences changing weight, fluid load, inertia and elastic effects.
- The polished rod then moves downward.
- The load changes again because the direction of motion, acceleration and pump-valve conditions change.

Therefore, the load is not constant.

A simplified representation is:

$$\mathbf{F_{rod}(t) = F_{static} + F_{fluid}(t) + F_{inertia}(t) + F_{elastic}(t) + F_{friction}(t)}$$

The dynamometer card is essentially the graphical representation of this changing load against position.

PROPOSED DESIGN

Our digital twin should continuously construct a **real-time dynamometer card** from:

- polished-rod position
- polished-rod load
- pumping speed

- stroke length
 - motor/current information where available
 - tubing pressure
 - casing pressure
 - production rate
 - temperature
 - reservoir-side state estimated by Layer 1.
-

2.5.2 How the Dynamometer Card Is Generated

PHYSICS

Suppose the polished rod position is:

$$x(t)$$

and polished-rod load is:

$$F(t)$$

Then one complete pumping cycle generates:

$$D = \{(x(t), F(t)) \mid \text{one complete cycle}\}$$

Plotting:

- X-axis $\rightarrow$ polished-rod position
- Y-axis $\rightarrow$ polished-rod load

produces the dynamometer card.

The important point is that **time itself is not the primary axis**.

Instead, we examine how load changes as the rod moves through its stroke.

EQUATION

Polished-rod velocity:

$$v(t) = dx/dt$$

Polished-rod acceleration:

$$a(t) = d^2x/dt^2$$

Rod inertial force:

$$F_{\text{inertia}} = m_{\text{eff}} a(t)$$

Therefore, acceleration around stroke reversal points can create significant load changes even if the fluid load itself has not changed dramatically.

ASSUMPTION

For the first digital-twin prototype, the surface position and load signals are assumed to be synchronized and sampled sufficiently fast to reconstruct one complete pumping cycle.

Real field sampling frequency and sensor quality must be obtained from actual Baghewala instrumentation/SCADA data.

2.5.3 What Does Each Part of the Card Represent?

The card should not be treated simply as an image.

Its shape is a **physical signature of the pumping system**.

PHYSICS

During the upstroke:

- the rod string moves upward
- rod weight and fluid load contribute to surface load
- pump pressure changes
- valve conditions change
- elastic deformation changes.

During the downstroke:

- rod acceleration reverses
- fluid and rod loading changes
- pump chamber conditions change
- valve states change.

Thus, different portions of the card correspond to different mechanical/hydraulic events.

PROPOSED DESIGN

Instead of feeding only the raw card to an AI model, divide every cycle into physically meaningful regions:

1. Upstroke

2. Upper reversal
3. Downstroke
4. Lower reversal

Then calculate features from each region.

This gives us **physics-informed ML features** rather than treating the card as an arbitrary image.

2.5.4 Relation Between Pump Fillage and Dynamometer Card

This is one of the most important connections in our system.

PHYSICS

Consider two wells with identical:

- stroke length
- pumping speed
- pump dimensions

but different reservoir inflow.

Case A — Good reservoir inflow

Reservoir supplies sufficient oil to the pump.

Therefore:

$$q_{in} \approx q_{pump\ demand}$$

The pump chamber can fill adequately.

This produces a healthier pumping cycle.

Case B — Poor reservoir inflow

Reservoir cannot supply fluid fast enough.

Therefore:

$$q_{in} \ll q_{pump\ demand}$$

The pump chamber becomes partially filled.

This is **pump underfill**.

Consequently, the fluid load carried by the pump/rod system changes, which modifies the dynamometer card.

Therefore:

Reservoir condition → Pump fillage → Rod load → Dynamometer card

This is extremely important for our architecture because Layer 1 can estimate reservoir inflow while Layer 2 can use the dynamometer card to infer whether the pump is actually receiving that inflow.

2.5.5 Pump-Fillage Signature

EQUATION

Pump fillage was defined previously as:

$$F = V_{\text{filled}} / V_{\text{available}}$$

where:

- **F** = pump fillage fraction
- **V_{filled}** = actual volume entering/filling the pump chamber
- **V_{available}** = pump chamber volume available during the relevant cycle.

Therefore:

$$0 \leq F \leq 1$$

Actual production can be represented approximately as:

$$Q_{\text{actual}} \approx A_p S N F \eta$$

where:

- **A_p** = plunger cross-sectional area
- **S** = stroke length
- **N** = pumping speed
- **F** = fillage
- **η** = other efficiency factors.

PHYSICS

If pumping speed increases while reservoir inflow remains approximately unchanged:

$$N \uparrow \rightarrow Q_{\text{theoretical}} \uparrow$$

but:

$$q_{\text{in}} \approx \text{constant}$$

Therefore:

$q_{in} / Q_{theoretical} \downarrow$

and fillage can decrease.

This explains an important optimization problem:

Increasing pump speed does not necessarily increase oil production.

The system may simply create more pump displacement than the reservoir can supply.

2.5.6 Dynamometer Card and Rod Loading

PHYSICS

The polished-rod load is affected by several components:

$$\mathbf{F_{rod} = F_{weight} + F_{fluid} + F_{inertia} + F_{elastic} + F_{friction}}$$

Therefore, changes in the dynamometer card may indicate changes in:

- pump loading
- fluid loading
- rod dynamics
- pump fillage
- friction
- valve behaviour
- mechanical condition.

EQUATION

A simplified dynamic model is:

$$\mathbf{M_{eff} \ddot{x} + C_{eff} \dot{x} + K_{eff} x = F_{ext}(t)}$$

where:

- **M_{eff}** = equivalent rod-system mass
- **C_{eff}** = equivalent damping
- **K_{eff}** = equivalent stiffness
- **x** = rod displacement
- **F_{ext}** = external force.

This model is deliberately simplified.

A real rod string is distributed rather than a single lumped mass.

2.5.7 Why the Card Changes After CSS

This is where our two layers become strongly coupled.

PHYSICS

CSS increases reservoir temperature:

$T_r \uparrow$

which reduces oil viscosity:

$\mu_o \downarrow$

which increases mobility:

$\lambda_o \uparrow$

which can increase reservoir inflow:

$q_{in} \uparrow$

which can increase pump fillage:

$F \uparrow$

which changes fluid loading:

F_{fluid} changes

which changes the dynamometer card.

Therefore:

CSS $\rightarrow$ $T_r \rightarrow \mu_o \rightarrow q_{in} \rightarrow F \rightarrow$ Rod Load $\rightarrow$ Dynamometer Card

This gives our digital twin a way to observe the effect of thermal stimulation through surface mechanical behaviour.

IMPORTANT

This does **not** mean that every change in a dynamometer card is caused by CSS.

Mechanical problems, valve problems, gas interference, rod friction, sensor errors and other factors can produce similar changes.

Therefore, the system must combine:

Reservoir model + wellbore model + SRP model + dynamometer analysis

rather than relying on the card alone.

2.5.8 Abnormal Dynamometer Cards

FACT / ENGINEERING KNOWLEDGE

Different downhole problems can produce characteristic changes in surface load-position behaviour.

Potential abnormal conditions include:

- incomplete pump fillage
- pump leakage
- valve problems
- gas interference
- rod/tubing friction
- mechanical restrictions
- rod-string problems
- abnormal loading
- sensor abnormalities.

PROPOSED DESIGN

Our AI system should therefore perform **multi-class pump-condition classification** rather than only predicting production.

Example output:

Condition	Digital Twin Interpretation
Normal	Pump operating within expected envelope
Underfill	Reservoir supply insufficient relative to pump demand
Possible valve issue	Abnormal valve-related loading pattern
Possible pump leakage	Reduced effective displacement
Mechanical anomaly	Rod/load behaviour outside expected envelope
Sensor anomaly	Physically inconsistent measurements

Condition	Digital Twin Interpretation
Unknown	Pattern does not match trained conditions

The **Unknown** category is important.

A winning system should not force every abnormal condition into a known class.

2.5.9 Dynamometer Card as an ML Input

PROPOSED DESIGN

We can represent the card in two complementary ways.

Approach A — Time-series representation

Input:

[position(t), load(t), velocity(t), acceleration(t)]

Models:

- 1D CNN
- LSTM
- GRU
- Transformer
- Temporal CNN

This preserves the temporal structure.

Approach B — Image representation

Convert the load-position cycle into an image.

Then use:

- CNN
- Vision Transformer
- autoencoder
- contrastive learning.

This is particularly useful because your team has **computer vision/deep learning expertise**.

BEST APPROACH FOR SIH

Do **not** choose only one.

Use a hybrid representation:

Raw signal → physical features + card image → multimodal ML

For example:

Dynamometer image features

-

time-series features

-

pressure

-

temperature

-

speed

-

production

-

Layer-1 reservoir state

→ **Pump-state classifier / fillage estimator**

This is much stronger than a simple CNN classifier.

2.5.10 Physics-Informed Feature Engineering

Instead of giving the AI hundreds of raw values, extract physically meaningful features.

PROPOSED FEATURES

Card geometry

- maximum load
- minimum load
- load range
- stroke width
- card area

- upper-stroke slope
- lower-stroke slope
- reversal sharpness.

Dynamic features

- velocity
- acceleration
- peak acceleration
- acceleration near reversals
- cycle time
- pumping speed.

Reservoir features

- estimated T_r
- estimated μ_o
- estimated q_{in}
- pressure differential
- cooling rate.

Pump features

- theoretical displacement
- estimated fillage
- supply-demand ratio
- estimated pump efficiency.

Mechanical features

- maximum rod load
- load amplitude
- cyclic loading
- estimated stress
- torque/power where available.

This creates a **physics-informed feature space**.

2.5.11 Dynamometer Card → Rod Floating Detection

This becomes especially important for the SIH problem.

Rod floating is not simply another production decline.

It is a **mechanical/dynamic abnormal condition**.

The key principle is:

Expected rod/pump dynamics $\neq$ observed rod/pump dynamics

Therefore, rod floating detection should compare:

Expected card / load trajectory

against

Observed card / load trajectory

PROPOSED DETECTION PIPELINE

Surface load + position



Cycle segmentation



Dynamometer card



Feature extraction



Expected-card model



Residual calculation



Anomaly detector



Rod-floating probability



Severity + confidence

This gives us a much stronger architecture than simply saying:

“Use ML to detect rod floating.”

2.5.12 Expected-Card Digital Twin

This is potentially one of our strongest innovations.

Instead of training an AI model only to classify abnormal cards, the digital twin first predicts:

What should the dynamometer card look like under the current operating conditions?

Inputs:

- reservoir inflow
- oil viscosity
- pressure
- pump speed
- stroke
- rod properties
- pump parameters
- tubing conditions.

Output:

Expected dynamometer card

Then compare:

Observed card – Expected card

The difference becomes the **physics-informed residual**.

PROPOSED DESIGN

Define:

$$R_{\text{card}} = D_{\text{observed}} - D_{\text{predicted}}$$

where:

- **D_observed** = measured dynamometer card
- **D_predicted** = digital-twin predicted card

- **R_card** = residual/error card.

If the residual becomes large or structurally abnormal:

Anomaly probability ↑

This is substantially more defensible than a black-box classifier.

2.5.13 Why This Is Valuable for SIH

The core problem is not simply:

“Predict oil production.”

The actual problem is:

Optimize the entire well-to-surface system while identifying mechanical and production abnormalities.

Dynamometer cards allow us to connect:

Reservoir

→ **Wellbore**

→ **Pump**

→ **Rod string**

→ **Surface measurement**

Therefore they become an important observability layer for the digital twin.

2.5.14 Layer 2.5 Output

At the end of this module, the digital twin should produce:

Input

Surface load + polished-rod position + operating conditions

↓

Dynamometer Card

↓

Physical feature extraction

↓

Pump/rod state estimation

↓

Outputs:

1. **Pump fillage estimate**
2. **Rod load estimate**
3. **Pump condition**
4. **Mechanical anomaly score**
5. **Rod-floating probability**
6. **Expected-vs-observed card residual**
7. **Confidence/uncertainty**

These outputs will feed the next modules.

Layer 2.5 Interface

The information flow becomes:

Layer 1 — Reservoir

$T_r \rightarrow \mu_o \rightarrow \lambda_o \rightarrow q_{in}$

↓

Layer 2.3 — Pump

$q_{in} \rightarrow \text{expected fillage}$

↓

Layer 2.4 — Rod Mechanics

Pump loading $\rightarrow$ rod dynamics

↓

Layer 2.5 — Dynamometer

Rod dynamics $\rightarrow$ surface load-position card

↓

ML

Observed card + physics features $\rightarrow$ pump/rod state



Optimization

State + constraints → optimal SRP/CSS operating recommendation

Key SIH Insight

The strongest architecture is therefore not:

“Dynamometer Card → CNN → Rod Floating”

It is:

Reservoir Physics → Inflow Prediction → SRP Physics → Expected Dynamometer Card → Observed Card → Physics-Informed Residual → ML Anomaly Detection → Optimization

That is the foundation for making the solution look like a **digital twin**, rather than just an ML prediction system.

Next in sequence: 2.6 Rod Floating Physics & Detection.

2.6 Rod Floating Physics & Detection

Rod floating is one of the most important mechanical failure/anomaly conditions for our digital twin because it directly connects **SRP operating conditions, rod dynamics, pump behaviour, and production loss**.

The objective is not merely to classify “rod floating = yes/no.” The system should estimate:

- whether rod floating is occurring,
 - its probability,
 - its severity,
 - its likely cause,
 - and the safest corrective operating action.
-

2.6.1 What Is Rod Floating?

FACT

The SRP rod string connects the surface pumping unit to the downhole pump. The rod string experiences changing loads during every pumping cycle.

Under abnormal dynamic conditions, the rod string can experience a loss of the expected tensile/compressive loading behaviour. This can cause the rod string to behave differently from the intended pumping motion.

For our digital twin, **rod floating is treated as a dynamic abnormality of the rod/pump system**, rather than simply a production-rate anomaly.

PHYSICS

The rod string is affected by:

- rod weight,
- buoyancy,
- fluid load,
- pressure forces,
- acceleration,
- elasticity,
- damping,
- friction,
- pumping speed,
- stroke length.

The instantaneous rod force can therefore be represented conceptually as:

$$F_{\text{rod}}(t) = F_{\text{weight}} + F_{\text{buoyancy}} + F_{\text{fluid}} + F_{\text{inertia}} + F_{\text{elastic}} + F_{\text{friction}}$$

When operating conditions change sufficiently, the balance between these forces changes.

The resulting rod motion may depart from the expected SRP trajectory.

2.6.2 Why Rod Floating Is Important at Baghewala

FACT

Baghewala produces heavy crude with high viscosity and operates with artificial lift and thermal stimulation. The SIH problem specifically identifies the need to optimize CSS and SRP operations and detect conditions such as rod floating.

PHYSICS

The Baghewala operating environment creates a strongly coupled system:

CSS

- temperature increases
- oil viscosity decreases
- reservoir inflow can increase
- pump loading changes
- rod loading changes
- rod dynamics change.

But the opposite situation can also occur:

Reservoir cooling

- viscosity increases
- inflow decreases
- pump fillage decreases
- fluid loading changes
- rod dynamics change.

Therefore, rod-floating detection should **not** operate independently of the reservoir state.

2.6.3 Force Balance Behind Rod Dynamics

EQUATION

A simplified rod-string model is:

$$\mathbf{M_eff} \ddot{\mathbf{x}} + \mathbf{C_eff} \dot{\mathbf{x}} + \mathbf{K_eff} \mathbf{x} = \mathbf{F_ext(t)}$$

where:

- **M_eff** = equivalent rod-string mass
- **C_eff** = equivalent damping
- **K_eff** = equivalent stiffness
- **x** = rod displacement
- **$\dot{\mathbf{x}}$** = rod velocity
- **$\ddot{\mathbf{x}}$** = rod acceleration
- **F_ext** = net external force.

The external force can be decomposed approximately as:

$$F_{\text{ext}} = F_{\text{weight}} + F_{\text{fluid}} + F_{\text{pressure}} + F_{\text{inertia}} + F_{\text{friction}}$$

The exact formulation depends on the rod-string and pump model.

ASSUMPTION

For V1, we use an equivalent mass-spring-damper model rather than a full distributed wave-equation model.

This keeps the digital twin computationally fast enough for optimization and real-time inference.

2.6.4 Why Pump Speed Matters

PHYSICS

Pump speed controls how rapidly the rod string moves through its stroke.

If:

$$N \uparrow$$

then the frequency of pumping cycles increases.

Acceleration also changes because:

$$a(t) = d^2x/dt^2$$

Therefore, higher pumping speed can increase dynamic/inertial loading.

At the same time:

$$Q_{\text{theoretical}} = A_p S N$$

so:

$$N \uparrow \rightarrow Q_{\text{theoretical}} \uparrow$$

But reservoir inflow does not necessarily increase at the same rate.

Therefore:

$$q_{\text{in}} / Q_{\text{theoretical}} \downarrow$$

can occur.

This can cause poorer pump fillage and alter the rod-load pattern.

KEY INSIGHT

A pump can be operated faster while actually becoming **less effectively supplied by the reservoir**.

That is why our optimizer cannot simply maximize SRP speed.

2.6.5 Rod Floating as a Coupled Reservoir–Pump–Rod Problem

The complete chain is:

Reservoir state

↓

q_{in}

↓

Pump fillage

↓

Fluid loading

↓

Rod dynamics

↓

Dynamometer card

↓

Rod-floating probability

This is one of the most important reasons to build a **well-to-surface digital twin** instead of independent ML models.

2.6.6 Expected vs Observed Rod Behaviour

This should be the core of our detection system.

PROPOSED DESIGN

The digital twin predicts the expected rod behaviour:

D_{predicted} = f(T_r, μ_o, q_{in}, P_{wf}, N, S, rod_properties, pump_properties)

The real well provides:

D_{observed} = measured dynamometer card

Then calculate:

$$R_card = D_observed - D_predicted$$

where:

- **D_predicted** = expected dynamometer card
- **D_observed** = measured card
- **R_card** = physics-informed residual.

If the residual remains within the expected uncertainty envelope:

Normal operation

If it moves substantially outside the envelope:

Potential abnormal condition

This approach is much stronger than using a static threshold.

2.6.7 Rod Floating Detection Pipeline

PROPOSED DESIGN

The complete pipeline should be:

SCADA / Sensors

↓

Data validation

↓

Cycle segmentation

↓

Dynamometer-card reconstruction

↓

Rod dynamics estimation

↓

Expected-card generation

↓

Observed – Expected residual



Feature extraction



Anomaly detection



Rod-floating classifier



Probability + severity



Recommended action

The output should look conceptually like:

Rod Floating Probability: 0.87

Severity: High

Confidence: 0.91

Primary suspected cause: Excessive pump speed / poor pump supply

Recommended action: Reduce SRP speed and reassess fillage

The actual threshold values must be calibrated using field data or validated synthetic/experimental data.

2.6.8 Physics-Based Features for Rod Floating

Instead of relying exclusively on a deep-learning model, extract physically meaningful features.

PROPOSED FEATURES

Load features

- maximum rod load
- minimum rod load
- mean load
- load range
- load variance

- cycle-to-cycle load variation.

Position features

- stroke length
- upper reversal position
- lower reversal position
- position asymmetry.

Dynamic features

- maximum velocity
- maximum acceleration
- acceleration at reversal
- velocity profile
- cycle frequency.

Dynamometer-card features

- card area
- card width
- card height
- slope changes
- curvature
- upper/lower stroke asymmetry
- deviation from expected card.

Hydraulic features

- tubing pressure
- casing pressure
- pump intake pressure
- estimated fluid level where available
- estimated q_{in}
- estimated pump fillage.

Thermal/reservoir features

- reservoir temperature estimate
- oil viscosity estimate
- cooling rate
- CSS cycle stage.

This is where our **data-engineering + deep-learning combination** becomes useful.

2.6.9 Rule-Based Detection vs ML Detection

We should not immediately use deep learning for everything.

Approach 1 — Rule-Based

Example:

If:

fillage < F_min

and

rod-load residual > R_threshold

then:

potential abnormal rod condition

Advantages:

- interpretable
- fast
- easy to validate
- useful when data is limited.

Disadvantage:

- fixed thresholds may fail under changing well conditions.
-

Approach 2 — Machine Learning

Inputs:

[card features, load, position, speed, pressure, fillage, q_in, μ_o , temperature]

Output:

P(rod floating | observations)

Potential models:

- Random Forest
- XGBoost
- LightGBM
- 1D CNN
- LSTM/GRU
- Transformer
- Autoencoder.

For SIH, tree-based models can be excellent for tabular features, while CNN/time-series models can analyze the dynamometer card.

2.6.10 Recommended Hybrid Model

PROPOSED DESIGN

Use three layers:

Layer A — Physics residual

Observed card – predicted card



Layer B — Feature-based ML

Use:

- residual features
- card geometry
- pump parameters
- reservoir state
- operating conditions.



Layer C — Deep-learning card model

Use the complete card as a time-series/image.



Fusion

Combine:

Physics score + ML score + DL score

to produce:

Rod Floating Probability

This is a much more defensible architecture than:

CNN → Rod Floating

2.6.11 Confidence and Uncertainty

This is essential for a field deployment system.

Suppose:

Rod floating probability = 0.92

but the sensor quality is poor.

The system should not present this as absolute truth.

Instead:

Rod Floating: HIGH RISK

Probability: 92%

Data confidence: LOW

This distinction is important.

PROPOSED OUTPUT

The digital twin should maintain:

State estimate

-

Prediction

-

Uncertainty

For example:

$q_{in} = 82 \pm 7 \text{ bbl/day}$

Fillage = 0.68 ± 0.06

Rod-floating probability = 0.84

This makes the system much more realistic for operational decision support.

2.6.12 Root-Cause Classification

Rod floating should not be the final answer.

The operator needs to know:

Why is this happening?

Potential causal categories:

Cause A — Reservoir-limited

$q_{in} \downarrow$

→ fillage $\downarrow$

→ abnormal pump/rod behaviour.

Cause B — Excessive SRP speed

$N \uparrow$

→ theoretical displacement $\uparrow$

→ supply-demand mismatch

→ fillage $\downarrow$

→ dynamic anomaly.

Cause C — High viscosity

$T_r \downarrow$

→ $\mu_o \uparrow$

→ mobility $\downarrow$

→ $q_{in} \downarrow$

→ pump underfill.

Cause D — Pump/valve problem

Reservoir inflow may be adequate, but actual pump performance is poor.

Cause E — Mechanical problem

Rod/string/pump behaviour deviates from expected mechanics.

Cause F — Sensor/data problem

The apparent anomaly may be caused by bad or misaligned measurements.

This causal hierarchy is very important for the optimization engine.

2.6.13 Counterfactual Diagnosis

This can become a major SIH differentiator.

Instead of only saying:

“Rod floating detected.”

the digital twin can test hypothetical actions.

For example:

Scenario 1

Current SRP speed = N

Predicted risk:

High

Scenario 2

Reduce speed by 10%.

Digital twin predicts:

Fillage ↑

Rod dynamic residual ↓

Production impact = –small

Scenario 3

Increase reservoir temperature through optimized CSS.

Digital twin predicts:

μ_o ↓

q_{in} ↑

Fillage ↑

Rod-floating risk ↓

This turns anomaly detection into **decision support**.

2.6.14 Rod-Floating Risk Score

PROPOSED DESIGN

We can define a composite risk score:

$$R_{RF} = w_1 R_{card} + w_2 R_{dynamic} + w_3 R_{fillage} + w_4 R_{load} + w_5 R_{operating}$$

where:

- **R_card** = card-residual abnormality
- **R_dynamic** = dynamic anomaly
- **R_fillage** = fillage-related risk
- **R_load** = load-related risk
- **R_operating** = operating-condition risk
- **w_i** = calibrated weights.

This is a **proposed system score**, not an established petroleum-engineering equation.

The score can then be mapped to:

- Low
 - Moderate
 - High
 - Critical.
-

2.6.15 Digital Twin Response to Rod Floating

The digital twin should not stop at detection.

The response loop should be:

Detect

↓

Diagnose

↓

Simulate corrective actions



Evaluate constraints



Recommend



Operator approval



Apply



Observe response



Update twin

For example:

Rod-floating probability ↑



Check:

- q_in
- fillage
- viscosity
- speed
- stroke
- load
- pressure



Determine likely cause



Simulate:

- reduce speed
- reduce/increase stroke within limits
- modify CSS timing
- modify soak period
- wait for thermal recovery
- inspect mechanical condition.



Choose safest feasible action.

2.6.16 Critical Safety Principle

PROPOSED DESIGN

The system should initially be **operator-in-the-loop**.

It should recommend:

“Reduce SRP speed from current operating point to recommended range.”

rather than directly controlling field equipment.

This is important because:

- incorrect model predictions can have operational consequences,
- sensor faults can create false alarms,
- equipment limits must be respected,
- field approval is required.

Autonomous control can be a future phase after sufficient validation.

2.6.17 How This Fits the Complete Digital Twin

The architecture now becomes:

CSS



Reservoir temperature



Oil viscosity



Reservoir mobility



Reservoir inflow



Wellbore



Pump fillage



Rod mechanics



Expected dynamometer card



Observed dynamometer card



Physics residual



ML/DL anomaly detection



Rod-floating probability



Root-cause diagnosis



Optimization engine



Recommended SRP/CSS action

This is the **well-to-surface optimization loop** the SIH problem is asking for.

2.6.18 Layer 2.6 Outputs

At the end of this module, our digital twin should produce:

Output	Meaning
Rod-floating probability	Likelihood of rod-floating condition
Severity	Low / Moderate / High / Critical
Fillage estimate	Estimated pump filling
Load anomaly	Deviation from expected rod loading
Card residual	Observed vs predicted card difference
Root-cause probability	Reservoir / speed / viscosity / pump / mechanical / sensor
Confidence	Reliability of the diagnosis
Recommended action	Candidate corrective operating change

The Major Innovation

The key idea is:

Do not train AI to recognize rod floating directly. First build a digital twin that predicts what normal rod behaviour should look like, then use the deviation between predicted and observed behaviour as the signal for ML-based anomaly detection.

That gives us:

Physics → Prediction → Residual → AI → Diagnosis → Optimization

instead of:

Data → Black-box classification

This distinction can become one of the strongest technical differentiators in the SIH pitch.

Next in sequence: 2.7 Wellbore Hydraulics & Pressure Dynamics.

2.7 Wellbore Hydraulics & Pressure Dynamics

Wellbore hydraulics is the missing physical link between **reservoir inflow** and **downhole pump behaviour**.

Layer 1 estimated:

Reservoir → q_{in}

Layer 2.3–2.6 established:

q_{in} → pump fillage → rod loading → dynamometer card → rod-floating risk

But the pump does not operate directly against reservoir conditions. Fluid must travel through the **wellbore, pump and tubing**, and pressure losses along this path strongly affect pump intake conditions.

Therefore:

Reservoir → Wellbore → Pump → Tubing → Surface

must be represented in the digital twin.

2.7.1 Why Wellbore Hydraulics Matters at Baghewala

FACT

The Baghewala field involves heavy crude with high viscosity and low reservoir pressure. The SIH problem identifies poor oil mobility as a major production challenge and requires optimization of CSS and SRP operations.

PHYSICS

For a heavy-oil system, viscosity is especially important because hydraulic pressure losses are strongly affected by fluid resistance.

The Layer 1 chain was:

CSS → $T \uparrow$ → $\mu_o \downarrow$ → mobility $\uparrow$ → $q_{in} \uparrow$

But once the fluid enters the wellbore, another chain becomes important:

μ_o → hydraulic resistance → pressure loss → pump intake pressure → inflow

Therefore, reservoir and wellbore cannot be treated as independent.

2.7.2 Basic Pressure Balance

The pressure at the downhole pump must be related to the reservoir and surface conditions.

A simplified pressure relationship is:

$P_r \rightarrow P_{wf} / P_{intake} \rightarrow P_{tubing} \rightarrow P_{wh}$

where:

- **P_r** = reservoir pressure
- **P_{wf}** = flowing bottomhole pressure
- **P_{intake}** = pump intake pressure
- **P_{tubing}** = pressure at a point in the tubing
- **P_{wh}** = wellhead pressure.

The pressure changes because of:

- hydrostatic head,
- frictional pressure loss,
- acceleration effects,
- pump pressure change,
- surface pressure constraints.

2.7.3 Hydrostatic Pressure

PHYSICS

A fluid column produces hydrostatic pressure because of its weight.

For a simplified liquid column:

$$\Delta P_h = \rho g h$$

where:

- **ΔP_h** = hydrostatic pressure difference
- **ρ** = fluid density
- **g** = gravitational acceleration
- **h** = vertical depth.

Therefore:

$$P_{\text{bottom}} \approx P_{\text{surface}} + \rho g h$$

for a simplified static liquid column.

IMPORTANT

This is only a first-order approximation.

Actual producing wells may involve:

- multiphase flow,
- changing fluid density,
- gas,
- temperature variation,
- pressure-dependent properties.

So the equation is appropriate for the reduced-order twin, but not a complete field hydraulic model.

2.7.4 Frictional Pressure Loss

PHYSICS

When fluid flows through tubing, energy is lost because of friction.

Conceptually:

Flow rate $\uparrow \rightarrow$ velocity $\uparrow \rightarrow$ frictional pressure loss $\uparrow$

and:

Viscosity $\uparrow \rightarrow$ hydraulic resistance $\uparrow$

For a simple pipe-flow representation:

$$\Delta P_f \propto f (L/D) (\rho v^2/2)$$

where:

- ΔP_f = frictional pressure loss
- f = friction factor
- L = flow length
- D = hydraulic diameter
- ρ = fluid density
- v = average fluid velocity.

The exact friction-factor relationship depends on the flow regime and fluid rheology.

2.7.5 Why Heavy Oil Makes This Difficult

FACT

Baghewala crude has very high viscosity. OIL public technical information reports viscosity in the approximate range of **10,000–13,000 cP at 50°C**, while the SIH problem describes the crude as highly viscous.

PHYSICS

For heavy oil:

$\mu_o \uparrow$

can lead to:

hydraulic resistance $\uparrow$

and therefore:

pressure losses $\uparrow$

for a given flow condition.

This creates an important feedback:

Reservoir cooling

→ **viscosity increases**

→ **inflow decreases**

and simultaneously:

→ **wellbore hydraulic resistance increases**

Therefore cooling can hurt production through **both reservoir mobility and wellbore hydraulics**.

2.7.6 Temperature Variation Along the Well

This is particularly important for CSS.

PHYSICS

Immediately after steam stimulation, the well/reservoir system may have a significantly different thermal state than during the later production period.

Therefore:

$T = T(z,t)$

rather than simply:

$T = \text{constant}$

where:

- z = depth
- t = time.

Oil viscosity consequently becomes:

$$\mu_o = \mu_o(T(z,t))$$

This means the fluid entering the pump may have a different viscosity from fluid farther up the tubing.

2.7.7 Reduced-Order Wellbore Model

A full transient multiphase thermal wellbore simulation would be computationally expensive and would require detailed field data.

For the SIH prototype, we should use a reduced-order model.

PROPOSED DESIGN

Represent the wellbore using a small number of hydraulic segments:

Reservoir

↓

Bottomhole segment

↓

Pump intake

↓

Tubing segment 1

↓

Tubing segment 2

↓

Wellhead

Each segment estimates:

- temperature
- pressure
- viscosity

- flow rate.

The digital twin therefore retains the dominant physics without attempting a full CFD-scale simulation.

2.7.8 Pressure-Flow Relationship

A useful reduced-order representation is:

$$P_{\text{intake}} = P_{\text{wh}} + \Delta P_{\text{h}} - \Delta P_{\text{f}} - \Delta P_{\text{other}}$$

where:

- **P_intake** = pump intake pressure
- **P_wh** = wellhead pressure
- **ΔP_h** = hydrostatic contribution
- **ΔP_f** = frictional pressure loss
- **ΔP_other** = other modeled pressure losses.

The exact sign convention should be fixed in the implementation.

PROPOSED DESIGN

Use this equation as the hydraulic backbone connecting the surface and downhole states.

2.7.9 Coupling With Reservoir Inflow

Recall Layer 1:

$$q_{\text{in}} = J_o(P_r - P_{\text{wf}})$$

But now:

$$P_{\text{wf}} \approx P_{\text{intake}}$$

and P_{intake} depends on wellbore hydraulics.

Therefore:

$$q_{\text{in}} = f(P_r, P_{\text{intake}}, \mu_o, \text{reservoir properties})$$

while:

$$P_{\text{intake}} = f(q_{\text{surface}}, \mu_o, \text{tubing geometry, depth, } P_{\text{wh}})$$

This creates a **coupled feedback system**.

2.7.10 The Important Feedback Loop

The complete relationship is:

Pump speed ↑

↓

Theoretical displacement ↑

↓

Flow demand ↑

↓

Tubing flow ↑

↓

Frictional pressure loss ↑

↓

Pump intake pressure changes

↓

Reservoir pressure differential changes

↓

Reservoir inflow changes

↓

Pump fillage changes

↓

Rod load changes

This is why optimizing SRP speed using only historical production data is insufficient.

2.7.11 Reservoir-Limited vs Wellbore-Limited Production

This distinction should become an important diagnostic capability.

Case A — Reservoir-limited

q_{in} is low

even though the wellbore can transport the produced fluid.

The solution may involve:

- CSS optimization,
- thermal recovery,
- reduced pump demand,
- optimized soak period.

Case B — Wellbore-limited

Reservoir can potentially supply fluid, but hydraulic pressure losses prevent efficient transport.

The solution may involve:

- operating-condition changes,
- hydraulic optimization,
- tubing/flow restrictions,
- surface pressure management.

Case C — Pump-limited

Reservoir and wellbore can supply fluid, but the SRP system cannot efficiently lift it.

Possible causes:

- pump fillage,
- speed,
- valve behaviour,
- mechanical condition.

Case D — Surface-limited

Downhole system is capable of producing, but surface constraints limit actual flow.

Examples:

- flowline pressure,
- separator capacity,
- facility constraints.

PROPOSED DESIGN

The digital twin should classify the current well state as:

Reservoir-limited

or

Wellbore-limited

or

Pump-limited

or

Surface-limited

or

Mixed limitation

This is highly useful for optimization.

2.7.12 Pressure Dynamics During a Pumping Cycle

The wellbore pressure is not necessarily static during an SRP cycle.

PHYSICS

During pumping:

plunger movement → pressure changes → valve transitions → fluid movement

Therefore:

$P(t)$

changes throughout the cycle.

This dynamic pressure response influences:

- standing-valve behaviour,
- traveling-valve behaviour,
- pump fillage,
- fluid loading,
- rod loading.

A high-fidelity model would therefore require transient pump/wellbore dynamics.

ASSUMPTION

For V1, we can represent the wellbore with cycle-averaged or quasi-steady hydraulic states and use the dynamometer model to capture the faster mechanical dynamics.

This creates a useful separation:

Slow dynamics

Reservoir temperature / viscosity / pressure

versus

Fast dynamics

Rod motion / load / pumping cycle.

This multi-timescale structure is very important for computational efficiency.

2.7.13 Multi-Timescale Digital Twin

Our digital twin should not treat every variable at the same update frequency.

Slow states

Updated over minutes/hours:

- reservoir temperature
- reservoir pressure
- viscosity
- thermal recovery state
- CSS cycle state.

Medium-speed states

Updated over minutes:

- pump intake pressure
- tubing pressure
- production rate
- pump fillage.

Fast states

Updated over each pumping cycle:

- polished-rod position

- rod load
- velocity
- acceleration
- dynamometer card.

This gives us:

Fast mechanical twin

-

Medium hydraulic twin

-

Slow reservoir/thermal twin

rather than one unnecessarily expensive monolithic model.

2.7.14 Wellbore Temperature Model

PROPOSED DESIGN

A reduced-order thermal model can represent the wellbore temperature as:

$$C_w \frac{dT_w}{dt} = Q_{in} - Q_{loss} - Q_{prod}$$

where:

- C_w = effective thermal capacity of the wellbore
- T_w = representative wellbore temperature
- Q_{in} = thermal energy entering the wellbore
- Q_{loss} = heat lost to surrounding formation
- Q_{prod} = thermal energy carried away by produced fluid.

For a spatial model:

$$\rho C_p \frac{\partial T}{\partial t} = k_T \frac{\partial^2 T}{\partial z^2} + Q$$

can be used as a simplified one-dimensional representation.

The parameters should be calibrated rather than invented.

2.7.15 Viscosity–Hydraulics Coupling

This is one of the strongest physics links in the entire system.

We already have:

$$T \uparrow \rightarrow \mu_o \downarrow$$

Now:

$$\mu_o \downarrow \rightarrow \text{hydraulic resistance} \downarrow$$

Therefore:

$$CSS \rightarrow T \uparrow \rightarrow \mu \downarrow \rightarrow \text{hydraulic losses} \downarrow$$

At the same time:

$$CSS \rightarrow T \uparrow \rightarrow \mu \downarrow \rightarrow \text{reservoir mobility} \uparrow \rightarrow q_{in} \uparrow$$

Thus thermal stimulation can potentially improve production through **two pathways**:

Reservoir pathway

$$T \uparrow \rightarrow \mu \downarrow \rightarrow \text{mobility} \uparrow \rightarrow \text{inflow} \uparrow$$

Wellbore pathway

$$T \uparrow \rightarrow \mu \downarrow \rightarrow \text{hydraulic resistance} \downarrow \rightarrow \text{pressure loss} \downarrow$$

This is an important insight for our well-to-surface digital twin.

2.7.16 Surface Pressure as a Control Variable

Surface pressure should not be treated merely as a measurement.

PHYSICS

Changing wellhead/surface pressure changes the pressure boundary condition for the wellbore.

This can influence:

P_intake

and consequently:

q_in

through:

$$q_{in} = J_o(P_r - P_{wf})$$

PROPOSED DESIGN

The optimization engine should therefore consider:

- SRP speed
- stroke length
- CSS steam parameters
- soak time
- wellhead pressure constraints

as coupled variables where field-operationally permissible.

2.7.17 Hydraulic Constraints

The optimization engine must respect constraints such as:

$P_{\text{intake}} \geq P_{\text{min}}$

P_{wellhead} within operating limits

$\Delta P_f \leq \text{allowable pressure loss}$

$N \leq N_{\text{max}}$

$S \leq S_{\text{max}}$

$F \geq F_{\text{min}}$

$F_{\text{rod}} \leq F_{\text{max}}$

The exact limits must come from actual Baghewala equipment/well specifications.

IMPORTANT

We should never hard-code arbitrary safety limits into the SIH prototype and present them as field limits.

They must be labelled:

Field specification / calibration parameter / assumed prototype constraint.

2.7.18 Wellbore Model as a Diagnostic Tool

The wellbore model can also detect inconsistencies.

Suppose:

Observed P_{intake}

is much different from:

Predicted P_{intake}

given the measured flow and pressure conditions.

Then:

Hydraulic residual = $P_{\text{observed}} - P_{\text{predicted}}$

A persistent large residual could indicate:

- changing fluid properties,
- incorrect viscosity estimate,
- restriction,
- sensor problem,
- unmodelled multiphase effects,
- pump/wellbore abnormality.

Therefore, just like the dynamometer card:

Hydraulic residual $\rightarrow$ anomaly signal

2.7.19 Physics + ML for Wellbore Prediction

The same hybrid strategy should be used here.

Physics model

Predict:

$P_{\text{intake}}, \Delta P_f, T_w$

ML correction

Learn:

Residual = Observed – Physics prediction

Then:

Corrected prediction = Physics prediction + ML residual

This has three advantages:

1. Physical consistency is retained.
2. Real field behaviour can be learned.

3. The model does not need to learn basic physics from scratch.

2.7.20 Layer 2.7 Outputs

The wellbore module should provide:

Output	Purpose
Pump intake pressure	Connects wellbore to reservoir
Tubing pressure profile	Hydraulic state
Pressure loss	Detects hydraulic restriction
Wellbore temperature	Viscosity estimation
Effective viscosity	Hydraulic/reservoir coupling
Hydraulic residual	Anomaly detection
Flow capacity	Wellbore limitation estimate
Limiting subsystem	Reservoir / wellbore / pump / surface

2.7.21 Updated Well-to-Surface Architecture

We now have:

LAYER 1 — RESERVOIR

CSS

↓

Thermal state

↓

Temperature

↓

Viscosity

↓

Mobility

↓

Reservoir inflow



LAYER 2 — WELL + SRP

Reservoir inflow



Wellbore pressure/temperature



Pump intake conditions



Pump fillage



Rod mechanics



Dynamometer card



Rod-floating/anomaly detection



LAYER 3 — OPTIMIZATION

Current state



Candidate CSS/SRP actions



Digital-twin simulation



Constraint checking



Objective evaluation



Optimal recommendation

This is the architecture we will build toward.

2.7.22 The Key SIH Insight

The most important point from this layer is:

The pump does not “see” reservoir pressure directly. It sees a dynamic pump-intake condition created by the coupled interaction of reservoir inflow, wellbore hydraulics, fluid viscosity, pump operation and surface pressure.

Therefore, the correct digital-twin chain is:

CSS

- **Reservoir temperature**
- **Oil viscosity**
- **Reservoir mobility**
- **Reservoir inflow**
- **Wellbore pressure/temperature**
- **Pump intake conditions**
- **Pump fillage**
- **Rod dynamics**
- **Dynamometer card**
- **Rod-floating risk**
- **Optimization**

That is the physical backbone of the **well-to-surface digital twin**.

Next in sequence: 2.8 Surface Pumping Unit & Motor/Power Dynamics.

2.8 Surface Pumping Unit & Motor/Power Dynamics

The previous layers established:

Reservoir → Wellbore → Pump → Rod String → Dynamometer → Rod-Floating Risk

Now we need to model the **surface energy system** that actually drives the rod string.

The surface system is important because changing SRP speed or stroke changes not only production but also:

- motor power,
- gearbox loading,
- torque,
- rod dynamics,
- pump displacement,
- energy consumption,
- mechanical risk.

Therefore, the optimization engine must optimize **production and energy together**, subject to equipment constraints.

2.8.1 Surface Pumping Unit Architecture

FACT

An SRP system uses a surface pumping unit to convert rotary motion from a motor into reciprocating motion of the polished rod.

The simplified energy path is:

Motor



Gearbox



Crank



Walking beam



Horsehead



Polished rod



Rod string



Downhole pump

This connects the surface electrical system to the downhole hydraulic system.

PHYSICS

The motor supplies rotational mechanical power.

The pumping unit converts:

Rotational motion → Reciprocating motion

The polished rod then transmits this mechanical motion to the rod string and downhole pump.

Therefore:

Electrical energy → Motor → Mechanical energy → Rod motion → Hydraulic work → Produced fluid

2.8.2 Pumping Speed

The primary surface operating variable is pumping speed.

Let:

N = pumping speed

typically represented in cycles per minute.

Then:

Cycle frequency $\propto$ N

Increasing N means more pumping cycles per unit time.

EQUATION

Theoretical pump displacement:

$$Q_{\text{theoretical}} = A_p S N$$

where:

- **A_p** = plunger area
- **S** = stroke length
- **N** = pumping speed.

Therefore:

N ↑ → theoretical displacement ↑

But actual production remains:

$$Q_{\text{actual}} \approx A_p S N F \eta$$

where:

- F = pump fillage
- η = other efficiency factors.

This distinction is critical.

2.8.3 Why Maximum Speed Is Not the Optimal Speed

PHYSICS

Suppose reservoir inflow is:

$$q_{\text{in}} = \text{constant}$$

while SRP speed increases:

$$N \uparrow$$

Then:

$$Q_{\text{theoretical}} \uparrow$$

but:

$$q_{\text{in}} \approx \text{constant}$$

Therefore the pump may become increasingly underfilled.

This can lead to:

- lower fillage,
- less incremental oil,
- altered rod loading,
- greater dynamic loading,
- higher energy consumption,
- potentially greater mechanical risk.

Hence:

Maximum pumping speed $\neq$ maximum economic production.

This should become one of the central principles of our optimization engine.

2.8.4 Surface Stroke Length

The other major control variable is stroke length.

Let:

S = polished-rod stroke length

Then:

$$Q_{\text{theoretical}} = A_p S N$$

Therefore:

S $\uparrow$ $\rightarrow$ theoretical displacement $\uparrow$

But the same supply-demand problem exists.

If:

$$q_{\text{in}} < A_p S N$$

then increasing stroke can also increase pump demand faster than reservoir supply.

PROPOSED DESIGN

The optimizer should jointly consider:

N + S

rather than optimizing speed alone.

2.8.5 Surface Motion Is Not Equal to Downhole Motion

PHYSICS

The rod string is elastic.

Therefore:

surface displacement $\neq$ exactly downhole plunger displacement

because of:

- rod stretch,
- elastic wave propagation,
- inertia,
- damping,

- friction.

A simplified relation is:

$$x_p(t) \approx x_{\text{surface}}(t) - \Delta x_{\text{rod}}(t)$$

where:

- x_p = downhole plunger position
- x_{surface} = polished-rod position
- Δx_{rod} = effective rod deformation.

This is why a surface dynamometer card must be interpreted using rod mechanics.

2.8.6 Motor Power

The motor provides the energy required to overcome:

- hydraulic load,
- rod weight,
- acceleration/inertia,
- friction,
- gearbox/pumping-unit losses.

EQUATION

Mechanical power can be expressed as:

$$P_{\text{mech}} = F v$$

where:

- F = instantaneous force
- v = instantaneous velocity.

For rotary motor operation:

$$P_{\text{rot}} = T \omega$$

where:

- T = torque
- ω = angular velocity.

Therefore:

P_motor → torque + speed → surface mechanical motion

2.8.7 Hydraulic Power

The useful hydraulic work associated with lifting fluid can be approximated by:

$$\mathbf{P_{hyd} \approx \Delta P Q}$$

where:

- ΔP = pressure increase provided to move the fluid
- Q = flow rate.

The total energy requirement is higher because the system has mechanical and hydraulic losses.

Conceptually:

$$\mathbf{P_{motor} > P_{mech} > P_{hyd}}$$

depending on how the boundaries and efficiencies are defined.

A more useful system-level relation is:

$$\mathbf{P_{hyd} = \eta_{total} P_{motor}}$$

where:

$$\mathbf{\eta_{total} = \eta_{motor} \eta_{gearbox} \eta_{unit} \eta_{rod} \eta_{pump} \eta_{hyd}}$$

as a simplified decomposition.

2.8.8 Energy Efficiency

The system should not optimize oil production alone.

A useful performance metric is:

$$\mathbf{Energy\ intensity = Energy\ consumed / Oil\ produced}$$

For example:

$$\mathbf{E_{oil} = kWh / bbl}$$

This allows us to distinguish between:

Scenario A

Production increases significantly while energy increases modestly.

→ desirable.

Scenario B

Energy consumption increases significantly but production barely changes.

→ inefficient.

Scenario C

Speed increases but fillage decreases.

→ potentially poor operating point.

This is exactly why the optimizer should be **multi-objective**.

2.8.9 Torque Dynamics

PHYSICS

The pumping unit experiences changing torque throughout the pumping cycle.

The torque requirement depends on:

- rod load,
- crank position,
- beam geometry,
- acceleration,
- counterbalance,
- fluid load.

Conceptually:

$$\mathbf{T(t) = f(F_{rod}(t), \theta(t), geometry, acceleration, counterbalance)}$$

where:

- $\mathbf{T(t)}$ = instantaneous torque
- $\mathbf{F_{rod}(t)}$ = rod load
- $\mathbf{\theta(t)}$ = crank angle.

Therefore, motor loading is inherently cyclic.

2.8.10 Counterbalance

PHYSICS

A pumping unit's counterbalance helps reduce the net torque and power requirement by balancing part of the rod/fluid load.

The system therefore does not simply lift the complete rod/fluid load from zero every cycle.

Counterbalance affects:

- motor torque,
- peak power,
- energy consumption,
- gearbox loading,
- motor operating efficiency.

PROPOSED DESIGN

Where counterbalance information is available, it should become an input to the surface pumping-unit model.

If it is unavailable in the prototype dataset, it should be treated as a fixed parameter rather than invented.

2.8.11 Motor Current as a Sensor

This is particularly useful for our data-engineering architecture.

PROPOSED DESIGN

Motor current can act as an indirect measurement of mechanical loading.

Potential inputs:

- motor current
- motor voltage
- motor speed
- torque estimate
- power
- polished-rod load
- polished-rod position.

These signals can be synchronized with the dynamometer card.

Therefore:

Electrical signature

-

Mechanical signature

can provide stronger anomaly detection.

2.8.12 Electrical–Mechanical–Hydraulic Coupling

This gives us another powerful chain:

Motor current



Motor torque



Pumping-unit torque



Rod load



Pump loading



Hydraulic production

At the same time:

Reservoir



q_{in}



Pump fillage



Fluid load



Rod load



Motor load

Therefore, motor measurements can indirectly contain information about downhole conditions.

This is extremely valuable when direct downhole measurements are unavailable.

2.8.13 Surface Power as an Anomaly Signal

Suppose the digital twin predicts:

P_{expected}

but measured motor power is:

P_{observed}

Then:

$$\mathbf{R_{power} = P_{observed} - P_{expected}}$$

A persistent abnormal residual can indicate:

- mechanical friction,
- pump loading changes,
- rod problems,
- hydraulic changes,
- abnormal operating conditions,
- sensor issues.

Again, this follows our general architecture:

Expected physics → observed data → residual → anomaly detection

2.8.14 Combining Three Surface Signals

We now have three highly valuable surface observables:

1. Dynamometer card

Load vs position

2. Motor/electrical signal

Current/power/torque vs time

3. Production/pressure

Flow + pressure vs time

Together they provide:

Mechanical + electrical + hydraulic observability

PROPOSED DESIGN

The ML system should therefore use multimodal data:

Dynamometer card

-

Motor current/power

-

Pressure

-

Production

-

Temperature

-

Layer-1 reservoir state

→

Well-state estimator

This is much more powerful than a single-sensor AI model.

2.8.15 Surface Operating Envelope

The pumping unit has a safe operating region.

Conceptually:

$$N_{\min} \leq N \leq N_{\max}$$

$$S_{\min} \leq S \leq S_{\max}$$

$$F_{\text{rod}} \leq F_{\text{max}}$$

$$T_{\text{motor}} \leq T_{\text{max}}$$

$$P_{\text{motor}} \leq P_{\text{max}}$$

The exact limits must come from equipment specifications.

IMPORTANT

For the SIH prototype:

- actual equipment limits → **FACT**
- published/standard engineering limits → **REFERENCE**
- assumed demonstration limits → **ASSUMPTION**

They must never be mixed.

2.8.16 Multi-Objective SRP Optimization

We can now define the optimization problem.

PROPOSED DESIGN

The optimizer should maximize something like:

$$J = w_q Q_o - w_E E - w_R R_{\text{mech}} - w_A R_{\text{anomaly}}$$

where:

- Q_o = predicted oil production
- E = energy consumption
- R_{mech} = mechanical-risk metric
- R_{anomaly} = anomaly/rod-floating risk
- w_q, w_E, w_R, w_A = user-defined or calibrated weights.

This is a **proposed objective function**, not an established field equation.

A more SIH-friendly implementation is to present a **Pareto frontier** rather than claiming one universal optimum.

2.8.17 Why Pareto Optimization Is Better

Consider two operating points:

Point A

- higher production
- higher energy
- higher mechanical risk.

Point B

- slightly lower production
- much lower energy
- much lower mechanical risk.

Which is better?

There is no universal answer.

The operator may choose based on:

- production priority,
- energy cost,
- equipment condition,
- field strategy,
- safety constraints.

Therefore the digital twin can generate:

Pareto-optimal operating points

and allow the operator to choose.

This makes the system more realistic.

2.8.18 CSS–SRP Interaction at Surface

This is where the full problem begins to become interesting.

Suppose CSS increases reservoir temperature.

Then:

$T_r \uparrow$

$\rightarrow \mu_o \downarrow$

$\rightarrow q_{in} \uparrow$

- pump fillage ↑
- fluid load changes
- rod load changes
- motor torque changes
- energy requirement changes

Thus, a CSS decision can affect the surface motor system.

Conversely:

SRP speed ↑

- fluid demand ↑
- pump intake conditions change
- reservoir inflow response changes

Therefore CSS and SRP cannot be optimized independently.

2.8.19 The Unified Control Vector

The digital twin can represent the controllable variables as:

$$\mathbf{u} = [\dot{m}_s, V_s, t_{\text{soak}}, N, S]$$

where:

- $\dot{m}_s$ = steam injection rate
- V_s = steam volume
- t_{soak} = soak time
- N = SRP speed
- S = stroke length.

Additional controllable variables may be added if actual field operations support them.

This is the first point where the problem becomes a true **joint optimization problem**.

2.8.20 State Vector

The corresponding system state can be represented as:

$$\mathbf{x} = [T_r, P_r, \mu_o, q_{\text{in}}, P_{\text{intake}}, F, F_{\text{rod}}, P_{\text{motor}}, R_{\text{RF}}]$$

where:

- T_r = reservoir temperature
- P_r = reservoir pressure
- μ_o = oil viscosity
- q_{in} = reservoir inflow
- P_{intake} = pump intake pressure
- F = pump fillage
- F_{rod} = rod load
- P_{motor} = motor power
- R_{RF} = rod-floating risk.

This is a conceptual state vector; the final implementation can use additional latent states.

2.8.21 Digital Twin Prediction

For every candidate operating point, the twin should simulate:

CSS input

↓

Reservoir state

↓

Inflow

↓

Wellbore pressure

↓

Pump fillage

↓

Rod dynamics

↓

Motor load

↓

Production



Energy



Risk

Then the optimizer compares all feasible candidates.

2.8.22 Example Decision

Imagine the current operating state is:

SRP speed = high

and the digital twin estimates:

- fillage = low
- rod-floating risk = high
- energy intensity = high.

The optimizer evaluates:

Option A

Increase speed.

Prediction:

Production ↑ slightly

Energy ↑ significantly

Fillage ↓

Risk ↑

Reject.

Option B

Reduce speed.

Prediction:

Production ≈ maintained

Energy ↓

Fillage ↑

Risk ↓

Potentially attractive.

Option C

First optimize CSS, then adjust SRP.

Prediction:

Reservoir inflow ↑

Fillage ↑

SRP can operate more efficiently

Potentially even better.

This is exactly the type of **counterfactual decision support** we want to demonstrate.

2.8.23 Surface Layer Outputs

The surface pumping-unit module should output:

Output	Purpose
Pumping speed	Current SRP operating state
Stroke length	Current displacement
Motor power	Energy consumption
Motor torque	Mechanical loading
Rod load	Mechanical state
Energy intensity	Efficiency
Mechanical risk	Equipment protection
Operating envelope	Feasible region
Candidate operating points	Optimization inputs

2.8.24 Updated Architecture

We now have:

LAYER 1

Reservoir + CSS



LAYER 2.7

Wellbore hydraulics



LAYER 2.3

Pump fillage



LAYER 2.4

Rod mechanics



LAYER 2.5

Dynamometer card



LAYER 2.6

Rod-floating detection



LAYER 2.8

Surface pumping unit + motor dynamics



LAYER 3

Joint CSS + SRP optimization

2.8.25 Key SIH Insight

The major insight from this layer is:

SRP speed and stroke are not merely production controls; they are energy, hydraulic, and mechanical controls simultaneously.

Therefore, a winning digital twin should never recommend:

“Increase RPM because production is low.”

It should evaluate:

Reservoir supply

-

Pump demand

-

Wellbore hydraulics

-

Pump fillage

-

Rod dynamics

-

Motor power

-

Mechanical risk

and then determine whether increasing, decreasing, or maintaining the operating point is optimal.

That is the transition from a **production prediction system** to a genuine **well-to-surface optimization system**.

Next in sequence: 2.9 Well-to-Surface Coupled Digital Twin State Model.

2.9 Well-to-Surface Coupled Digital Twin State Model

We now have enough individual physics modules to combine them into a **single coupled digital twin**.

The central idea is that the well should no longer be represented as independent modules:

Reservoir model

SRP model

Rod model

Wellbore model

Surface model

Instead, all of them must exchange states continuously.

2.9.1 Why a Coupled Digital Twin Is Needed

FACT

The SIH problem asks for a digital twin that supports optimization of **CSS and SRP operations** for heavy-oil wells at Baghewala.

The physical system is inherently coupled.

A CSS decision changes the reservoir thermal state.

That changes viscosity.

Viscosity affects mobility and inflow.

Inflow affects pump fillage.

Pump fillage affects rod loading.

Rod loading affects motor torque and power.

Therefore:

A decision made at one layer propagates through the entire well-to-surface system.

A digital twin that models only one layer cannot correctly evaluate these interactions.

2.9.2 Complete Physical Chain

The complete physical chain developed so far is:

CSS

↓

Steam thermal input

↓

Reservoir temperature

↓

Oil viscosity

↓

Oil mobility



Reservoir inflow



Wellbore pressure & temperature



Pump intake conditions



Pump fillage



Rod dynamics



Dynamometer card



Rod-floating risk



Surface pumping-unit load



Motor torque/power



Surface production

This becomes the core physics architecture of the project.

2.9.3 State Variables

A digital twin needs to maintain an estimate of the **current state of the physical well**.

PROPOSED STATE VECTOR

We can represent the state as:

$$\mathbf{x}(t) = [T_r, P_r, \mu_o, q_{in}, P_{intake}, T_w, F, x_r, v_r, F_{rod}, D_{card}, P_{motor}, R_{RF}, q_o]$$

where:

Variable Meaning

T_r	Reservoir temperature
P_r	Reservoir pressure
μ_o	Oil viscosity
q_{in}	Reservoir inflow
P_{intake}	Pump intake pressure
T_w	Wellbore temperature
F	Pump fillage
x_r	Rod displacement
v_r	Rod velocity
F_{rod}	Rod load
D_{card}	Dynamometer-card state
P_{motor}	Motor power
R_{RF}	Rod-floating risk
q_o	Surface oil production

Not every state needs to be directly measured.

Some are **latent states** estimated by the digital twin.

2.9.4 Measured vs Estimated States

This distinction is essential.

DIRECTLY MEASURABLE

Depending on field instrumentation:

- polished-rod position
- polished-rod load
- pumping speed

- stroke
- motor current
- motor voltage
- wellhead pressure
- tubing pressure
- casing pressure
- temperature
- production rate.

ESTIMATED

Potentially:

- reservoir temperature
- reservoir pressure
- oil viscosity
- reservoir inflow
- pump fillage
- pump intake pressure
- rod stress
- mechanical condition
- rod-floating probability.

PROPOSED DESIGN

The twin should maintain a **state confidence** alongside every estimated state.

For example:

Estimated $q_{in} = 85 \text{ bbl/day}$

Confidence = 88%

rather than presenting the estimate as a measured fact.

2.9.5 Control Variables

The twin also needs to know what the operator can change.

Define:

$$\mathbf{u}(t) = [\dot{m}_s, V_s, t_{\text{soak}}, N, S]$$

where:

- $\dot{m}_s$ = steam injection rate
- V_s = steam volume
- t_{soak} = soak duration
- N = SRP speed
- S = SRP stroke length.

These are the principal control variables for the first prototype.

Additional controls can be added only when actual field operations support them.

2.9.6 Disturbances

Not everything affecting the well is controllable.

Define:

$\mathbf{d}(t)$ as external disturbances.

Examples:

- reservoir property changes
- ambient temperature
- steam-quality variation
- water quality
- changing fluid composition
- sensor noise
- unexpected mechanical degradation
- surface pressure variation
- unmodelled reservoir heterogeneity.

This distinction is important:

Control = what we intentionally change

Disturbance = what happens to the system

2.9.7 State Transition Model

The complete digital twin can be represented conceptually as:

$$\mathbf{x}(t+1) = \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), \boldsymbol{\theta}] + \mathbf{w}(t)$$

where:

- $\mathbf{x}(t)$ = current system state
- $\mathbf{u}(t)$ = control inputs
- $\mathbf{d}(t)$ = disturbances
- $\boldsymbol{\theta}$ = physical/model parameters
- $\mathbf{w}(t)$ = model uncertainty/process noise.

Measurements are:

$$\mathbf{y}(t) = \mathbf{h}[\mathbf{x}(t), \mathbf{u}(t)] + \mathbf{v}(t)$$

where:

- $\mathbf{y}(t)$ = measured sensor data
- $\mathbf{h}(\cdot)$ = measurement relationship
- $\mathbf{v}(t)$ = sensor/measurement noise.

This gives us the mathematical foundation of the digital twin.

2.9.8 What Does $\mathbf{f}(\cdot)$ Actually Contain?

The function $\mathbf{f}(\cdot)$ is not one black-box equation.

It is the combination of our physical modules.

Reservoir transition

$$\mathbf{x}_{\text{res}}(t+1) = \mathbf{f}_{\text{res}}(\mathbf{x}_{\text{res}}, \mathbf{u}_{\text{CSS}})$$

produces:

$$T_r, P_r, \mu_o, q_{\text{in}}$$

Wellbore transition

$$\mathbf{x}_{\text{wb}}(t+1) = \mathbf{f}_{\text{wb}}(q_{\text{in}}, \mu_o, P_{\text{wh}}, \text{geometry})$$

produces:

P_intake, T_w

Pump transition

$x_{\text{pump}}(t+1) = f_{\text{pump}}(q_{\text{in}}, P_{\text{intake}}, N, S, \mu_o)$

produces:

F

Rod transition

$x_{\text{rod}}(t+1) = f_{\text{rod}}(F, N, S, \text{rod_properties})$

produces:

F_rod, x_r, v_r

Surface transition

$x_{\text{surface}}(t+1) = f_{\text{surface}}(F_{\text{rod}}, N, S, \text{motor_properties})$

produces:

P_motor, torque, energy

Together:

$f = f_{\text{res}} + f_{\text{wb}} + f_{\text{pump}} + f_{\text{rod}} + f_{\text{surface}}$

Conceptually, these modules form one coupled system.

2.9.9 Bidirectional Coupling

A major design principle is that information should flow in **both directions**.

Forward direction

Reservoir

→ **Wellbore**

→ **Pump**

→ **Rod**

→ **Surface**

But the reverse influence also exists.

Reverse direction

SRP speed/stroke

→ **pump demand**

→ **pump intake pressure**

→ **flowing pressure**

→ **reservoir inflow**

Therefore:

The reservoir affects the pump, and the pump operating condition affects the reservoir boundary condition.

This is a feedback system.

2.9.10 Coupling Equations

Layer 1 gave:

$$\mathbf{q_in = J_o(P_r - P_wf)}$$

Layer 2.7 gives approximately:

$$\mathbf{P_wf \approx P_intake}$$

Layer 2.3 gives:

$$\mathbf{Q_theoretical = A_p S N}$$

and:

$$\mathbf{Q_actual \approx A_p S N F \eta}$$

Therefore the coupling becomes:

q_in

↓

P_intake

↓

F

↓

Q_actual

while simultaneously:

N,S

↓

Q_theoretical

↓

P_intake

↓

q_in

This creates a closed physical loop.

2.9.11 The Most Important Digital-Twin Loop

The real-time loop should be:

1. Measure

↓

2. Clean data

↓

3. Estimate hidden states

↓

4. Update digital twin

↓

5. Predict future state

↓

6. Simulate candidate actions

↓

7. Check constraints

↓

8. Optimize

↓

9. Recommend action

↓

10. Operator approval



11. Apply action



12. Measure response



13. Update model

This is the operational heart of the system.

2.9.12 State Estimation

The digital twin cannot directly measure everything.

Therefore it needs a **state estimator**.

PROPOSED DESIGN

Inputs:

- load
- position
- speed
- pressure
- temperature
- production
- motor current
- CSS history.



State estimator



Outputs:

- T_r
- P_r

- μ_o
- q_{in}
- P_{intake}
- F
- mechanical state.

Possible methods include:

- Extended Kalman Filter
- Unscented Kalman Filter
- Particle Filter
- moving-horizon estimation
- neural state-space models.

For the SIH prototype, we should avoid using an unnecessarily complex estimator if the available data is limited.

A **physics-based estimator + ML correction** is likely the most defensible approach.

2.9.13 Physics Model + ML Residual

This becomes the recurring architecture throughout our twin.

Physics model

Predict:

$\hat{y}_{physics}$

Actual observation

y_{actual}

Residual

$$r = y_{actual} - \hat{y}_{physics}$$

Then ML learns:

$$r \approx f_{ML}(inputs)$$

Corrected prediction:

$$\hat{y}_{corrected} = \hat{y}_{physics} + r_{ML}$$

This can be applied to:

- reservoir temperature
- viscosity
- inflow
- pump fillage
- pressure
- dynamometer card
- motor power.

This is a **hybrid digital twin**.

2.9.14 Why Not Use a Pure Deep-Learning Model?

A pure neural network could attempt:

Sensor data → Production

But that creates several problems.

Problem 1 — Limited field data

SIH teams generally will not have enormous labelled Baghewala datasets.

Problem 2 — Explainability

An operator needs to understand why a recommendation was generated.

Problem 3 — Physical inconsistency

A black-box model might predict a physically unrealistic state.

Problem 4 — Extrapolation

The model may fail when operating conditions differ from training data.

Problem 5 — Safety

A control recommendation should have physically meaningful constraints.

Therefore:

Physics should define the feasible behaviour; ML should learn what physics misses.

2.9.15 Digital Twin Hierarchy

Our proposed twin can have four levels.

Level 1 — Physics core

- thermal model
- viscosity model
- mobility
- Darcy inflow
- wellbore hydraulics
- pump displacement
- rod mechanics
- surface power.

Level 2 — Data assimilation

Use real measurements to continuously update the model.

Level 3 — ML residual/anomaly layer

Learn:

- systematic model error
- pump fillage
- abnormal cards
- rod-floating risk
- equipment degradation.

Level 4 — Optimization

Search for:

best CSS + SRP operating policy

under constraints.

This hierarchy is extremely important for the pitch.

2.9.16 Digital Twin State at Any Moment

At any point in time, the dashboard should conceptually be able to show:

Reservoir

Temperature: estimated

Viscosity: estimated

Pressure: estimated

Inflow: estimated

Wellbore

Pump intake pressure

Wellbore temperature

Hydraulic pressure loss

SRP

Speed

Stroke

Pump fillage

Rod

Maximum load

Dynamic loading

Rod-floating probability

Surface

Motor power

Energy intensity

Production

Current oil rate

Predicted oil rate

This gives the operator a **single well state**, rather than multiple disconnected dashboards.

2.9.17 Digital Twin State Vector With Uncertainty

A stronger representation is:

$$\hat{\mathbf{X}}(t) = [\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n]$$

along with:

$\Sigma(t)$

where:

- $\hat{X}$ = estimated state
- Σ = uncertainty/covariance representation.

For example:

$T_r = 78 \pm 3^{\circ}\text{C}$

$q_{in} = 92 \pm 8 \text{ bbl/day}$

$F = 0.76 \pm 0.05$

Rod-floating risk = $0.12 \pm$ confidence estimate

The exact uncertainty method can be selected during implementation.

2.9.18 Physics-Based Digital Twin vs Data-Driven Twin

Aspect	Physics-only	ML-only Our Hybrid Twin	
Explainability	High	Low	High
Field data requirement	Moderate	High	Moderate
Handles nonlinear patterns	Limited	Strong	Strong
Physical constraints	Strong	Weak	Strong
Anomaly detection	Moderate	Strong	Strong
Extrapolation	Better if physics valid	Weak	Better
SIH suitability	Good	Risky	Best fit

This hybrid architecture should be one of our major differentiation points.

2.9.19 Digital Twin as a Graph

Another useful conceptual representation is a **physics dependency graph**:

CSS

→ **Thermal Energy**

→ **Temperature**

- **Viscosity**
- **Mobility**
- **Inflow**
- **Wellbore Pressure**
- **Pump Fillage**
- **Rod Load**
- **Dynamometer**
- **Motor Power**
- **Production**

And feedback:

SRP controls

- **Pump demand**
- **Pump intake**
- **Reservoir inflow**

This graph can also help explain the system to judges.

2.9.20 Multi-Timescale Architecture

One of the most technically valuable design decisions is to use different simulation timescales.

Reservoir/CSS

minutes → hours → days

Wellbore

seconds → minutes

Pumping cycle

seconds

Rod dynamics

milliseconds → seconds

Optimization

minutes → hours

This means we should not run a full high-frequency reservoir simulation for every rod movement.

Instead:

Fast twin

handles mechanical dynamics.

Medium twin

handles hydraulic behaviour.

Slow twin

handles reservoir thermal evolution.

This drastically reduces computational requirements.

2.9.21 Digital Twin Synchronization

The real field and virtual field must remain synchronized.

PROPOSED DESIGN

At time **t**:

Real well measurements

↓

Data ingestion

↓

Quality control

↓

State estimation

↓

Twin update

↓

Prediction

↓

Decision

At the next measurement interval:

New observation

↓

Compare against prediction

↓

Update model

This creates a continuously learning operational twin.

2.9.22 Twin Health Score

We can also monitor the digital twin itself.

Define conceptually:

Twin Health = f(data quality, residuals, model uncertainty, sensor availability)

If:

- sensor data missing,
- residuals become very large,
- model uncertainty becomes high,

then the twin should say:

“Low-confidence prediction”

rather than generating aggressive recommendations.

This is important for field safety.

2.9.23 Fail-Safe Behaviour

PROPOSED DESIGN

If:

sensor failure

or

model uncertainty > threshold

or

prediction outside validated operating domain

then:

Optimization disabled

and system switches to:

Monitoring / advisory-only mode

This is much safer than forcing an optimization result.

2.9.24 What Makes This a Real Digital Twin?

A dashboard showing predicted production is **not enough**.

Our system qualifies conceptually as a digital twin because it contains:

1. **Physical representation**
2. **Real-time data connection**
3. **State estimation**
4. **Dynamic prediction**
5. **Model updating**
6. **What-if simulation**
7. **Decision optimization**
8. **Feedback from real system**

The key distinction is:

The twin should evolve with the physical well.

2.9.25 Final Coupled Mathematical Representation

The complete system can be summarized as:

State transition

$$\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t, \mathbf{d}_t, \boldsymbol{\theta}) + \mathbf{w}_t$$

Measurement

$$\mathbf{y}_t = \mathbf{h}(\mathbf{x}_t, \mathbf{u}_t) + \mathbf{v}_t$$

Prediction

$$\hat{\mathbf{x}}_{t+k} = \mathbf{F}(\mathbf{x}_t, \mathbf{u}_{t:t+k-1})$$

Optimization

$$u = \operatorname{argmax} J(x, u)^*$$

subject to:

$$\mathbf{g}(\mathbf{x}, \mathbf{u}) \leq \mathbf{0}$$

and

$$\mathbf{h}(\mathbf{x}, \mathbf{u}) = \mathbf{0}$$

where:

- $\mathbf{x}$ = system state
- $\mathbf{u}$ = control variables
- $\mathbf{d}$ = disturbances
- $\boldsymbol{\theta}$ = model parameters
- $\mathbf{w}$ = process uncertainty
- $\mathbf{v}$ = measurement noise
- J = optimization objective
- $\mathbf{g}$ = inequality constraints
- $\mathbf{h}$ = equality constraints.

This becomes the mathematical foundation for Layer 3.

2.9.26 Layer 2.9 Outputs

At the end of this module, we have:

State

Reservoir + wellbore + pump + rod + surface

Controls

CSS + SRP

Measurements

SCADA/sensors

Hidden-state estimation

Temperature, viscosity, inflow, fillage, etc.

Prediction

Future production + mechanical behaviour

Residuals

Observed – expected

Anomaly detection

Rod floating + other abnormalities

Optimization interface

Current state → candidate actions → predicted outcomes

2.9.27 The Complete Well-to-Surface Twin

Our architecture is now:

LAYER 1 — RESERVOIR & CSS

CSS controls



Thermal model



Temperature



Viscosity



Mobility



Reservoir inflow



LAYER 2 — WELL + SRP

Wellbore hydraulics



Pump intake pressure



Pump fillage



Rod mechanics



Dynamometer card



Rod-floating detection



Surface pumping unit



Motor torque/power



Production



LAYER 3 — DECISION ENGINE

State estimation



Prediction



What-if simulation



Constraint checking



Multi-objective optimization



Recommended CSS + SRP action



OPERATOR



REAL WELL



NEW DATA



DIGITAL TWIN UPDATE

This closes the loop.

Key SIH Insight

The most important conceptual transition is this:

We are no longer building separate models for CSS, SRP, rod floating and production. We are building one coupled state-space representation of the well in which every subsystem influences the others.

That gives us the foundation for the actual **optimization engine**.

Next in sequence: 2.10 Digital Twin State Estimation, Data Assimilation & Real-Time Synchronization.

2.10 Digital Twin State Estimation, Data Assimilation & Real-Time Synchronization

We now have the coupled physical twin from Section 2.9.

The next question is:

How does the virtual well know what is actually happening in the real Baghewala well right now?

The answer is **state estimation + data assimilation**.

The physical well provides measurements, but many of the most important variables are not directly measurable continuously.

For example:

- reservoir temperature,
- reservoir inflow,
- oil viscosity,
- pump fillage,
- pump intake condition,
- rod condition,
- rod-floating risk.

Therefore, the digital twin must continuously combine:

Physics model + sensor data + historical data + ML

to estimate the current state.

2.10.1 Why State Estimation Is Necessary

FACT

The SIH problem requires a digital twin for optimization of CSS and SRP operations.

A digital twin cannot simply copy sensor values because sensors only observe part of the physical system.

For example:

Measured

→ polished-rod load

but what we actually want to know may be:

Estimated

→ pump fillage

Similarly:

Measured

→ temperature at some location

but what we need may be:

Estimated

→ reservoir thermal state.

Therefore:

Measured variables are observations; the digital-twin state is an inferred representation of the whole well.

2.10.2 Observed State vs Hidden State

We divide variables into two categories.

OBSERVABLE VARIABLES

Potentially available from SCADA/instrumentation:

- polished-rod position
- polished-rod load
- SRP speed
- stroke length
- motor current

- motor power
- wellhead pressure
- tubing pressure
- casing pressure
- temperature
- oil production rate
- water production rate
- steam injection information.

The exact instrumentation available at Baghewala must be confirmed from field data.

HIDDEN / LATENT VARIABLES

The twin may estimate:

- reservoir temperature
- reservoir pressure
- oil viscosity
- reservoir inflow
- pump intake pressure
- pump fillage
- rod mechanical state
- hydraulic restriction
- rod-floating probability.

These hidden states are what make the twin valuable.

2.10.3 Mathematical State Estimation Framework

We already defined:

$$\mathbf{x}_{t+1} = \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t, \mathbf{d}_t, \boldsymbol{\theta}) + \mathbf{w}_t$$

and:

$$\mathbf{y}_t = \mathbf{h}(\mathbf{x}_t, \mathbf{u}_t) + \mathbf{v}_t$$

Now we need to estimate:

$\hat{\mathbf{x}}_t$

from measurements:

$\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_t$

Conceptually:

$\hat{\mathbf{x}}_t = \text{Estimate}(\mathbf{y}_{0:t}, \mathbf{u}_{0:t}, \text{model})$

where:

- $\hat{\mathbf{x}}_t$ = estimated current state
 - $\mathbf{y}_{0:t}$ = all observations up to time t
 - $\mathbf{u}_{0:t}$ = control history
 - **model** = digital twin.
-

2.10.4 Prediction + Correction

The most important idea is that state estimation has two stages.

Stage 1 — Prediction

The physics model predicts what the state should become.

$\hat{\mathbf{x}}_t |_{t-1} = \mathbf{f}(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_{t-1})$

Stage 2 — Correction

The actual sensor measurements arrive.

Compare:

Predicted measurement

with:

Observed measurement

Then correct the state.

Conceptually:

Prediction

↓

Measurement

↓

Prediction error



State correction



Updated twin

This process repeats continuously.

2.10.5 Innovation / Measurement Residual

Define:

$$\mathbf{r}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$$

where:

- $\mathbf{y}_t$ = observed measurement
- $\hat{\mathbf{y}}_t$ = predicted measurement
- $\mathbf{r}_t$ = residual/innovation.

For example:

Predicted wellhead pressure:

$$\hat{\mathbf{y}}_P = 20 \text{ bar}$$

Observed:

$$\mathbf{y}_P = 23 \text{ bar}$$

Then:

$$\mathbf{r}_P = +3 \text{ bar}$$

A small residual may be normal measurement/model noise.

A persistent large residual can indicate:

- model error,
- changing well condition,
- sensor problem,
- unmodelled physics,
- equipment abnormality.

2.10.6 Why Residuals Are Extremely Valuable

We already used residuals for:

- dynamometer cards,
- motor power,
- wellbore pressure.

Now we can generalize the concept.

The twin continuously asks:

Does the real well behave as the model expects?

If yes:

Twin confidence ↑

If no:

Residual ↑

and the system investigates.

This makes residuals a common language between:

Physics + ML + anomaly detection

2.10.7 State Estimation Methods

Several methods can be used.

Option 1 — Kalman Filter

Suitable for approximately linear systems with Gaussian noise.

Option 2 — Extended Kalman Filter

Useful when the state equations are nonlinear.

Our system is nonlinear because:

$$\mu_o = f(T)$$

and:

$$q_{in} = f(P, \mu_o)$$

and:

$$Q_{\text{actual}} = f(F, N, S)$$

Therefore EKF could be considered.

Option 3 — Unscented Kalman Filter

Useful for nonlinear state-transition systems without explicitly calculating complex Jacobians.

Option 4 — Particle Filter

Useful for highly nonlinear/non-Gaussian systems, but computationally heavier.

Option 5 — Moving Horizon Estimation

Uses a recent window of measurements and optimizes the estimated state.

This can be attractive for a digital twin because it naturally incorporates constraints.

2.10.8 Recommended SIH Approach

PROPOSED DESIGN

Do not make the project dependent on a sophisticated estimator that requires large amounts of field data.

Use a layered approach:

Physics-based state prediction

-

constraint-based correction

-

ML residual correction

For example:

Physics model

↓

Predicted q_{in}

↓

Compare with production/pressure

↓

ML learns systematic residual

↓

Corrected q_{in} estimate.

This is easier to explain and demonstrate.

2.10.9 Data Assimilation

State estimation tells us **what the current state is**.

Data assimilation is the broader process of continuously incorporating new observations into the digital twin.

The loop is:

Real well

↓

New measurements

↓

Data validation

↓

State estimation

↓

Twin update

↓

Prediction

↓

New measurement

↓

Repeat.

This makes the digital twin dynamic rather than static.

2.10.10 Data Engineering Layer

This is where the team's **data-engineering expertise** becomes important.

A practical pipeline is:

Sensors / SCADA



Data ingestion



Timestamp synchronization



Data quality checks



Feature engineering



State estimation



Digital twin



ML/anomaly detection



Optimization



Dashboard

2.10.11 Time Synchronization

This is a surprisingly important engineering issue.

Suppose:

- pressure is sampled at 1 Hz,
- load at 50 Hz,
- motor current at 10 Hz,
- production every few minutes.

These signals do not naturally align.

Therefore, the system needs a common time base.

PROPOSED DESIGN

Create:

master timestamp

and resample/aggregate signals appropriately.

For example:

High-frequency stream

Rod load + position

→ cycle-level processing.

Medium-frequency stream

Motor/pressure

→ minute-level processing.

Slow stream

Production/CSS state

→ minute/hour-level processing.

This connects naturally with our multi-timescale twin.

2.10.12 Data Quality Layer

Before data enters the twin:

Check 1 — Missing values

Is data absent?

Check 2 — Range violations

Is a measurement physically impossible?

Check 3 — Sensor freezing

Has a sensor remained exactly constant for too long?

Check 4 — Spikes

Is there a sudden isolated measurement inconsistent with the surrounding signal?

Check 5 — Timestamp errors

Are measurements delayed or duplicated?

Check 6 — Cross-sensor consistency

Does the measurement agree with related variables?

For example:

Production ↑

but:

pump speed = 0

may require investigation.

2.10.13 Physics-Based Data Validation

This can become another differentiator.

Instead of validating data only with statistical rules, validate it against physics.

For example:

If:

T ↑ significantly

but:

μ_o ↑ dramatically

the system should flag a potential inconsistency if the viscosity model predicts the opposite trend.

Similarly:

If:

N ↑

but:

Q_{theoretical} ↓

something may be wrong with the data pipeline or parameter mapping.

Thus:

Physics itself becomes a data-quality validator.

2.10.14 Sensor Confidence

Every measurement should have a quality/confidence score.

For example:

Pressure sensor

Quality = 0.98

Temperature sensor

Quality = 0.72

Load sensor

Quality = 0.95

The estimator can then give more weight to reliable measurements.

PROPOSED DESIGN

Each observation becomes:

measurement = value + timestamp + quality + source

This is a strong data-engineering practice for the prototype.

2.10.15 State Confidence

The same concept applies to estimated states.

Example:

Reservoir temperature

Estimated value:

T_r = 78°C

Confidence:

0.84

Why?

Because the estimate may depend on:

- sparse temperature measurements,
- steam history,

- production response,
- thermal model uncertainty.

Therefore:

State + uncertainty

should always travel together.

2.10.16 Data Assimilation Example

Consider a CSS production cycle.

Step 1 — Before steam

Twin estimates:

$T_r = 47^\circ\text{C}$

$\mu_o = \text{high}$

$q_{in} = \text{low}$

Step 2 — Steam injection

The model predicts:

$T_r \uparrow$

Step 3 — Actual data arrives

Temperature/pressure/production observations arrive.

Suppose observed response is weaker than predicted.

Then:

Observed response < predicted response

Therefore:

Residual $\uparrow$

The twin updates:

effective thermal efficiency $\downarrow$

or identifies another possible cause.

Step 4 — New prediction

The next cycle prediction uses the corrected state.

This prevents the twin from repeatedly making the same error.

2.10.17 Online Parameter Calibration

The digital twin contains parameters such as:

- effective heat-loss coefficient
- effective thermal capacity
- viscosity-model parameters
- inflow productivity coefficient
- hydraulic resistance
- rod damping
- pump efficiency.

These parameters should not simply be guessed.

PROPOSED DESIGN

Use historical/current observations to calibrate them.

Conceptually:

$$\vartheta = \operatorname{argmin} \Sigma(y_{\text{observed}} - y_{\text{predicted}})^2*$$

subject to physical constraints.

This is parameter estimation.

2.10.18 Why Online Calibration Matters at Baghewala

A real field is not perfectly homogeneous.

Different wells can have:

- different reservoir properties,
- different completion conditions,
- different pump characteristics,
- different rod strings,
- different thermal responses.

Therefore:

One global model parameter set may not represent every well accurately.

PROPOSED DESIGN

Use:

Global model

-

Well-specific calibration

This gives us a reusable platform without assuming all wells behave identically.

2.10.19 Digital Twin Warm Start

When a new well enters the system, there may not be enough data for a fully personalized model.

Therefore:

Stage 1

Initialize from:

field-level prior

Stage 2

Use available well information.

Stage 3

Assimilate historical operating data.

Stage 4

Continuously calibrate.

This allows the twin to improve over time.

2.10.20 Handling Missing Sensors

A robust twin must not collapse when one sensor fails.

Example:

Suppose the temperature sensor becomes unavailable.

The twin can temporarily estimate temperature from:

- CSS history,
- steam input,
- production response,
- pressure,
- thermal model.

But uncertainty increases.

Therefore:

Sensor unavailable

↓

Physics-based estimation continues

↓

Confidence decreases

↓

Optimization becomes more conservative

This is preferable to simply stopping the entire system.

2.10.21 Handling Conflicting Sensors

Suppose:

Sensor A says pressure = 20 bar

Sensor B says pressure = 31 bar

The twin should not blindly choose one.

It should evaluate:

- sensor reliability,
- historical bias,
- physics consistency,
- temporal consistency.

Then assign confidence.

Potential output:

Pressure estimate = 24 ± 5 bar

Data quality = degraded

2.10.22 Real-Time Synchronization Architecture

The proposed architecture is:

PHYSICAL WELL

Sensors / SCADA



DATA ENGINE

Ingestion



Timestamp alignment



Quality control



Feature extraction



STATE ESTIMATOR



Estimated well state



DIGITAL TWIN

Physics simulation



ML LAYER

Residual correction + anomaly detection



OPTIMIZER

Candidate action simulation



OPERATOR DASHBOARD

Recommendation + confidence



Operator decision



PHYSICAL WELL

2.10.23 Update Frequency

The twin should operate at different update frequencies.

Layer	Suggested conceptual update
Rod/load signal	Every sample / cycle
Dynamometer features	Every pumping cycle
Motor/power	Seconds
Pressure	Seconds–minutes
Pump fillage	Cycle/minute
Reservoir state	Minutes–hours
CSS thermal state	Minutes–hours
Optimization	Event-driven / periodic

These are **system-design targets**, not claims about Baghewala's actual SCADA rates.

2.10.24 Event-Driven Synchronization

Optimization does not need to run continuously at maximum frequency.

It can be triggered by important events.

Examples:

Event 1

Rod-floating risk suddenly increases.

→ Recalculate.

Event 2

Reservoir temperature falls below expected range.

→ Recalculate.

Event 3

Production deviates significantly from prediction.

→ Recalculate.

Event 4

CSS cycle enters production stage.

→ Run optimization.

Event 5

Sensor quality degrades.

→ Switch to conservative/advisory mode.

This reduces computational cost.

2.10.25 Digital Twin Error Monitoring

The system should continuously monitor prediction errors.

Define:

$$e_i(t) = y_i(t) - \hat{y}_i(t)$$

for each observable variable.

Examples:

$$e_q = q_{\text{actual}} - q_{\text{predicted}}$$

$$e_P = P_{\text{actual}} - P_{\text{predicted}}$$

$$e_F = F_{\text{estimated}} - F_{\text{expected}}$$

$$e_{P_{\text{motor}}} = P_{\text{actual}} - P_{\text{predicted}}$$

Then monitor:

- mean error

- absolute error
 - RMSE
 - bias
 - drift
 - residual distribution.
-

2.10.26 Model Drift

Over time, the physical system can change.

For example:

- pump wear,
- rod degradation,
- changing fluid properties,
- reservoir depletion,
- altered thermal behaviour.

Therefore a model that worked six months ago may become less accurate.

PROPOSED DESIGN

Monitor residual drift.

If:

|residual| increases persistently

then:

model recalibration / retraining trigger

This makes the twin adaptive.

2.10.27 Digital Twin Trust Layer

This can become an important dashboard feature.

Instead of displaying only:

Predicted production = 720 bbl/day

display:

Predicted production = 720 bbl/day

Confidence = 86%

Model health = Good

Data quality = Good

Operating regime = Within validated range

This is much more credible for an industrial application.

2.10.28 Out-of-Distribution Detection

The twin should know when it is outside the conditions on which it was trained/calibrated.

For example:

- extremely unusual temperature,
- unusual pressure,
- abnormal pumping speed,
- unfamiliar dynamometer-card pattern.

Then:

OOD probability ↑

and the system should avoid aggressive recommendations.

This can be implemented using:

- distance-based methods,
- autoencoders,
- ensemble uncertainty,
- density estimation,
- conformal prediction.

The exact method can be selected later based on available data.

2.10.29 Recommended Hybrid State Estimator

For the SIH prototype, I recommend:

Layer A — Physics predictor

Predict:

$T_r, P_r, \mu_o, q_{in}, P_{intake}$

Layer B — Data assimilation

Correct state using observations.

Layer C — ML residual model

Learn systematic deviations.

Layer D — Uncertainty model

Estimate confidence.

So:

Physics

-

Measurements

-

ML

→

Estimated state + uncertainty

This is a very strong digital-twin architecture.

2.10.30 Example Real-Time Cycle

Suppose the system receives:

SRP speed

rod load

rod position

motor current

pressure

temperature

production

The pipeline performs:

Step 1

Validate data.

Step 2

Segment pumping cycle.

Step 3

Construct dynamometer card.

Step 4

Estimate pump fillage.

Step 5

Update rod dynamics.

Step 6

Update hydraulic state.

Step 7

Update reservoir state.

Step 8

Compare expected vs observed behaviour.

Step 9

Estimate rod-floating risk.

Step 10

If significant deviation exists, trigger optimization.

Step 11

Simulate candidate CSS/SRP actions.

Step 12

Recommend the best feasible action.

This is the complete **real-time digital-twin loop**.

2.10.31 Layer 2.10 Outputs

At the end of this layer, we have:

Output	Purpose
Current state estimate	Unified view of the well
State uncertainty	Confidence in estimates
Sensor quality	Trust in observations
Model residuals	Physics/data mismatch
Parameter estimates	Well-specific calibration
Model health	Twin reliability
OOD status	Whether model is operating within validated domain
Updated digital twin	Current virtual representation

2.10.32 The Critical SIH Differentiator

The important idea is:

The digital twin should not merely simulate Baghewala. It should continuously reconcile its simulation with the actual well.

That means:

Predict → Measure → Compare → Correct → Predict again

rather than:

Build model → run once → show result

This is what turns the model into a **living digital twin**.

2.10.33 Updated Architecture

The complete architecture now becomes:

REAL BAGHEWALA WELL

↓

Sensors / SCADA

↓

Data Engineering



Data quality + synchronization



STATE ESTIMATION



Reservoir + wellbore + pump + rod + surface states



COUPLED DIGITAL TWIN



Physics prediction



ML RESIDUAL / ANOMALY LAYER



Rod floating + equipment/state anomalies



OPTIMIZATION ENGINE



What-if simulation



CSS + SRP recommendation



OPERATOR



Real-world action



NEW DATA



Twin synchronization

Key SIH Insight

The deepest architectural idea so far is:

The digital twin should maintain a continuously updated belief about the well, not just a continuously updated set of sensor readings.

That “belief” contains:

State + prediction + uncertainty + model health

and becomes the input to the optimization engine.

Next in sequence: 2.11 Digital Twin Validation, Calibration & Synthetic Data Strategy.

2.11 Digital Twin Validation, Calibration & Synthetic Data Strategy

We have now built the conceptual digital twin:

Reservoir → Wellbore → Pump → Rod → Surface

and connected it to:

Sensors → State Estimation → ML → Optimization

The next critical question is:

How do we prove that our digital twin is actually correct enough to trust?

This is especially important for SIH because we are unlikely to have unrestricted access to complete, labelled, real-time Baghewala SCADA data.

Therefore, our validation strategy must distinguish between:

1. **Physics validation**
 2. **Data validation**
 3. **ML validation**
 4. **Digital-twin validation**
 5. **Optimization validation**
 6. **Synthetic-data validation**
-

2.11.1 Why Validation Is Critical

FACT

The SIH problem requires an operational digital twin for optimizing CSS and SRP operations at Baghewala.

A visually impressive dashboard is not sufficient.

If the twin predicts:

Production ↑

but the actual system would experience:

Fillage ↓ + rod load ↑ + energy ↑

then the recommendation is unsafe or economically poor.

Therefore:

Every recommendation must be traceable to validated physics, validated data relationships, or explicitly identified assumptions.

2.11.2 Validation Hierarchy

Our validation should follow a hierarchy.

Level 1 — Equation validation

Are the equations mathematically implemented correctly?

Level 2 — Physics validation

Does the model reproduce expected physical trends?

Examples:

$T \uparrow \rightarrow \mu \downarrow$

$\mu \downarrow \rightarrow \text{mobility} \uparrow$

$q_{\text{in}} \uparrow \rightarrow \text{fillage generally} \uparrow$

Level 3 — Component validation

Does each module behave correctly?

- reservoir
- wellbore
- pump
- rod
- surface.

Level 4 — Coupled validation

Do the modules behave correctly when connected?

Level 5 — Historical validation

Does the twin reproduce observed well behaviour?

Level 6 — Decision validation

Does the optimizer select actions that improve the desired objectives without violating constraints?

This hierarchy is much stronger than reporting one overall model accuracy.

2.11.3 Calibration vs Validation

These two must not be confused.

CALIBRATION

Used to determine model parameters.

Example:

Heat-loss coefficient

Productivity coefficient

Rod damping

Pump efficiency

VALIDATION

Used to test the calibrated model on data it has **not seen during calibration**.

Therefore:

A model should never be validated only on the same data used to calibrate it.

2.11.4 Parameter Calibration

The digital twin contains parameters:

$\theta = [C_{\text{eff}}, U_{\text{eff}}, J_o, A, B, C_r, K_r, \eta_{\text{pump}}, \dots]$

where these represent effective thermal, reservoir, rod and pump parameters.

We can estimate them by minimizing model error.

EQUATION

A simple calibration objective:

$$\vartheta = \operatorname{argmin} \sum_t ||y_t - \hat{y}_t(\vartheta)||^2$$

where:

- θ^* = calibrated parameter set
- y_t = observed measurements
- $\hat{y}_t(\theta)$ = model prediction
- Σ = summation over observations.

This is a proposed calibration formulation.

2.11.5 Multi-Objective Calibration

A better approach is to calibrate against several observations simultaneously.

For example:

$$J_{\text{cal}} = w_1 E_q + w_2 E_P + w_3 E_T + w_4 E_{\text{load}} + w_5 E_{\text{power}}$$

where:

- E_q = production error
- E_P = pressure error
- E_T = temperature error
- E_{load} = rod-load/card error
- E_{power} = motor-power error.

This prevents the model from becoming accurate for one variable while becoming unrealistic for another.

2.11.6 Baghewala Parameter Strategy

FACT

The SIH problem provides Baghewala-specific baseline conditions including:

- Jodhpur Sandstone reservoir
- heavy crude

- approximately 17–19° API
- reservoir temperature around 46–48°C
- high viscosity
- low reservoir pressure.

Other publicly reported OIL technical information can provide additional operating ranges, but those values must remain clearly separated from the SIH baseline.

PROPOSED DESIGN

Build a parameter registry:

Parameter	Value	Unit	Source	Confidence	Calibration
Reservoir temperature	46–48	°C	SIH	Very High	No
API gravity	17–19	°API	SIH	Very High	No
Viscosity	field-specific	cP	OIL/SIH	High/Very High	Yes
Productivity coefficient	TBD	—	Field data	TBD	Yes
Thermal-loss coefficient	TBD	—	Calibration	TBD	Yes
Rod stiffness	TBD	—	Equipment data	TBD	Yes

The exact source hierarchy established earlier should be retained.

2.11.7 Synthetic Data Is Necessary

PROBLEM

We may not have sufficient labelled Baghewala data for:

- rod floating
- pump fillage
- abnormal dynamometer cards
- pump failures
- counterfactual CSS operations.

Training a deep-learning model directly on random synthetic data would be dangerous.

Instead, synthetic data should be generated from our **physics-based twin**.

2.11.8 Physics-Based Synthetic Data Generation

PROPOSED PIPELINE

Baghewala parameter ranges



Physics model



CSS simulation



Reservoir response



Wellbore simulation



SRP simulation



Rod dynamics



Dynamometer card



Motor/power response



Production



Synthetic dataset

This produces internally consistent synthetic observations.

2.11.9 Synthetic Data Must Be Clearly Labelled

This is extremely important.

We should never present synthetic values as:

“Actual Baghewala SCADA data.”

Instead:

Data type = Physics-generated synthetic

or:

Data type = Public-parameter-calibrated synthetic

or:

Data type = Field data

The dashboard and pitch should explicitly communicate this.

That protects the technical credibility of the project.

2.11.10 What Synthetic Data Should Contain?

Each synthetic sample should contain:

Reservoir

- temperature
- pressure
- viscosity
- inflow.

CSS

- injection rate
- steam volume
- soak duration
- production stage
- cooling stage.

Wellbore

- pump intake pressure
- tubing pressure
- temperature

- pressure loss.

SRP

- speed
- stroke
- pump displacement
- fillage.

Rod

- load
- displacement
- velocity
- acceleration.

Surface

- motor current
- motor power
- torque.

Production

- oil rate
- water rate if modelled.

Labels

- normal/abnormal
- fillage
- rod-floating state
- root-cause category.

2.11.11 Generate the Entire CSS Cycle

A synthetic dataset should not contain isolated random operating points.

It should represent the actual temporal nature of CSS.

Cycle:

Injection



Soak



Production



Cooling



Next cycle

The simulation should generate a time series across all these phases.

This allows the ML system to learn **transitions**, not just static states.

2.11.12 Synthetic Scenario Families

We should create several scenario families.

Scenario 1 — Normal CSS

Normal injection → soak → production → cooling.

Scenario 2 — Insufficient heating

Low thermal response.

Scenario 3 — Excessive cooling

Viscosity rises rapidly.

Scenario 4 — Excessive SRP speed

Pump demand exceeds inflow.

Scenario 5 — Pump underfill

Low fillage.

Scenario 6 — Rod-floating condition

Abnormal dynamic response.

Scenario 7 — Valve/pump abnormality

Pump performance deviates from expected behaviour.

Scenario 8 — Hydraulic restriction

Pressure loss becomes excessive.

Scenario 9 — Sensor failure

One or more sensors become unreliable.

Scenario 10 — Combined anomaly

For example:

High viscosity + excessive speed + underfill

These combined cases are especially valuable.

2.11.13 Domain Randomization

A major danger of synthetic data is that the ML model learns the exact simulator rather than the real physical system.

To reduce this:

PROPOSED DESIGN

Randomize physically meaningful parameters within validated ranges:

- viscosity
- reservoir pressure
- thermal properties
- productivity
- rod properties
- pump efficiency
- friction
- sensor noise.

This creates many physically plausible realizations.

2.11.14 Sensor Noise

Real measurements are imperfect.

Therefore synthetic signals should include realistic measurement noise.

For example:

$$y = y_{\text{true}} + \epsilon$$

where:

$\epsilon \sim$ measurement-noise distribution

The distribution should ideally be based on actual sensor specifications/data.

If actual noise characteristics are unavailable:

Use a clearly labelled prototype assumption.

Do not claim the noise model is Baghewala-specific unless evidence exists.

2.11.15 Sensor Fault Simulation

We should deliberately generate bad data.

Examples:

Spike

One measurement suddenly jumps.

Drift

Sensor gradually moves away from true value.

Dropout

Sensor disappears.

Stuck sensor

Value remains constant.

Time shift

Sensor timestamps become misaligned.

Bias

Sensor consistently over/under-reports.

This lets us test whether the digital twin remains safe under imperfect data.

2.11.16 Synthetic-to-Real Gap

This is one of the biggest ML risks.

A model trained on synthetic data may perform extremely well on synthetic test data but poorly on real field data.

Therefore:

Synthetic accuracy $\neq$ field accuracy

This must be explicitly acknowledged.

2.11.17 How to Reduce the Synthetic-to-Real Gap

Strategy 1

Use physics-generated data rather than arbitrary synthetic data.

Strategy 2

Use public Baghewala operating ranges.

Strategy 3

Inject realistic noise.

Strategy 4

Use domain randomization.

Strategy 5

When real field data becomes available, fine-tune/calibrate.

Strategy 6

Use unsupervised/weakly supervised anomaly detection where labels are unavailable.

Strategy 7

Use physics constraints during inference.

2.11.18 ML Dataset Split

Never randomly mix nearly identical adjacent time samples between training and testing.

For time-series data, that can cause data leakage.

Instead:

Training

Earlier cycles/wells.

Validation

Different cycles/wells.

Testing

Entirely held-out cycles/wells.

Ideally:

Train wells $\neq$ test wells

if enough wells are available.

This tests whether the model generalizes to unseen wells.

2.11.19 Validation Metrics

Different components require different metrics.

Reservoir prediction

Use:

- MAE
- RMSE
- MAPE where appropriate
- R^2
- trend consistency.

Pressure prediction

Use:

- MAE
- RMSE
- maximum error
- bias.

Pump fillage

Use:

- MAE
- RMSE

- calibration error.

Rod-floating detection

Use:

- precision
- recall
- F1
- ROC-AUC
- PR-AUC.

For rare failures, **PR-AUC** is particularly useful.

Production prediction

Use:

- MAE
 - RMSE
 - relative error
 - cumulative production error.
-

2.11.20 Why Accuracy Alone Is Not Enough

Suppose:

Rod-floating classifier accuracy = 98%

Sounds excellent.

But if rod floating occurs in only 1% of samples, a model that predicts:

“Normal” every time

would already achieve approximately 99% accuracy.

Therefore, accuracy is misleading for rare-event detection.

We need:

Recall + Precision + F1 + PR-AUC

and ideally:

False-negative rate

because missing a dangerous condition is more serious than generating an additional advisory.

2.11.21 Cost-Sensitive Validation

Not every error has equal cost.

For example:

False negative

Actual rod floating

but system says:

Normal

Potentially serious.

False positive

System says:

Possible rod floating

but well is normal.

Operationally inconvenient, but usually less dangerous.

Therefore the decision system can use:

Cost(FN) > Cost(FP)

for safety-critical anomalies.

This should influence threshold selection.

2.11.22 Digital Twin Validation

The entire twin should be validated at multiple levels.

Component validation

Does each model work?

Coupled validation

Do components work together?

Temporal validation

Does the twin reproduce the evolution of a CSS cycle?

Counterfactual validation

Do predicted responses to operating changes make physical sense?

Operational validation

Does the recommendation improve the objective without violating constraints?

2.11.23 Counterfactual Validation

This is particularly important for the optimizer.

Suppose the twin predicts:

Reduce SRP speed → fillage increases

This should be physically plausible.

Similarly:

Increase temperature → viscosity decreases

should occur.

If the optimizer produces:

Increasing viscosity improves oil mobility

then the system should automatically reject the result.

Therefore:

The optimizer should be validated against physics, not only against ML predictions.

2.11.24 Conservation and Constraint Checks

Every simulation should run basic physical checks.

Examples:

$$0 \leq F \leq 1$$

$$\mu_o > 0$$

$$q_o \geq 0$$

$$N \geq 0$$

$$S \geq 0$$

Energy balance should remain physically meaningful.

Pressure should remain within valid model ranges.

Temperatures should remain physically plausible.

These are **hard sanity constraints**.

2.11.25 Scenario-Based Validation

Create predefined benchmark scenarios.

For example:

Benchmark A — Cold reservoir

Expected:

μ high

q_{in} low

Benchmark B — Heated reservoir

Expected:

μ lower

q_{in} higher

Benchmark C — Pump overspeed

Expected:

fillage decreases

Benchmark D — Reduced speed

Expected:

pump demand decreases

Benchmark E — Hydraulic restriction

Expected:

pressure loss increases

Benchmark F — Mechanical anomaly

Expected:

card residual increases

The model must reproduce these directional behaviours.

2.11.26 Digital Twin Golden Cases

For the SIH demonstration, create a set of **golden scenarios**.

These are predefined cases where the expected outcome is known.

For example:

Golden Case 1

Healthy heated well.

Golden Case 2

High-viscosity/cooling well.

Golden Case 3

SRP overspeed.

Golden Case 4

Rod-floating event.

Golden Case 5

Hydraulic restriction.

Golden Case 6

Sensor failure.

The judges can see that the twin responds appropriately to each.

2.11.27 Calibration Dashboard

The development dashboard should show:

Observed

vs

Physics prediction

vs

ML-corrected prediction

for:

- temperature
- pressure
- production
- fillage
- motor power.

This makes model improvement visible.

2.11.28 Confidence Bands

Instead of plotting:

Predicted production = 700

plot:

Predicted production = $700 \pm \text{uncertainty}$

Then compare observed production.

If actual measurements consistently fall outside the uncertainty band:

Model needs recalibration.

This is a much more mature digital-twin presentation.

2.11.29 Validation of Rod-Floating Model

For rod floating specifically:

Stage 1

Generate normal cards.

Stage 2

Generate abnormal cards.

Stage 3

Add sensor noise.

Stage 4

Train model.

Stage 5

Test on unseen scenarios.

Stage 6

Test on unseen parameter combinations.

Stage 7

Measure:

- recall
- precision
- PR-AUC
- false-negative rate.

Stage 8

Test robustness under sensor failures.

This demonstrates that the model isn't simply memorizing card shapes.

2.11.30 Validation of the Optimization Engine

The optimizer should be tested against known scenarios.

Example:

Situation

High reservoir temperature

High inflow

Low fillage

Expected recommendation:

Increase pump demand cautiously, if mechanical constraints allow.

Situation

Low reservoir temperature

High viscosity

Low inflow

Expected recommendation:

Do not aggressively increase SRP speed.

Instead:

Consider thermal recovery / CSS optimization.

Situation

High rod-floating risk

Expected recommendation:

Move away from the risky operating point.

These directional outcomes should be part of the validation suite.

2.11.31 A Strong SIH Validation Matrix

Component	Input	Output	Validation
Thermal model	Steam	Temperature	MAE/RMSE + trend
Viscosity model	Temperature	μ	physical trend + calibration
Inflow model	P, μ	q_in	MAE/RMSE
Wellbore	q, μ , P	P_intake	MAE/RMSE
Pump	q_in, N, S	Fillage	MAE/RMSE
Rod	load/motion	dynamics	card error
Rod floating	card + state	risk	PR-AUC/F1
Surface	load + speed	power	MAE/RMSE
Production	complete state	q_o	MAE/RMSE
Optimizer	state	action	constraint + scenario validation

2.11.32 Validation Levels for the Final Demo

For the SIH presentation, show three levels.

LEVEL 1 — Physics validation

Show:

Temperature $\uparrow \rightarrow$ Viscosity $\downarrow \rightarrow$ Mobility $\uparrow$

LEVEL 2 — Digital twin validation

Show:

Predicted production vs simulated/observed production

LEVEL 3 — Decision validation

Show:

Current operating point

vs

Recommended operating point

with:

- production
- energy
- fillage
- mechanical risk.

This makes the demo understandable even to judges who are not petroleum specialists.

2.11.33 What We Should NOT Claim

This is extremely important for credibility.

Do not say:

“Our model predicts actual Baghewala production with 98% accuracy”

unless actual Baghewala validation data supports it.

Do not say:

“Our synthetic data represents actual OIL SCADA.”

It does not.

Do not say:

“Our rod-floating classifier is field validated”

unless field-labelled data exists.

Instead say:

“The prototype is physics-calibrated to publicly reported Baghewala operating ranges and validated using synthetic scenario testing; field deployment requires calibration and validation against OIL data.”

That statement is honest and technically defensible.

2.11.34 Synthetic Data Architecture

The complete synthetic-data engine becomes:

Baghewala parameter ranges



Physics parameter sampler



Reservoir/CSS simulator



Wellbore simulator



SRP simulator



Rod simulator



Surface/motor simulator



Sensor/noise simulator



Fault injection



Synthetic labelled dataset



ML training



Held-out scenario validation

This can itself become a valuable component of the project.

2.11.35 Fault Injection Engine

A particularly strong innovation is to make abnormal conditions controllable.

For example:

Fault type = Rod floating

Severity = 0.7

Start time = t_0

Then the simulator modifies the physical parameters accordingly.

Similarly:

Fault type = pump leakage

Fault type = hydraulic restriction

Fault type = sensor failure

This allows us to systematically test the AI.

2.11.36 Digital Twin Benchmark Mode

We can create two modes.

MODE 1 — Live/Field Mode

Uses actual data when available.

MODE 2 — Simulation Mode

Uses physics-generated synthetic data.

Both feed the same:

State Estimator → Twin → ML → Optimizer

architecture.

This means our software can be demonstrated even without live OIL data.

2.11.37 Layer 2.11 Outputs

At the end of this layer we have:

1. **Calibration methodology**
 2. **Physics validation**
 3. **Synthetic-data generator**
 4. **Fault-injection engine**
 5. **ML validation framework**
 6. **Digital-twin validation framework**
 7. **Optimization validation**
 8. **Uncertainty/confidence framework**
 9. **Field-data transition plan**
-

2.11.38 Final Validation Philosophy

The complete philosophy should be:

FACT

→ What Baghewala/OIL actually reports.

PHYSICS

→ What engineering equations predict.

CALIBRATION

→ Parameters adjusted using observations.

ML

→ Learns systematic residuals and complex patterns.

VALIDATION

→ Tests on unseen data/scenarios.

OPTIMIZATION

→ Searches feasible actions.

SAFETY

→ Rejects physically impossible or high-risk recommendations.

This separation is essential.

Key SIH Insight

The strongest validation philosophy for our project is:

We do not claim that synthetic data proves field performance. Instead, we use physics to generate plausible scenarios, use those scenarios to develop and stress-test the system, and design the twin so that real Baghewala data can later calibrate and validate it.

That makes the project **credible, deployable, and honest** rather than artificially claiming field accuracy.

Next in sequence: 2.12 Digital Twin Optimization Engine — Joint CSS + SRP Optimization.

2.12 Digital Twin Optimization Engine — Joint CSS + SRP Optimization

Now we move from **prediction** to **decision-making**.

Up to Layer 2.11, the digital twin can answer:

“What is the current state of the well, and what is likely to happen?”

Layer 2.12 answers:

“Given the current state, what should we do next?”

This is where our project becomes an **optimization system**, rather than only a monitoring or prediction dashboard.

2.12.1 Why Joint Optimization Is Necessary

FACT

Baghewala uses thermal stimulation through CSS and artificial lift through SRP because heavy oil has poor mobility under primary recovery. The SIH problem specifically asks for optimization of CSS and SRP operations through a well-to-surface digital twin.

Therefore, treating them as two independent optimization problems is a major weakness.

PHYSICS

CSS changes:

Temperature → viscosity → mobility → reservoir inflow

while SRP changes:

Pump demand → pump intake pressure → reservoir drawdown → actual production

Therefore, the two systems interact.

Core coupling

CSS

Steam ↑

↓

Reservoir temperature ↑

↓

Viscosity ↓

↓

Mobility ↑

↓

Reservoir inflow ↑

↓

Pump supply ↑

while simultaneously:

SRP speed ↑

↓

Pump displacement ↑

↓

Wellbore flow demand ↑

↓

Pump intake conditions change

↓

Reservoir drawdown changes

↓

Reservoir inflow changes

Hence:

CSS controls the supply side; SRP controls the lifting/demand side.

The optimum must consider both simultaneously.

2.12.2 Optimization Inputs

The optimizer should not directly use raw sensor values.

It should use the **estimated digital-twin state**.

State

We can represent the current state as:

$$\hat{\mathbf{x}} = [\mathbf{T}_r, \mathbf{P}_r, \mu_o, \mathbf{q}_{in}, \mathbf{P}_{intake}, \mathbf{F}, \mathbf{F}_{rod}, \mathbf{P}_{motor}, \mathbf{q}_o, \mathbf{R}_{RF}]$$

where:

- $\mathbf{T}_r$ = reservoir temperature
- $\mathbf{P}_r$ = reservoir pressure
- μ_o = oil viscosity
- $\mathbf{q}_{in}$ = reservoir inflow
- $\mathbf{P}_{intake}$ = pump-intake pressure
- $\mathbf{F}$ = pump fillage
- $\mathbf{F}_{rod}$ = rod load
- $\mathbf{P}_{motor}$ = motor power
- $\mathbf{q}_o$ = oil production
- $\mathbf{R}_{RF}$ = rod-floating risk.

These are current **estimated states**, not necessarily all directly measured.

2.12.3 Optimization Controls

The main decision vector is:

$$\mathbf{u} = [\dot{\mathbf{m}}_s, \mathbf{V}_s, \mathbf{t}_{soak}, \mathbf{N}, \mathbf{S}]$$

where:

- $\dot{\mathbf{m}}_s$ = steam injection rate
- $\mathbf{V}_s$ = steam volume
- $\mathbf{t}_{soak}$ = soak duration
- $\mathbf{N}$ = SRP speed
- $\mathbf{S}$ = stroke length.

Depending on field operating practice, additional controls could eventually include:

- CSS cycle timing
- production restart timing
- surface pressure

- pumping schedule.

For the SIH prototype, we should keep the control space manageable.

2.12.4 The Optimization Problem

At its core:

$$\mathbf{u}^* = \operatorname{argmax} J(\mathbf{x}, \mathbf{u})$$

subject to:

$$\mathbf{g}(\mathbf{x}, \mathbf{u}) \leq \mathbf{0}$$

and physical/model constraints.

In simple terms:

Find the operating conditions that give the best overall outcome without violating safety, equipment, or process limits.

2.12.5 What Does “Best” Mean?

Maximum oil production alone is not sufficient.

Suppose:

Option A

Production = 800 BPD

Energy = very high

Rod risk = high.

Option B

Production = 770 BPD

Energy = significantly lower

Rod risk = low.

Option B may actually be the better operational choice.

Therefore we need a **multi-objective optimizer**.

2.12.6 Proposed Objective Function

A first formulation:

$$J = w_q Q_o - w_E E - w_R R_{\text{mech}} - w_A R_{\text{anomaly}} - w_C C_{\text{operational}}$$

where:

- Q_o = predicted oil production
- E = energy consumption
- R_{mech} = mechanical risk
- R_{anomaly} = anomaly/rod-floating risk
- $C_{\text{operational}}$ = operating cost
- w_q, w_E, w_R, w_A, w_C = configurable weights.

This is a **proposed design**, not a Baghewala-established equation.

2.12.7 Why Weighted Optimization Alone Is Not Ideal

A fixed weighted objective creates a problem.

For example:

If:

$$w_q = 0.8$$

the optimizer may sacrifice too much mechanical safety for production.

If:

$$w_R = 0.8$$

it may become excessively conservative.

Therefore, the stronger design is:

Use hard safety constraints + multi-objective optimization for performance.

2.12.8 Hard Constraints

The optimizer should first reject unsafe actions.

Examples:

Pump fillage

$$F \geq F_{\text{min}}$$

Rod load

$$F_{\text{rod}} \leq F_{\text{max}}$$

Motor power

$$P_{\text{motor}} \leq P_{\text{max}}$$

Pump speed

$$N_{\text{min}} \leq N \leq N_{\text{max}}$$

Stroke

$$S_{\text{min}} \leq S \leq S_{\text{max}}$$

Pressure

$$P_{\text{intake}} \geq P_{\text{min}}$$

Temperature

Stay within the validated thermal operating domain.

Rod-floating risk

$$R_{\text{RF}} \leq R_{\text{RF,max}}$$

Actual numerical limits must come from equipment/field specifications.

We must not invent them.

2.12.9 Feasible vs Optimal

This distinction is important.

Suppose we generate 100 candidate operating points.

Maybe:

60 are physically feasible

and:

40 violate constraints.

We first eliminate the 40.

Then optimize among the 60.

So:

Physics/safety constraints → feasible region

↓

Optimization → best point

This is much safer than asking an ML model to directly output a control action.

2.12.10 Candidate Action Simulation

This is where our digital twin becomes powerful.

Instead of immediately changing the well operation:

Step 1

Estimate current state.

Step 2

Generate candidate actions.

Example:

- current speed
- speed -5%
- speed -10%
- speed +5%
- speed +10%

and similarly for CSS parameters.

Step 3

Run each candidate through the digital twin.

Step 4

Predict:

- production
- fillage
- energy
- rod load
- pump intake pressure
- rod-floating probability.

Step 5

Reject unsafe candidates.

Step 6

Rank remaining candidates.

Step 7

Present recommendation.

This is **what-if simulation**.

2.12.11 Example Candidate Matrix

Suppose the current condition is:

High viscosity + low inflow + moderate SRP speed

The optimizer could evaluate:

Candidate SRP Speed CSS Action			Predicted Production	Fillage	Risk
A	Current	None	Medium	Low	Medium
B	-5%	None	Medium	Higher	Low
C	-10%	None	Medium	High	Low
D	Current	More thermal input	High	Higher	Medium
E	+10%	None	Low/Medium	Lower	High
F	+10%	More thermal input	High	Medium	High

The optimizer may select D or B depending on the actual model state and objective weights.

The important point:

The system explains why the recommendation was selected.

2.12.12 Joint CSS-SRP Interaction

Consider a cold reservoir.

Current condition

T_r ↓

↓

$\mu_o \uparrow$

↓

$q_{in} \downarrow$

↓

pump fillage ↓

Increasing SRP speed may therefore make the situation worse:

$N \uparrow \rightarrow \text{pump demand} \uparrow \rightarrow \text{fillage} \downarrow$

Instead, thermal stimulation may be more beneficial:

$CSS \rightarrow T_r \uparrow \rightarrow \mu_o \downarrow \rightarrow q_{in} \uparrow$

Then, after the reservoir response improves:

SRP speed can be increased cautiously.

This produces an important control philosophy:

Do not optimize SRP speed without considering reservoir thermal state.

2.12.13 Dynamic Optimization

The optimum today may not be the optimum tomorrow.

CSS changes reservoir state over time.

Therefore:

$u^*t \neq u^*\{t+1\}$

The optimizer should continuously reconsider the operating point.

This leads naturally to **Model Predictive Control (MPC)** as a future/high-value architecture.

2.12.14 Model Predictive Control Concept

MPC works conceptually as:

Current state

↓

Predict future trajectory

↓

Evaluate multiple control sequences



Select best sequence



Apply only first action



Measure new state



Re-optimize

This is ideal for a digital twin because the twin already provides the predictive model.

2.12.15 Why MPC Is Attractive for SIH

A static optimizer says:

“Set SRP speed to 5 SPM.”

An MPC-style system says:

“Given the current thermal state and expected cooling, operate at this speed now; reassess after the next measurement window.”

This is much closer to real operational decision-making.

However:

PROPOSED DESIGN

For the SIH prototype:

Use MPC conceptually

but implement a simpler:

receding-horizon candidate simulation optimizer

unless sufficient real-time data and computational infrastructure are available.

This reduces implementation risk.

2.12.16 Multi-Timescale Optimization

Our system has several timescales.

Fast

Rod dynamics:

milliseconds → seconds

Medium

Pump/wellbore:

seconds → minutes

Slow

Reservoir thermal response:

minutes → hours → days

Therefore one optimizer should not blindly operate at one frequency.

Proposed hierarchy

Level 1 — Safety

Continuous/cycle-level.

Level 2 — SRP optimization

Minutes/cycle-level.

Level 3 — CSS optimization

Hours/cycle-stage-level.

This is more realistic.

2.12.17 Hierarchical Optimization

We can implement:

Layer A — Safety Controller

Checks:

- rod load
- motor load
- pressure
- fillage

- anomaly risk.

Layer B — SRP Optimizer

Optimizes:

- speed
- stroke.

Layer C — CSS Optimizer

Optimizes:

- steam rate
- steam volume
- soak duration.

Layer D — Joint Coordinator

Ensures that CSS and SRP decisions are mutually compatible.

This hierarchical architecture is highly suitable for our project.

2.12.18 Why Not Let One Neural Network Control Everything?

This is a major design decision.

A black-box deep-learning model could theoretically map:

Sensors → control action

But it has major disadvantages:

- difficult to explain
- difficult to validate
- unsafe outside training distribution
- difficult to enforce hard physical constraints
- difficult to justify to an industrial operator.

Therefore:

AI should recommend within a physics-constrained optimization framework, not replace the engineering model.

This is one of our strongest differentiators.

2.12.19 Role of AI/ML in Optimization

ML should help estimate what physics alone cannot easily capture.

Examples:

ML Model 1

Pump fillage estimation.

ML Model 2

Rod-floating probability.

ML Model 3

Production residual prediction.

ML Model 4

Equipment degradation prediction.

ML Model 5

Thermal-response correction.

The optimizer then uses these predictions.

So:

Physics → state

ML → uncertainty/residual/complex behaviour

Optimizer → decision

2.12.20 Residual-Based Optimization

Suppose the physics model predicts:

Production = 750 BPD

but the actual system gives:

700 BPD

Residual:

$R_q = 700 - 750 = -50$ BPD

If this residual persists, the optimizer should not blindly trust the physics model.

The ML residual model can learn:

$$\mathbf{R_q} = \mathbf{f}(\mathbf{data})$$

and produce:

$$\mathbf{Corrected\ production\ prediction} = \mathbf{Physics\ prediction} + \mathbf{ML\ residual}$$

This gives a hybrid model.

2.12.21 Optimization Under Uncertainty

The digital twin does not know the state perfectly.

For example:

$$\mathbf{T_r} = \mathbf{62^{\circ}C} \pm \mathbf{5^{\circ}C}$$

rather than exactly 62°C.

Similarly:

$$\mathbf{q_in} = \mathbf{80} \pm \mathbf{10\ BPD}$$

Therefore, the optimizer should account for uncertainty.

Conceptually:

maximize expected performance

while controlling:

risk under uncertainty.

2.12.22 Conservative Optimization

If uncertainty is high:

recommendations should become more conservative.

For example:

Twin confidence = 95%

→ normal optimization.

Twin confidence = 60%

→ restricted operating changes.

Twin confidence = 30%

→ advisory/monitoring only.

This connects directly with the model-health system developed in Layer 2.10.

2.12.23 Trust-Aware Optimization

We can define:

$$J_{\text{eff}} = J_{\text{performance}} - \lambda U$$

where:

- **$J_{\text{performance}}$** = expected operational value
- **U** = uncertainty
- **λ** = penalty on uncertainty.

Again, this is a proposed formulation.

The optimizer therefore prefers:

Good performance with high confidence

over:

Excellent predicted performance with very high uncertainty.

2.12.24 Explainable Recommendations

The dashboard should never simply say:

“Increase SRP speed.”

Instead:

Recommendation

Increase SRP speed by 5%

Why?

- Reservoir temperature is high.
- Estimated viscosity is low.
- Inflow capacity has increased.
- Pump fillage is currently 92%.
- Rod-floating risk remains low.

- Motor load remains within limits.

Expected impact

Production: +X%

Energy: +Y%

Risk: +Z%

where X/Y/Z come from the model and are shown with uncertainty.

This is far more convincing to judges.

2.12.25 Counterfactual Explanation

Another powerful feature:

“What happens if I don't follow the recommendation?”

For example:

Current plan

Maintain current SRP speed.

Predicted:

Production decline over next period

Recommended plan

Reduce speed by 5%.

Predicted:

Production slightly lower immediately but fillage improves and mechanical risk decreases.

Alternative

Increase speed by 10%.

Predicted:

Short-term production increase but high underfill/rod risk.

This turns the digital twin into a **decision-support simulator**.

2.12.26 Pareto Front

Because production, energy and risk conflict, there may not be one universal optimum.

Instead we can produce a **Pareto front**.

Example:

Operating Point	Production	Energy	Risk
A	High	High	High
B	Medium-high	Medium	Low
C	Medium	Low	Very low

The operator can choose depending on operational priorities.

This is more realistic than claiming:

“There is one mathematically perfect operating point.”

2.12.27 Operator-in-the-Loop

PROPOSED DESIGN

The prototype should remain:

Decision Support System

rather than:

Fully Autonomous Control System

Flow:

Twin



Optimization



Recommendation



Operator review



Approval



Action



Measurement

This is safer and more credible for an industrial prototype.

2.12.28 Optimization Failure Handling

If:

- sensor quality is poor
- twin residual is too high
- uncertainty is too high
- state is outside training domain
- constraints cannot be evaluated
- critical sensor fails

then:

Optimization = disabled

and system switches to:

Monitoring / advisory mode

This should be hard-coded into the safety architecture.

2.12.29 Optimization Objective Hierarchy

The decision hierarchy should be:

Priority 1

Safety

Priority 2

Equipment integrity

Priority 3

Process stability

Priority 4

Production

Priority 5

Energy/cost optimization

The exact business priorities may later be configurable.

This avoids a dangerous situation where the optimizer sacrifices equipment safety to maximize production.

2.12.30 Joint CSS-SRP Optimization Example

Consider a hypothetical current state:

Reservoir temperature: declining

Viscosity: increasing

Inflow: declining

Fillage: 70%

SRP speed: high

Rod-floating risk: medium-high

The optimizer evaluates:

Action A

Increase SRP speed.

Prediction:

pump demand ↑

fillage ↓

rod risk ↑

Rejected.

Action B

Reduce SRP speed.

Prediction:

pump demand ↓

fillage ↑

rod risk ↓

Feasible.

Action C

Increase thermal stimulation.

Prediction:

$T \uparrow$

$\mu \downarrow$

$q_{in} \uparrow$

fillage potentially $\uparrow$

Feasible.

Action D

Increase SRP speed + thermal stimulation.

Potentially highest production but increased mechanical risk.

The optimizer may choose:

C first $\rightarrow$ allow reservoir response $\rightarrow$ then reassess SRP speed.

This is exactly the kind of reasoning we want the judges to see.

2.12.31 Optimization Algorithm Options

Several algorithms are possible.

Option 1 — Grid Search

Evaluate predefined combinations.

Pros: simple, transparent.

Cons: computationally expensive as dimensions increase.

Excellent for SIH prototype.

Option 2 — Bayesian Optimization

Useful when simulations are expensive.

Pros: sample efficient.

Cons: more complex.

Option 3 — Genetic Algorithm / NSGA-II

Excellent for multi-objective Pareto optimization.

Pros: handles multiple objectives and nonlinear constraints.

Cons: computationally heavier.

Option 4 — Model Predictive Control

Excellent for dynamic systems.

Pros: explicitly handles future trajectories and constraints.

Cons: requires reliable dynamic model.

Option 5 — Reinforcement Learning

Potentially powerful.

Pros: learns sequential decision policies.

Cons: risky, data-hungry, difficult to validate, unsafe for direct field deployment.

2.12.32 Recommended SIH Strategy

We should **not** attempt to build a fully autonomous RL controller.

Instead:

V1

Physics-constrained grid/candidate search

V2

Multi-objective optimizer / NSGA-II

V3

MPC

Future

Safe RL / hybrid RL-MPC, only after adequate validation.

This is both technically defensible and achievable.

2.12.33 Our Strongest Optimization Architecture

The proposed system becomes:

Real-time data



State estimator



Digital twin



Generate candidate CSS + SRP actions



Physics simulation



ML residual correction



Uncertainty estimation



Hard safety constraints



Multi-objective ranking



Pareto-optimal actions



Explainable recommendation



Operator approval



Action



New data

This is the core decision loop of our project.

2.12.34 Optimization Outputs

For every recommendation, output:

Recommended action

Example:

SRP speed: 4.5 → 4.2 SPM

Expected production

+X BPD

Expected energy

+Y kWh/day

Fillage

Z%

Rod risk

Low / Medium / High

Confidence

XX%

Reason

Natural-language explanation.

Alternatives

Show 2–3 feasible alternatives.

This makes the system operator-friendly.

2.12.35 Key Innovation

The strongest conceptual innovation is:

Jointly optimize thermal stimulation and artificial lift using a physics-constrained, uncertainty-aware digital twin rather than optimizing CSS and SRP independently.

The chain is:

CSS

- reservoir state
- inflow
- pump supply
- SRP operating envelope
- production/risk

and simultaneously:

SRP

- pump demand
- pressure boundary
- reservoir inflow
- production.

This creates a genuine **well-to-surface closed-loop optimization system**.

2.12.36 Layer 2.12 Final Architecture

Physical Well



SCADA / Sensors



Data Engineering



State Estimation



Well-to-Surface Digital Twin



Candidate CSS + SRP Actions



Physics Simulation



ML Residual / Anomaly Models



Uncertainty Estimation



Hard Constraints



Multi-Objective Optimization



Pareto Ranking



Explainable Recommendation



Operator Approval



Physical Well



New Data

This is the complete decision loop.

Most Important SIH Takeaway

Our project should **not** be pitched as:

“AI predicts oil production.”

It should be pitched as:

“A physics-informed, data-assimilated well-to-surface digital twin that jointly simulates and optimizes CSS thermal stimulation and SRP artificial lift while predicting pump fillage, rod dynamics, rod-floating risk, production and energy under operational constraints.”

That is a substantially stronger technical proposition.

Next in sequence: **2.13 Machine Learning Architecture** — exactly which ML models we should use, what each model predicts, what data it needs, and why.

2.13 Machine Learning Architecture — ML Models, Inputs, Outputs & Roles

We now have the **physics-based digital twin** and the **optimization engine**.

The next question is:

Where exactly should AI/ML be used, and where should it NOT be used?

This distinction is extremely important for SIH. We should not put deep learning everywhere just because the team has AI/ML expertise.

Our strongest architecture is:

Physics handles what we understand.

ML handles what physics cannot easily capture.

Optimization decides what action to take.

2.13.1 Core ML Philosophy

PROPOSED DESIGN

Our architecture should follow:

Physical system

↓

Physics model

↓

Residual / hidden-state problem

↓

ML model

↓

Corrected prediction / anomaly

↓

Optimization

This is a **hybrid physics + ML architecture**.

The ML system should therefore not replace the digital twin.

It should make the twin:

- more accurate
 - adaptive
 - anomaly-aware
 - capable of estimating hidden states
 - better at handling nonlinearities.
-

2.13.2 Where ML Is Actually Needed

There are several areas where first-principles physics becomes difficult.

1. Pump fillage estimation

Fillage may not be directly measured continuously.

2. Rod-floating detection

The relationship between dynamometer-card shape and abnormal behaviour can be complex.

3. Production residual prediction

Physics may capture the dominant trend but miss field-specific behaviour.

4. Equipment anomaly detection

Mechanical degradation can produce subtle patterns.

5. Sensor anomaly detection

Bad sensors need to be identified automatically.

6. Parameter adaptation

Well behaviour changes over time.

These are excellent ML targets.

2.13.3 ML Architecture Overview

Our ML layer can contain:

Model A

Pump Fillage Estimator

Model B

Rod-Floating Detector

Model C

Dynamometer Card Representation Model

Model D

Production Residual Model

Model E

Sensor Anomaly Detector

Model F

Equipment Health / Remaining Useful Life model

The first four are the most important for the SIH prototype.

2.13.4 Model A — Pump Fillage Estimator

PROBLEM

Pump fillage is a hidden state.

We know:

$$F = V_{\text{filled}} / V_{\text{available}}$$

but direct continuous measurement may not be available.

Therefore we estimate it.

Inputs

Potential inputs:

- reservoir inflow estimate
- pump speed
- stroke
- pump displacement
- pump-intake pressure
- tubing pressure

- temperature
 - estimated viscosity
 - polished-rod load
 - polished-rod position
 - motor current
 - production rate.
-

Output

$\hat{F}$ = estimated pump fillage

with:

$$0 \leq \hat{F} \leq 1$$

and uncertainty:

σ_F

2.13.5 Candidate Models for Fillage

Random Forest / XGBoost

Good baseline.

Advantages:

- works well with tabular data
- interpretable
- fast
- works with limited datasets.

MLP

Useful for nonlinear relationships.

LSTM/GRU

Useful when historical time-series context matters.

Temporal Convolutional Network

Good for sequential signals with efficient computation.

Transformer

Potentially powerful for long sequences but probably excessive for V1.

2.13.6 Recommended Fillage Model

For SIH:

Baseline

XGBoost

Advanced

TCN/LSTM

Final

Hybrid:

Physics estimate + XGBoost residual correction

For example:

F_{physics}

↓

ML predicts:

$\Delta F = F_{\text{actual}} - F_{\text{physics}}$

Then:

$\hat{F} = F_{\text{physics}} + \Delta F_{\text{ML}}$

This is more robust than asking ML to predict fillage from scratch.

2.13.7 Model B — Rod-Floating Detection

This is one of our most valuable AI components.

Input

Dynamometer card + time-series + operating state.

Output

P(rod floating)

For example:

Rod-floating probability = 0.87

along with:

- severity
 - confidence
 - likely cause.
-

2.13.8 Multi-Modal Rod-Floating Model

Instead of using only the dynamometer card:

Input stream 1

Dynamometer card

Input stream 2

Time-series

- load
- position
- velocity
- acceleration
- motor current.

Input stream 3

Hydraulic variables

- pressure
- temperature
- flow.

Input stream 4

Reservoir state

- T_r
- μ_o
- q_{in} .

Then combine them.

2.13.9 Dynamometer Card as an Image

A pumping cycle can be represented as:

Load vs Position

This creates a 2D curve/image.

A CNN can learn characteristic geometric patterns.

Possible architectures:

- ResNet
- EfficientNet
- Vision Transformer
- lightweight CNN.

For SIH, a lightweight CNN is sufficient.

2.13.10 Dynamometer Card as Time Series

Instead of converting it to an image:

[position(t), load(t)]

can be treated directly as a sequence.

Possible models:

- 1D CNN
- TCN
- LSTM
- GRU
- Transformer.

Recommended

TCN/1D CNN

for V1 because it is computationally efficient.

2.13.11 Hybrid Card Model

Our strongest design is:

Physics-generated expected card

compared with:

Observed card

to produce:

Card residual

Then feed:

Observed card + residual features + operating state

into the ML model.

This means the AI learns:

What the physics model fails to explain.

That is much stronger than pure image classification.

2.13.12 Rod-Floating Feature Engineering

We should calculate physics-informed features.

Load features

- maximum load
- minimum load
- load range
- average load
- load variance.

Position features

- stroke length
- cycle duration
- reversal points.

Dynamic features

- velocity
- acceleration

- jerk.

Card geometry

- enclosed area
- slope
- curvature
- asymmetry
- upper/lower transition behaviour.

Hydraulic features

- pump-intake pressure
- pressure differential
- temperature
- viscosity.

Reservoir features

- q_{in}
- T_r
- μ_o .

Operational features

- speed
- stroke
- cycle number
- CSS stage.

These features give the model engineering context.

2.13.13 Model C — Production Prediction

Production is a key output of the digital twin.

Instead of using ML alone:

Physics model

Predicts:

q_physics

ML model

Predicts:

Δq

Then:

$q_{pred} = q_{physics} + \Delta q$

This is our preferred approach.

2.13.14 Production Residual Model

The residual can depend on:

- temperature
- viscosity
- pressure
- inflow
- pump fillage
- SRP speed
- stroke
- wellbore pressure
- historical production
- CSS cycle number
- time since stimulation.

Therefore:

$\Delta q = f_{ML}(x, u, \text{history})$

Possible models:

- XGBoost
- Random Forest
- MLP
- LSTM/GRU

- TCN.

Recommendation

Use:

XGBoost baseline

and:

TCN/LSTM for temporal refinement.

2.13.15 Model D — Sensor Anomaly Detection

Before feeding sensor data to the twin, we need to determine:

Can we trust this sensor?

This is critical because bad data can cause bad optimization.

Types of sensor faults

Spike

Sudden abnormal value.

Drift

Gradual change.

Freeze

Constant value.

Dropout

Missing data.

Bias

Persistent offset.

Time misalignment

Incorrect timestamp.

Cross-sensor inconsistency

One sensor disagrees with physically related measurements.

2.13.16 Sensor Anomaly Model

We can combine:

Rule-based checks

- range limits
- rate-of-change limits
- missingness.

Statistical methods

- z-score
- rolling statistics
- change-point detection.

ML

- Isolation Forest
- Autoencoder
- One-Class SVM.

Physics consistency

Compare:

observed

vs

physics-predicted

values.

This hybrid approach is much more reliable than an ML-only detector.

2.13.17 Model E — Equipment Health

Eventually we can estimate equipment degradation.

Inputs:

- motor current
- motor power
- torque

- rod load
- dynamometer card
- speed
- vibration if available
- pressure
- temperature.

Output:

Equipment Health Score

Example:

Health = 82/100

This can support predictive maintenance.

However:

IMPORTANT

This should be a **secondary/future feature**, not the core SIH feature.

We should not spread development effort too thin.

2.13.18 Model F — Thermal Response Correction

The physics model predicts:

$T_r(t)$

but the actual reservoir response may differ because of:

- heterogeneity
- heat losses
- steam distribution
- changing thermal properties
- imperfect parameterization.

Therefore ML can predict:

$\Delta T = T_{\text{observed}} - T_{\text{physics}}$

Then:

$$T_{\text{pred}} = T_{\text{physics}} + \Delta T_{\text{ML}}$$

This creates an adaptive thermal model.

2.13.19 Which Models Should We Actually Build?

For the SIH MVP, I recommend only **four core ML models**.

ML-1

Pump Fillage Estimator

ML-2

Rod-Floating Detector

ML-3

Production Residual Predictor

ML-4

Sensor/Model Anomaly Detector

Everything else can be future expansion.

This keeps the system focused.

2.13.20 Complete ML Pipeline

The architecture becomes:

Raw data



Data quality



Feature engineering



Physics state estimation



Physics predictions



Residual calculation



ML models



State correction / anomaly probability



Uncertainty estimation



Digital twin state



Optimizer

This creates a very clean separation of responsibilities.

2.13.21 ML Input Feature Groups

Instead of one giant feature table, use feature groups.

Group A — Reservoir

- T_r
- P_r
- μ_o
- q_{in}
- thermal state.

Group B — CSS

- steam rate
- steam volume
- steam pressure
- steam temperature
- soak time
- cycle number

- stage.

Group C — Wellbore

- pump-intake pressure
- tubing pressure
- wellhead pressure
- temperature
- pressure drop.

Group D — SRP

- speed
- stroke
- cycle time.

Group E — Rod

- load
- position
- velocity
- acceleration.

Group F — Surface

- motor current
- motor power
- torque.

Group G — Historical

- previous production
- previous fillage
- previous cycle
- previous anomaly score.

This makes feature provenance clear.

2.13.22 Temporal Context

A major mistake would be to predict everything from the current timestamp alone.

Oil wells have memory.

For example:

Current production

depends partly on:

- previous CSS cycle
- previous temperature
- previous production
- previous pumping conditions.

Therefore ML models should use historical windows.

Example:

$X(t-K:t)$

rather than only:

$X(t)$.

2.13.23 Multi-Timescale ML

Because the physical system operates at different timescales:

Rod model

High-frequency.

Pump model

Cycle-level.

Wellbore model

Seconds/minutes.

Reservoir model

Minutes/hours/days.

Therefore we should not force all signals into one sampling frequency.

Instead:

High-frequency stream

→ rod/card model.

Medium-frequency stream

→ pump/hydraulic model.

Low-frequency stream

→ reservoir/CSS model.

Then fuse their outputs.

2.13.24 Feature Fusion

We can create:

z_rod

from dynamometer data.

z_hyd

from pressure/flow.

z_res

from reservoir state.

z_surface

from motor/power.

Then:

$z = \text{Fusion}(z_{\text{rod}}, z_{\text{hyd}}, z_{\text{res}}, z_{\text{surface}})$

and use this for anomaly classification.

This is a strong multimodal architecture.

2.13.25 Physics-Informed ML

We should explicitly enforce physical constraints.

For example:

Constraint

$F \in [0,1]$

Constraint

$$\mu > 0$$

Constraint

$$q \geq 0$$

Trend

Generally:

$$T \uparrow \rightarrow \mu \downarrow$$

The ML model should not be allowed to produce obviously impossible predictions.

2.13.26 Physics-Informed Loss

For a model predicting temperature, we can conceptually define:

$$L = L_{\text{data}} + \lambda_1 L_{\text{physics}} + \lambda_2 L_{\text{constraint}}$$

where:

- **L_{data}** = prediction error
- **L_{physics}** = violation of governing physics
- **$L_{\text{constraint}}$** = physical constraint violation.

This is a proposed design.

For SIH V1, we do not necessarily need a full PINN.

A simpler residual-learning approach is more practical.

2.13.27 Why We Should NOT Build a Full PINN

Physics-Informed Neural Networks are attractive academically.

But for this project:

- data is heterogeneous
- system is multi-timescale
- multiple coupled subsystems exist
- field data is limited
- optimization requires fast inference.

A complete PINN could consume too much development time.

Therefore:

Use physics-based reduced-order models + ML residual learning instead of a monolithic PINN.

This is more achievable and easier to explain.

2.13.28 Model Selection Strategy

Every ML problem should have:

Baseline

Simple interpretable model.

Advanced model

Deep learning.

Final selection

Whichever performs better **and remains explainable/robust**.

Example:

Problem	Baseline	Advanced
Fillage	XGBoost	TCN
Rod floating	XGBoost	CNN/TCN
Production residual	XGBoost	LSTM/TCN
Sensor anomaly	Isolation Forest	Autoencoder

This gives us a proper benchmarking framework.

2.13.29 Explainability

Industrial operators need to know:

Why did the model say this?

For tabular models:

SHAP

can identify influential features.

Example:

Rod-floating probability = 0.88

Top contributors:

1. High pump speed
2. Low estimated fillage
3. Abnormal card residual
4. High dynamic load
5. Low reservoir inflow.

This is far better than:

“AI detected rod floating.”

2.13.30 Uncertainty for ML Predictions

Every prediction should ideally include:

Prediction

-

Confidence

-

Model health

-

Out-of-distribution status

For example:

Rod-floating probability: 0.86

Confidence: 0.91

Model health: Good

Data quality: 96%

OOD: No

This is a major trust feature.

2.13.31 Out-of-Distribution Detection

Suppose the ML model was trained on:

40–100 BPD

but receives:

300 BPD

The model may produce a confident-looking but unreliable prediction.

Therefore:

OOD detector

should identify whether the current state is outside the training domain.

If OOD:

Reduce recommendation aggressiveness or switch to advisory mode.

2.13.32 Online Learning

Well conditions change.

Therefore we eventually need:

Model monitoring → drift detection → recalibration → retraining

But we should **not** automatically retrain models on every new observation.

Instead:

1. Monitor residuals.
2. Detect systematic drift.
3. Check data quality.
4. Determine whether physical parameters changed.
5. Recalibrate.
6. Retrain if necessary.
7. Validate before deployment.

This avoids model degradation from bad data.

2.13.33 ML Model Registry

For an industrial system, every model should have metadata:

- model version

- training data range
- training date
- features
- validation metrics
- applicable wells
- applicable operating range
- uncertainty
- deployment status.

This can become part of our data-engineering architecture.

2.13.34 Model Governance

For every prediction, we should know:

Which model generated it?

Which version?

What data was used?

Was the state inside the training domain?

What was the confidence?

This is especially important when recommendations influence physical operations.

2.13.35 Final ML Architecture

The complete ML subsystem becomes:

Raw SCADA / Sensor Data

↓

Data Quality Engine

↓

Physics-Based State Estimation

↓

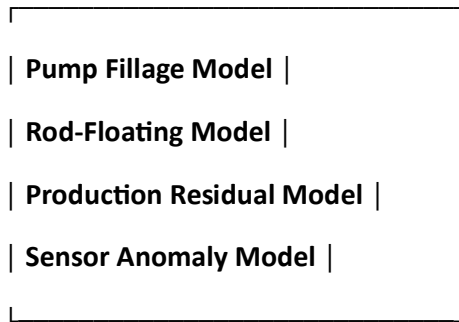
Physics Predictions



Residual Generation



Feature Engineering



Multimodal Fusion



Uncertainty + OOD Detection



Corrected Digital Twin State



Optimization Engine

This is the architecture I would recommend for the SIH prototype.

2.13.36 Mapping ML to Team Skills

Your team's existing skills fit this architecture very well.

AI/ML

- XGBoost
- anomaly detection
- residual learning
- uncertainty

- optimization interfaces.

Deep Learning

- CNN for dynamometer cards
- TCN/LSTM for time-series
- multimodal fusion.

Computer Vision

- dynamometer-card image representation
- visual anomaly classification
- AR/VR visual overlays later.

Data Engineering

- SCADA pipeline
- time synchronization
- feature store
- data quality
- model registry
- historical data management.

This gives each team specialization a meaningful role.

2.13.37 What We Should Build First

Priority order:

P0 — Essential

Pump fillage estimator

Rod-floating detector

Production residual model

Physics residual engine

P1 — Strong addition

Sensor anomaly detection

SHAP/explainability

OOD detection

P2 — Future

Equipment health

Remaining useful life

Advanced deep-learning models

Safe reinforcement learning

2.13.38 What We Should NOT Build

To maximize SIH execution probability, avoid:

✗ **Giant end-to-end neural network**

Too difficult to validate.

✗ **Fully autonomous RL controller**

Too risky and data-hungry.

✗ **Full 3D reservoir simulator**

Too computationally heavy for the prototype.

✗ **PINN for the entire well**

Unnecessary complexity.

✗ **Ten unrelated ML models**

Creates an impressive-looking but incoherent system.

Instead:

Build fewer models that directly contribute to the well-to-surface optimization loop.

2.13.39 The One-Line ML Story for Judges

If a judge asks:

“Where exactly is AI in your system?”

The answer should be:

“Physics predicts the expected well behaviour; AI learns the residuals and hidden states that physics cannot reliably capture—especially pump fillage, rod-floating risk and

production deviations—and the optimizer uses these predictions to select the safest high-value CSS and SRP operating point.”

That is a very strong answer.

2.13.40 Final Layer 2.13 Architecture

The entire intelligence stack is:

Physics

→ expected behaviour



State Estimation

→ current hidden state



ML

→ residuals + anomalies + hidden states



Uncertainty/OOD

→ trust level



Digital Twin

→ predicted future states



Optimization

→ candidate actions



Safety Constraints

→ reject unsafe actions



Explainable Recommendation

→ operator decision.

Key SIH Insight

Our biggest AI differentiation is **not** “using deep learning.”

It is:

Using AI exactly where the physics model is insufficient, while keeping the AI constrained by physics and feeding its uncertainty into the optimizer.

That gives us a **Physics + Data + AI + Optimization** architecture rather than a generic AI dashboard.

Next in sequence: 2.14 Data Engineering Architecture — SCADA ingestion, time-series storage, feature engineering, data quality, data lake/warehouse, streaming, feature store, and how all data flows through the digital twin.

ERRATA

Added 12 September 2026

This page corrects four points in the research document. It was added after a review of all project documents against the merged technical reference (SIH26120), which is the source of truth for the model. The pages before this one are unchanged.

1. Viscosity law — Sections 3.4, 3.8, 3.11 and 7.2

Written: $\ln(\mu_o) = A + B/T_K$.

Correction: The model uses the ASTM D341 (Walther) form instead: $\log_{10}(\log_{10}(v + 0.7)) = A - B \cdot \log_{10}(T)$, with T in kelvin, v in cSt and $\mu = v \times SG$. It is anchored at 11,500 cP at 50 °C (the midpoint of the reported 10,000–13,000 cP), with $SG = 0.963$, $B = 3.5$ (assumed) and $A = 9.3933$. See the technical reference, equation 3. The physical statement $T \uparrow \Rightarrow \mu_o \downarrow$ is unchanged.

2. Baseline values — Sections 5.3 and 5.12

Written: the model baseline is 46–48 °C and 17–19° API, from the SIH problem statement.

Correction: The model uses the Oil India field values: native reservoir temperature $T_i = 50$ °C (the reported bottomhole temperature) and oil $SG = 0.963$ (from $\rho_o = 963$ kg/m³, about 15.4° API). The SIH values are kept for reference, and the difference is cited openly. See the technical reference, section 5, "On the conflicting values".

3. Cause of rod floating — Sections 2.6.4, 2.6.12, 7.6 and 8.3

Written: rod floating is linked to low pump fillage and low reservoir inflow (Section 2.6.4, and Section 2.6.12, causes A to C). Cause C traces high viscosity to floating through lower inflow and pump underfill. Layer 1 is said not to capture rod floating (Sections 7.6 and 8.3).

Correction: Rod floating happens when the rods cannot fall through the oil in the tubing as fast as the polished rod moves down. It occurs when the required fall speed, $S \cdot N/30$, is greater than the available fall speed, which drops as oil viscosity rises. Its causes are high oil viscosity, high pump speed and a long stroke. Low fillage causes fluid pound, which is a different fault with a different dynamometer-card shape. The two can occur together, but one does not cause the other.

The technical reference (equation 5) computes the floating limit $N_{\max}$ directly from viscosity, so the physics layer does predict it. Dynamometer-card analysis (Layer 2) is still needed to confirm floating in the field. Section 2.3.14 ("Low fillage should not automatically be labelled rod floating") agrees with this correction.

4. CSS stages — Section 6.10

Written: four explicit stages — injection, soak, production and cooling. Section 3.11 gives the same sequence.

Correction: The model uses three phases: injection, soak and production. Cooling is not a separate stage. It happens during production, and heat lost during soak is included in the thermal efficiency. This agrees with Section 4.6, which says cooling is not necessarily a separate operational step but the natural result of heat input stopping. See the technical reference, equation 7.